

Md Tanvir Hasan

INTEGRATING USER-CENTERED DESIGN INTO SCRUM: A CANVAS APPROACH

ACTA UNIVERSITATIS LAPPEENRANTAENSIS 1268

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Abstract

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Scrum is a widely adopted Agile framework in the software industry due to its focus on speed and communication. Despite its popularity, Scrum often falls short in addressing end-user needs, usability is often overlooked, and there is a lack of structured methods and techniques for incorporating design activities. Scrum mainly focuses on delivering functional requirements and allocates limited time for design activities. The result is functional products that may not meet user expectations. However, are these Scrum challenges? According to the Scrum Guide, Scrum is a framework—and not a methodology—that enables the integration of various practices and techniques. Scrum's effectiveness largely depends on how practitioners implement it. Therefore, this dissertation approaches the problem from a different angle, arguing that these are not Scrum challenges, but challenges related to the application of Scrum.

With this perspective, the research was conducted in two phases: the first phase focused on identifying and understanding usability-related issues in organizations practicing Scrum, while the second phase centered on exploring, developing and validating solutions to address them. The research first employed semi-structured interviews with Scrum practitioners across various organizations to identify the key barriers they face in integrating usability. Four anti-patterns were identified, which served as the foundation for the next phase of exploration. To investigate how user-centered design (UCD) could be applied or adapted to overcome the identified barriers, the study utilized a combination of methods, including case studies, surveys, additional semi-structured interviews, and think-aloud protocols. The result was a UCD canvas, a lightweight canvas approach to enable Scrum teams to integrate usability activities without disrupting Scrum's existing processes. The canvas was evaluated with Scrum practitioners and found to be easy to use, relevant, effective, and time efficient.

The dissertation makes several contributions. First, the identification of four anti-patterns in Scrum-driven organizations. These anti patterns can help practitioners reflect on and improve their current processes. Second, it provides insights into how UCD can be applied or adapted to fit Scrum workflow. Third, the development and evaluation of a UCD canvas. The canvas enables Scrum teams to incorporate usability concerns and share requirements with stakeholders using UCD techniques. The canvas can be used by anyone regardless of their background and it is also time efficient. However, the use of the canvas in a real-world Scrum project has yet to be evaluated. Future research could explore the canvas's applicability and effectiveness in a real-life Scrum project.

Keywords: scrum, UCD, lightweight UCD, agile UCD

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Abstract

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Publications

List of publications

This dissertation is based on the following papers. The rights have been granted by publishers to include the papers in dissertation.

I. Hasan, M. T., Wolff, A. and Knutas, A. (2024) 'Involvement of end user in Scrum-driven organizations: Four anti-patterns', *Proceedings of TKTP 2024: Annual Doctoral Symposium of Computer Science, Vaasa, Finland, 10–11 June 2024.* CEUR Workshop Proceedings, 3776, pp. 12–21.

II. Hasan, M. T., Wolff, A. and Porras, J. (2021) 'Co-designing recycling solutions on campus: A case study exploring openness, realism and empowerment of users in a Living Lab', *Proceedings of the 10th International Conference on Communities & Technologies – Wicked Problems in the Age of Tech (C&T '21)*, pp.289–295. ACM. doi: 10.1145/3461564.3461591.

III. Hasan, M.T., Wolff, A., Knutas, A., Pässilä, A. and Kantola, L. (2022) 'Playing games through interactive data comics to explore water quality in a lake: A case study exploring the use of a data-driven storytelling method in co-design', *Proceedings of the CHI Conference on Human Factors in Computing Systems Ex-tended Abstracts (CHI '22),* pp. 1–7. ACM. doi:10.11453491101.3503561.

IV. Hasan, M.T., Wolff, A. and Knutas, A. (2025), June. 'Integrating Lightweight UCD into Scrum: A Canvas Approach for Non-Designers'. In 2025 International Conference on Computer Systems and Technologies (CompSysTech) (pp. 1-7). IEEE.

Author's contribution

Md Tanvir Hasan is the principal author and investigator in Publications I, II, III and IV. No publication in this list has been used by other researchers for their doctoral dissertation.

Publication I: Md Tanvir Hasan is the lead author and investigator for this paper. Md Tanvir Hasan was responsible for all aspects of this research including designing the study, recruiting participants, conducting interviews, recording all sessions, and performing data analysis. Annika Wolff provided guidance on developing interview questions and participant recruitment. Antti Knutas advised on data analysis approaches. Both Annika Wolff and Antti Knutas reviewed the paper.

Publication II: Md Tanvir Hasan is the lead author and investigator for this paper. Md Tanvir Hasan was responsible for developing and conducting surveys, interviews, a workshop, and data analysis. Annika Wolff and Jari Porras provided guidance on developing the personas. Annika Wolff provided guidance on workshop activities. Both Annika Wolff and Jari Porras reviewed the paper.

Publication III: Md Tanvir Hasan is the lead author and investigator for this paper. Md Tanvir Hasan was responsible for developing the comic visualizations, card games, planning workshop activities, and conducting the survey and data analysis. Annika Wolff provided guidance on developing the card games and workshop activities. Antti Knutas contributed on the data analysis. Anne Pässilä and Lasse Kantola recruited participants as well as organized and conducted the workshop. Both Annika Wolff and Antti Knutas reviewed the paper.

Publication IV: Md Tanvir Hasan is the lead author and investigator of this paper. Md Tanvir Hasan was responsible for all aspects of this research including designing the study, developing the lean user-centered design (UCD) canvas, recruiting participants, conducting interviews, conducting the think-aloud protocol, recording all sessions, evaluating the canvas, and performing data analysis. Annika Wolff provided expert guidance on developing the canvas and contributed to participant recruitment. Both Annika Wolff and Antti Knutas contributed methodological expertise and reviewed the paper.

Declarations

Turnitin

The originality of this dissertation has been reviewed with the Turnitin similarity checking service.

Commercial software

This dissertation was prepared with the help of the following commercial software:

NVivo 12 - a qualitative data analysis software that used for data coding, classification and analysis. NVivo licensed software is provided by LUT University for the researchers. I utilized NVivo 12 for the qualitative data coding and analysis in Publications IV.

AI use

During the preparation of this doctoral dissertation, Md Tanvir Hasan—the author of the dissertation—used Grammarly in order to develop the presentation and grammar of textual information in the summary of the dissertation. After using Grammarly, the author reviewed and edited the content and takes full responsibility for the content of the doctoral dissertation.

1 Introduction

Agile is a popular approach in software development (Herdika and Budiardjo, 2020). For decades, software engineers have debated on the issue of how software development should be organized to deliver faster, better, and cheaper solutions. The development of Agile software emerged from the insights of experienced software practitioners who sought to improve software development. The benefits of Agile can be understood through its twelve guiding principles outlined in the Agile Manifesto (Beck et al., 2001). A few examples include prioritizing customer satisfaction through the early and continuous delivery of valuable software, embracing changing requirements even in later stages of development, ensuring frequent delivery of working software, emphasizing collaboration between developers and other stakeholders and strong focus on technical excellence and good design (Beck et al., 2001).

Since the publication of the Agile Manifesto, a substantial amount of research has been conducted to investigate Agile (Dybå and Dingsøyr, 2008; Schön, Thomaschewski, and Escalona, 2017; Curcio et al., 2018; Inayat et al., 2015; Almeyda, Zapata Del Río and Cohn, 2021), and studies have shown high rates of Agile project failures (Curcio et al., 2018), with a few key challenges such as minimal documentation, customer availability, inappropriate architecture, budget and time estimation, neglecting non-functional requirements (NRFs), customer inability and agreement, contractual limitations, requirements change and its evaluation (Inayat et al., 2015).

But why do Agile projects fail, even though the software is built on practices recommended by experienced software professionals? Compared to the traditional software development approach, Agile is more lightweight. It focuses on incremental delivery as well as organizing and making coding more efficient. Many studies have identified requirement engineering (RE) challenges in Agile. Agile RE activities are iterative and not sequential. They are informal and performed during several short development cycles (Dingsøyr et al., 2012). Schön et al. (2017) conducted a Delphi study with 26 Agile experts to identify challenges in Agile RE. Their research revealed six challenges for organizations practicing Agile; two of these challenges are related to stakeholders and end users. An overview of these challenges includes a lack of involvement of direct end users and difficulties in involving stakeholders in the development process. Challenges such as these can result in a software with bad usability, as end-user perspectives may not be considered or validated without involving the stakeholders. Further, Inayat et al. (2015) performed a systematic literature review in which they identified eight challenges posed by Agile requirements engineering. One of these challenges was ignoring nonfunctional requirements, such as external quality usability (Inayat et al., 2015). Thus, using Agile methodologies in software development cannot ensure a product that will satisfy end-user needs (Argumanis, Moquillaza, and Paz, 2021). To ensure product quality, it is essential to address the issue related to nonfunctional requirements in Agile (Helmy, Kamel, and Hegazy, 2012). This can be achieved by emphasizing usability and ensuring the consideration of the end user's perspective in Agile processes.

Given the range of available Agile methods, a comprehensive exploration of each method is beyond the scope of this dissertation. This dissertation focuses on the most widely adopted and practiced Agile framework in organizations—Scrum (Kikitamara and Noviyanti, 2018; Salah, Paige, and Cairns, 2014; Kuusinen and Väänänen-Vainio-Mattila, 2012). Scrum is a lightweight Agile framework mostly used for software development. There are several reasons why Scrum is widely adopted in industries, a couple of which are that Scrum focuses on short releases and frequent stakeholder involvement. Since Scrum is an Agile approach, it also faces requirements engineering challenges when applied to real-world settings (Alhazmi and Huang, 2020). Previous studies have revealed specific challenges, such as lack of time to consider end-user needs (Almeyda, Zapata Del Río and Cohn, 2021), the absence of an adequate description of methods and techniques to be used in the design process (Teka, Dittrich, and Kifle, 2018) and usability is often overlooked (Rafi et al., 2015). In practice, Scrum mainly focuses on achieving functional requirements and allocates minimum time and resources for the design phase. Consequently, the product fails to reach its full potential, thus risking end-user frustration (Lundström, Åberg, and Blomkvist, 2015). The output is often a product that meets the functional requirements but carries a high risk of market failure, as end users are unable to effectively achieve their goals using the product (Argumanis, Moquillaza, and Paz, 2021).

1.1 Research motivation and objectives

Previous studies (Inayat et al., 2015; Schön et al., 2017; Almeyda, Zapata Del Río, and Cohn, 2021, Al-Razgan et al., 2022; Curcio et al., 2019) have highlighted several challenges associated with Agile. Due to its lightweight nature and specific focus on fast delivery, it faces challenges related to involving end users and integrating usability. Furthermore, there is limited research that summarizes the concept of UX management or outlines how it can be integrated into Agile (Hinderks et al., 2022). Scrum, as one of the most widely adopted Agile frameworks, encounters challenges, such as the lack of consideration of end-user needs (Almughram and Alyahya, 2017; Cajander, Larusdottir and Gulliksen, 2013), no instructions to perform design activities (Teka, Dittrich, and Kifle, 2018), and the ignoring of usability requirements (Singh, 2008; Rafi et al., 2015; Al-Razgan et al., 2022).

To address these challenges various solutions have been proposed, including new models (Prasad et al., 2018; Singh 2008; Rafi et al. 2015), modified Scrum frameworks (Xiong and Wang, 2010; Ferreira, Noble, and Biddle, 2007; Rahayu et al., 2016; Aranda Muñoz et al., 2016; Argumanis, Moquillaza, and Paz, 2021), and initiatives to educate developers on user-centered design (UCD) activities (Øvad et al., 2015; Øvad and Larsen 2016). While these approaches are commendable, this dissertation seeks to extend the line of research from a different perspective. Rather than considering the identified challenges as Scrum challenges, this thesis argues that they are Scrum application challenges. Scrum is not a methodology but a framework, and its effectiveness depends on how practitioners implement it. Instead of altering the Scrum process, this dissertation aims to adapt

usability activities in a manner that aligns and fits with the Scrum as closely as possible. Thus, this thesis has the following objectives:

- Objective 1: To identify the organizational barriers faced by Scrum practitioners that hinder the integration of usability activities.
- Objective 2: To explore how UCD can be applied or adapted to address the barriers faced by Scrum practitioners
- Objective 3: To propose a solution to integrate UCD techniques into Scrum while minimizing changes to existing processes.

1.2 Research questions

To achieve the objectives of this dissertation, the following principal research question was attempted to be answered:

How can UCD be integrated into Scrum to ensure that the end-user's perspective is considered?

To address the principal research question, the following two sub-research questions were considered:

1. What are the main barriers to integrating usability activities in Scrum? (RQ1)
2. How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum? (RQ2)

To integrate usability activities into the Scrum workflow, it is essential to understand the specific challenges that practitioners in Scrum-driven organizations encounter in relation to usability integration. Therefore, RQ1 aims to uncover the existing barriers to integrate usability in Scrum-driven organizations. This question helps to assess Scrum practices in organizations, uncover practical challenges, and identify reasons why usability is often overlooked in real-world Scrum implementation.

RQ2 focuses on the barriers identified through the first research question (RQ1) and explores how UCD techniques can be applied or adapted to overcome these challenges. The goal is to find ways to integrate UCD activities into the Scrum process by considering the identified Scrum practices in organizations, the underlying reasons for neglecting usability, and the practical constraints faced by Scrum practitioners.

Table 1.1 presents the research questions, corresponding publications, and research methods utilized to address the research questions:

Table 1.1: Research questions and publication mapping

Principal research question	Sub-research questions	Publication	Research method
How can UCD be integrated into Scrum to ensure that the end-user's perspective is considered?	RQ1: What are the main barriers to integrate usability activities in Scrum?	Publication I	Semi-structured interviews
	RQ2: How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?	Publication II	Case study Survey Semi-structured interviews
	RQ2: How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?	Publication III	Case study Open-ended survey
	RQ2: How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?	Publication IV	Semi-structured interviews Open-ended survey Think aloud protocols

1.3 Thesis structure

In this dissertation, a constructivist also known as an interpretivist philosophical stance is adopted to address the research questions as it posits that knowledge is deeply attached to human context. A detailed discussion on why this philosophical stance was chosen over alternatives is provided in Chapter 3. The research was conducted by combining multiple qualitative methods, such as semi-structured interviews, case study, survey, and think-aloud protocols across different phases of the study. The work performed in this

dissertation is structured across two main stages. Figure 1.1 illustrates how these stages align with the research questions, publications, and objectives of the dissertation.

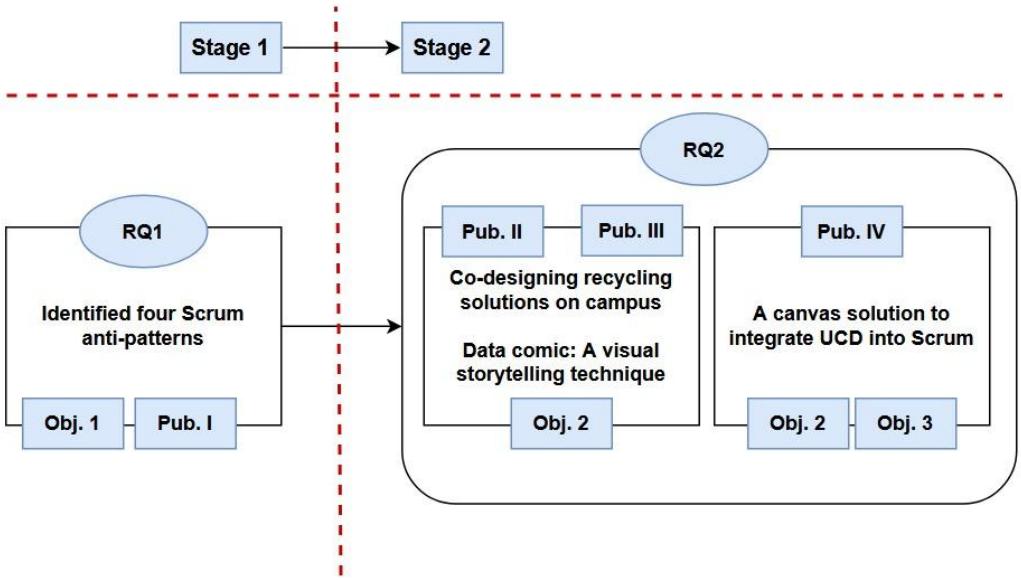


Figure 1.1: Structure of the thesis

Stage 1: Identifying usability barriers faced by Scrum practitioners in organizations

Stage 1 of this dissertation addresses RQ1, fulfills objective 1, and resulted in the publication I. In the first stage of this dissertation, semi-structured interviews were conducted with Scrum practitioners from organizations to identify, in practice, how end users are involved in Scrum; if they are not involved, then what are the reasons; how usability issues are handled; and why usability is often ignored. Four anti-patterns were identified from the interviews. The rationales underlying these anti-patterns were also explored from the semi-structured interviews, which served as the foundation for the subsequent exploration in Stage 2.

Stage 2: Applying or adapting UCD techniques to overcome the identified barriers

Stage 2 of the dissertation addresses RQ2 and is divided into two main sections. Both sections explore different approaches to overcoming the identified barriers from Stage 1, thereby aligning with the second and third objective of the dissertation.

The first section focuses on two of the barriers of involving end users identified in publication I:

- User characteristics, expertise, and background
- Time limitation

The section also addresses two anti-patterns: the involvement of the end user is not mandatory, a customer can represent an end user and product demo can occur with no end-user involvement.

Upon further investigation, it was found that these anti-patterns are common practices because end users are not always domain experts, thus making it difficult to communicate technical concepts or solutions to them. Also involving them directly throughout the Scrum is not feasible and formal usability techniques are ignored due to time limitation. Therefore, a method is needed to represent their perspectives indirectly. In publication II, personas and storyboards proved to be effective tools for communicating with diverse stakeholders during a time sensitive co-design session to ideate solution based on user needs. The personas were also used to represent a broad range of end-user perspectives without requiring their direct involvement. This addresses the issue of representing the end user's perspective in events without involving them directly, in a clear and easily understandable way.

Customers who often represent end users are also not always domain experts and sometimes tend to show less interest in technical details. Our goal is to have a UCD solution that can be used by anyone. But for Scrum team, who are not often familiar with UCD, applying UCD techniques requires accessible instructions and tools. A potential solution to these issues is to present information to any stakeholders such as end users, customers or Scrum team in a simple and easy-to-understand format. In publication III a co-design workshop was conducted with non-domain experts to make sense of complex issues and predict future with the help of comic visualization. Participant feedback was collected through an open-ended survey. This highlighted the potential of comic visualization and data comic, a visual data-driven storytelling technique that is effective for representing complex issues in a clear and understandable manner.

In both publication II and III, co-design workshops were conducted to evaluate the techniques. While co-design is not directly related to Scrum, there are similarities such as the need to communicate ideas among diverse stakeholders within time-constrained settings. Therefore, evaluating concepts within a co-design context can provide valuable insights into how similar approaches can be applied within Scrum.

In the second section, and in response to the identified barriers and anti-patterns, a canvas solution was proposed, which aimed at integrating UCD techniques into Scrum and resulted in publication IV. This corresponds to the second and third objective of the dissertation. To evaluate the canvas, we performed think-aloud protocols and interviews with Scrum practitioners, thereby assessing the usability and practicality of the canvas.

The remainder of the thesis is organized in the following manner:

Chapter 1 introduces the background and motivation underlying the research, outlines the research objectives and research questions, presents the selected research methodology, and provides an overview of the dissertation structure.

Chapter 2 provides a comprehensive review of the theoretical background and related work relevant to the research topic. The chapter introduces key concepts such as Agile, Scrum, usability, UCD, comic visualization, and usability issues in Scrum. It also summarizes the relevant extant studies in this domain.

Chapter 3 explains the philosophical stance of the research and the research methods employed to collect information. This chapter encompasses the overall research design, including data collection and analysis activities. It also explains the rationale underlying the selection of qualitative methods to conduct the research.

Chapter 4 presents the scientific publications related to this dissertation and summarizes the key findings from the publications. Each publication is summarized in terms of background, objective, key findings, and the contributions of the findings to the dissertation. The chapter reflects how the publications address the research question. It further discusses the practical implications of the outcomes.

Chapter 5 provides an in-depth discussion of the key findings of the dissertation, acknowledges limitations, and provides directions for future research.

Chapter 6 concludes the dissertation by summarizing the key contributions.

2 Background and related work

2.1 The development of Agile software and Scrum

Over the last 25 years, a large number of different approaches have emerged to develop software, but only a few remain in use today (Abrahamsson et al., 2017). Industries have become skeptical regarding the use of new solutions that are complex (Wiegers 1998). It is against this background that Agile software development methodologies have gained popularity (Abrahamsson et al., 2017).

2.1.1 The development of Agile software

Software development has evolved over the years, transitioning from traditional, plan-driven models (waterfall) to flexible, iterative approaches. The launch of Extreme Programming (XP) (Beck 1999a; Beck 1999b) is considered the starting point of different Agile software development approaches. Unlike traditional approaches, Agile methods are lightweight and avoid excessive processes. In general, Agile methods either avoid upfront planning or allocate limited time for upfront design activities, as the goal is to adapt quickly to changing requirements (Salah, Paige, and Cairns, 2014). The "Agile Movement" began with the publication of the Agile Software Development Manifesto in 2021 (Beck et al., 2001; Cockburn, 2002a). According to the manifesto, the main values of Agile are listed below (Beck et al., 2001).

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

The Agile Manifesto further outlines twelve guiding principles that form the foundation of Agile software development (Beck et al., 2001). Agile emphasizes that the highest priority is to satisfy the customer through early and continuous delivery of valuable software. Agile welcomes change even in the later stages of development and delivering working software frequently. It promotes regular communication between developers and other stakeholders, building projects around motivated individuals, and recognizing face-to-face conversation as the most efficient and effective way to convey information. Working software is considered the primary measure of progress in Agile. Agile also promotes sustainable development, continuous attention to technical excellence and good design, and values simplicity. Furthermore, Agile believes that self-organizing teams produce the best architectures, requirements, and designs, and that teams should regularly reflect on how to become more effective and adjust accordingly (Beck et al., 2001).

According to Miller (2001), Agile has the following characteristics from the fast delivery point of view:

- Modular development processes
- Iterative short cycles (Typically operates in one-to-six-week iteration cycles)
- Focuses only on essential activities within the development process
- Adaptive and incremental approach
- Building software in small steps to reduce risks
- People-oriented
- Emphasize collaboration and effective communication

A development method is considered Agile if it is incremental (small releases), cooperative (collaboration between customer and developer), straightforward (easy to learn and modify) and adaptive (open to changes) (Abrahamsson et al., 2017). The main focus of the Agile development team is to deliver essential functions as quickly as possible, collect feedback, and adapt on the basis of this feedback.

2.1.2 **Scrum**

Scrum was developed in the early 1990s by Ken Schwaber and Jeff Sutherland. Scrum is the most widely used and popular agile framework in the software industry (Kikitamara and Noviyanti, 2018; Salah, Paige, and Cairns, 2014; Kuusinen and Väänänen-Vainio-Mattila, 2012). It is a lightweight, iterative, incremental agile project management framework that helps to generate value by implementing flexible solutions (Schwaber and Sutherland, 2020). Scrum organizes work into sprints over a fixed duration event of one month or less. It is expected that there should be a shippable product increment by the end of each sprint. A sprint includes four other events: sprint planning, daily Scrum, sprint review, and sprint retrospective (Schwaber and Sutherland, 2020).

Further, a Scrum team consists of three distinct roles: a Scrum master (SM), a product owner (PO), and developers (Schwaber and Sutherland, 2020). The product owner is responsible for the product backlog, which is a list of tasks that need to be completed for the product. In sprint planning, developers select tasks from the product backlog to complete a sprint and work on them during the sprint. At the end of the sprint, there is a sprint review session in which the team shows what they have achieved in front of the stakeholders to collect their feedback. In the sprint retrospective, the entire team reflects on what went well, what went wrong, and how to improve in the next sprint. The entire process begins again with planning for a new sprint (see Figure 2.1). The Scrum master's job is to help everyone understand the rules of Scrum (Schwaber and Sutherland, 2020).

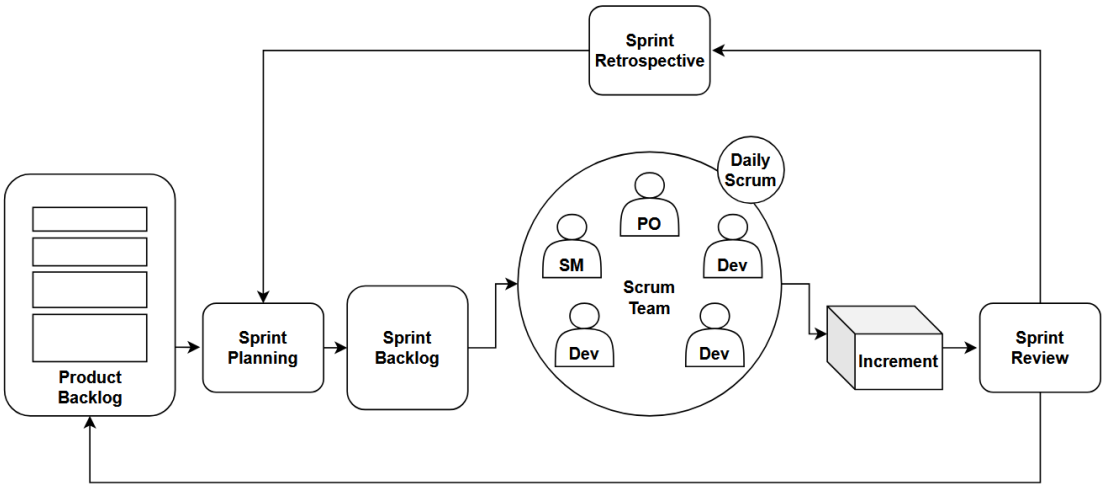


Figure 2.1: The Scrum process

Scrum teams are cross functional. The team needs to be small, often including 10 or fewer people, to have better communication and more productivity (Schwaber and Sutherland, 2020). The Scrum team is responsible for all product-related activities. However, the Scrum framework is incomplete, as the Scrum guide only defines the parts that are necessary. Different methods and techniques can be employed within this framework (Schwaber and Sutherland, 2020), and the appropriate utilization of the framework depends on professionals.

2.2 Usability and user-centered design

The term usability has been interpreted in various ways by different standards and researchers, thus leading to a range of meanings and perspectives (Sagar and Saha, 2017; Seffah and Metzker, 2004). The idea of usability emerges from the notion of being user friendly. In software engineering, usability is defined as how easily a software can be used (Sagar and Saha, 2017). According to IEEE Std. 610.12, usability is described as "the ease with which a user can learn to operate, prepare inputs for, and interpret outputs of a system or component." Nielsen (1994) defined five attributes of usability: efficiency, errors, learnability, memorability, and satisfaction. ISO 9241-11 defined three attributes of usability: effectiveness, efficiency, and satisfaction. Over the years, numerous usability models have been developed to evaluate the usability of software, and different models have different attributes (Sagar and Saha, 2017). According to a systematic literature review study (Sagar and Saha, 2017), the most commonly addressed attributes are:

- Learnability: Easy to learn
- Efficiency: Efficient to use

- Satisfaction: Pleasant to use
- Effectiveness: Accuracy and completeness that help to achieve the goal

Usability is an essential part of software quality. The quality of a software can be measured by several internal and external criteria. There are six categories of software quality: functionality, reliability, usability, efficiency, maintainability, and portability (Sagar and Saha, 2017) (see Figure 2.2).

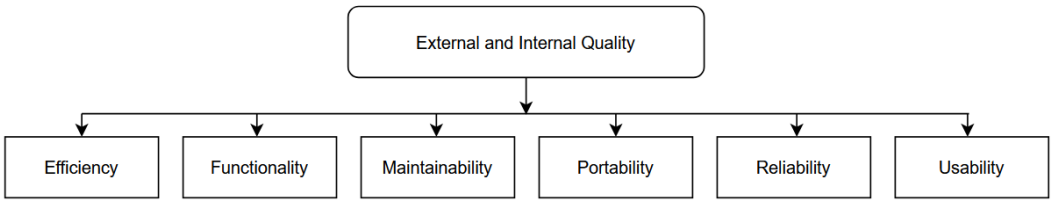


Figure 2.2: Software quality model (adapted from Sagar and Saha, 2017)

Usability is one of the most important software quality attributes. Studies have revealed that bad usability has directly or indirectly caused software quality-related issues (Sagar and Saha, 2017). In practice, poor usability is one of the prime causes of software application failure (Seffah and Metzker, 2004).

2.2.1 User-centered design (UCD)

The term UCD emerged from Donald Norman's research laboratory at the University of California, San Diego (UCSD) during the 1980s. It was widely used after the release of the co-authored book entitled "User-Centered System Design: New Perspectives on Human-Computer Interaction" (Norman & Draper, 1986). UCD is a design process in which end users can influence design decisions (Abras, Maloney-Krichmar, and Preece, 2018). UCD ensures that the end product is based on end-users' goals and needs (Brhel *et al.*, 2015). It is a combination of various types of methods. In human–computer interaction (HCI), terms such as UCD, usability engineering, and interaction design are often used interchangeably (Brhel *et al.*, 2015). UCD techniques are rather useful in the collection, visualization, representation, and validation of software requirements.

According to ISO 13407, which is an international standard established in 1999, there are four main iterative activities of UCD (Jokela *et al.*, 2003) (see Figure 2.3):

- Understand and specify the context of use: Identify the end users, product environment, and the tasks end users perform with the product.
- Specify the user and organizational requirements: Defining design guidelines and constraints to ensure the product meets usage requirements.

- Produce design solutions: Turn knowledge into design solutions.
- Evaluate design against requirements: Evaluate the design against the identified user requirements.

Various techniques can be used at different phases of UCD. According to a systematic literature review (Hasani, Sensuse, and Suryono, 2020), surveys and interviews are the most commonly used methods for collecting requirements during the initial phase. Other commonly used techniques include observation, focus groups, personas, and user journey mapping. In the design phase, the most frequently used techniques are high and low fidelity prototyping; moreover, wireframing and paper prototyping are also popular choices (Hasani, Sensuse, and Suryono, 2020). For the evaluation phase, usability testing and questionnaires are the most commonly used methods, followed by think-aloud protocols and field testing (Hasani, Sensuse, and Suryono, 2020).

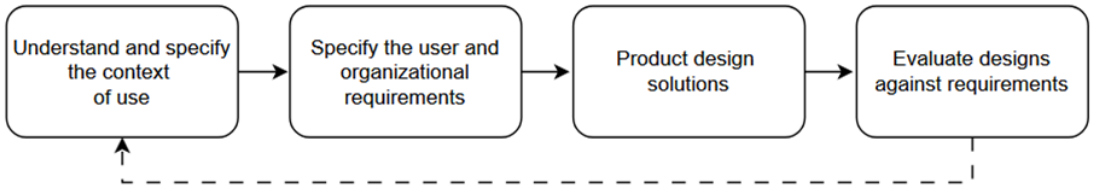


Figure 2.3: UCD activities (adapted from Jokela et al., 2003)

In their research, Jia *et al.* identified a set of techniques commonly employed by Scrum practitioners. Among these, the most frequently used techniques involve direct engagement with end users, including workshops, interviews, user meetings, and questionnaires (Jia *et al.*, 2012). The most effective and frequently used UCD techniques, according to Agile practitioners, are listed below (Schön, Thomaschewski, and Escalona, 2017; Øvad and Larsen, 2015; Schön, Thomaschewski, and Escalona, 2017):

- user story
- prototype
- story card
- persona
- journey map
- storyboard

Unlike incremental process models, such as waterfall, Agile methodologies do not require heavy documentation. User story is a tool used in Agile software development to capture the functional requirements of the software. Thus, for obvious reasons, user story and story cards are used by most Agile practitioners. In this thesis, user stories will not be

prioritized, as they are an integral part of Agile methodology for representing functional requirements and are commonly utilized by practitioners. Prototyping will also not be considered, as it is a technique utilized during the design phase. Since the focus of this thesis is on requirement collection and analysis, we examine the other techniques we identified—persona, storyboarding, and journey mapping.

Persona concept was introduced by Alan Cooper (Cooper, 1999) to guide designers and inform their design decisions. Persona is a fictional character that represents end users with similar goals, motivations, and frustrations. It helps to define software requirements based on what an end user wants rather than on what developers think an end user might want. Persona helps developers to build empathy for the end user (Cooper, 1999). Through personas, developers are able to understand end-user behavior, their needs, and how they will interact with the software. There is no strict template for persona. An example template has been illustrated in Figure 2.4. A persona can be represented in a textual description or as a visual representation. Narrative personas present end-user needs from the proposed product, while visual personas include certain human aspects such as emotional expressions and pictures (Karolita *et al*., 2023).



Figure 2.4: An example of a persona template

A journey map is the visual representation of the step-by-step experience an end user has with the product or service. A user journey map captures the goals of end users, different stages of the journey, tasks performed by the end users, the challenges the encounter, and opportunities for improvement (Rosenbaum, Otalora, and Ramírez, 2017).

Storyboards are scenarios presented in graphical language to enhance recognition and understanding of user interaction and functional flow. While there are a variety of styles for graphical representations, comic visualizations are widely used for producing storyboards (Haesen *et al*., 2010) (see Figure 2.5). This study combined journey map and storyboard by utilizing the elements of a journey map and presenting them through a storyboard panels.



Figure 2.5: An example storyboard template

These three UCD artifacts (personas, journey maps, and storyboards) have often been utilized concurrently or in conjunction with one another in many research (McQuaid, Goel and McManus, 2003; Sedeño *et al.,* 2017; Kantola and Jokela, 2007; Kalbach, 2016; Samson, Granath and Alger, 2017); this work continues that line of research by using these artifacts together.

While the involvement of end users in UCD can vary, the main concept is that users play a role in the design decisions (Abras, Maloney-Krichmar, and Preece, 2018). Certain UCD approaches involve users mainly during the earlier stages, such as requirement collection, analysis, and usability testing. In certain cases, users are actively involved in the design process through collaboration with designers and have the power to influence design decisions from start to finish (Abras, Maloney-Krichmar, and Preece, 2018). To clear the confusion within the design research space, Dubberly and Sanders (2006) developed a design practice and design research map consisting of two opposing mindsets (see Figure 2.6). The left side of the map represents an expert-driven mindset. In this context, researchers focus on developing solutions for people. They view people as subjects. The right side of the map is characterized as a participatory mindset. In this context, rather than creating solutions *for* people, researchers create solutions *with* people (Dubberly and Sanders, 2006); here, people are viewed as co-creators in the design process. UCD primarily falls within the left zone, with a focus on creating products or services based on user needs (Dubberly and Sanders, 2006). This is achieved by collecting, analyzing, and representing user data to establish design principles.

In UCD, it is important to identify who is the end user and how to involve them in the design process (Abras, Maloney-Krichmar, and Preece, 2018). In the normal sense, end users are the people who will ultimately use the product or service to accomplish a task or goal. But there are other types of users as well, such as people who manage these end users, people who are going to be affected by the use of the product or service (Abras, Maloney-Krichmar, and Preece, 2018). There are three types of users (Eason, 1987): primary, secondary, and tertiary. The primary users are the people who will actually use the product. The secondary users are the group of people who will occasionally use the

product, and the tertiary users are those who will be affected by the use of the product or make decisions regarding its purchase. This dissertation focuses on the primary users.

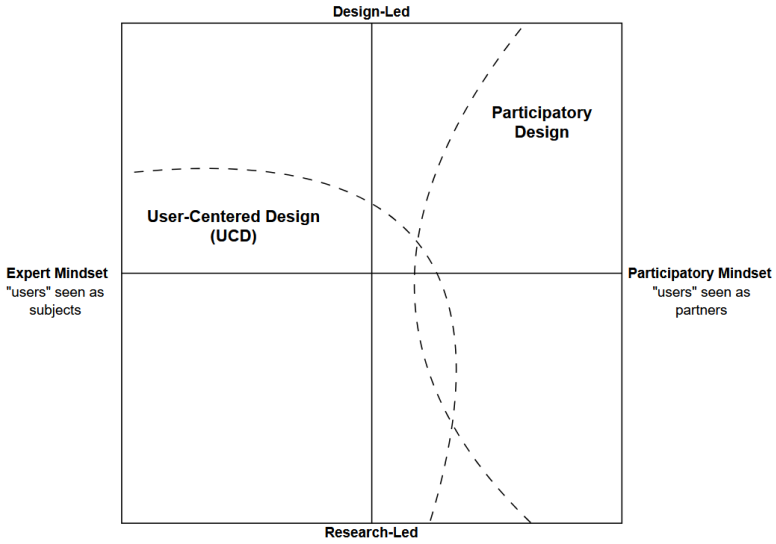


Figure 2.6: Design research map (adapted from Dubberly and Sanders, 2006)

The major drawbacks of UCD are that it can be very costly and time-consuming. The process requires resources (both human and financial) and time to collect data from end users (Abras, Maloney-Krichmar, and Preece, 2018). According to Nielsen, developers avoid using these types of techniques because they are complex, expensive, and too time-consuming (Nielsen, 1994).

2.3 Comic visualization and data comic

Engaging stakeholders in co-design sessions to ideate and brainstorm on solutions is increasingly becoming a common practice, with data serving as a valuable resource to inform these creative processes (Hasan *et al.*, 2022). But most often it is difficult for the general population to understand data. Co-design sessions are often time-constrained, as people tend to prefer shorter sessions (e.g., a two-hour session) (Kjørstad *et al.*, 2021). This limited duration makes it challenging to teach participants new skills for interpreting or analyzing data (Hasan *et al.*, 2022). The solution could be making the data simpler so that people can interact with it without being a domain expert (Hasan *et al.*, 2022). This is where data comics come into play. These data comics can simplify the process of understanding data by merging the visual language and storytelling techniques of traditional comics with data visualization (Wang, Dingwall, and Bach, 2019). Comics serve as a medium for sequential art. They are often perceived as light entertainment, but capable of tackling serious subjects (graphic documentary novels are an example of this) (Adams and Owens, 2015). Transforming a subject matter into a comic strip not only

enables readers to immerse in the narrative but also transport them into a fictional world. This unique ability has led to the integration of comics into various HCI research contexts (Sturdee and Lindley, 2019), such as explaining ongoing work with technology and research through design (Dylan *et al.*, 2016), data-driven storytelling (Bach *et al.*, 2017), and provocation (Sturdee *et al.*, 2018). Additionally, it has proven to be valuable in co-design processes (Ardito *et al.*, 2021). Similarly, comic boarding a participatory design technique has been shown to be effective, particularly when working with children (Hiniker, Sobel, and Lee, 2017).

This dissertation draws on the data comic visualization approach of Bach *et al.* (2018). In his research, he introduced a few design patterns to generate comics from data. These design patterns were utilized to develop the comic visualizations presented in this study. The narrative power of comics comes from the combination of words and pictures. Pictures can be used as the visual evidence of the data. For better understanding, a picture should ideally be combined with text. According to Bach (Bach *et al.*, 2017), this combination can be a) paragraph with the picture, b) putting text into the picture, or c) placing a picture into text (see Figure 2.7).

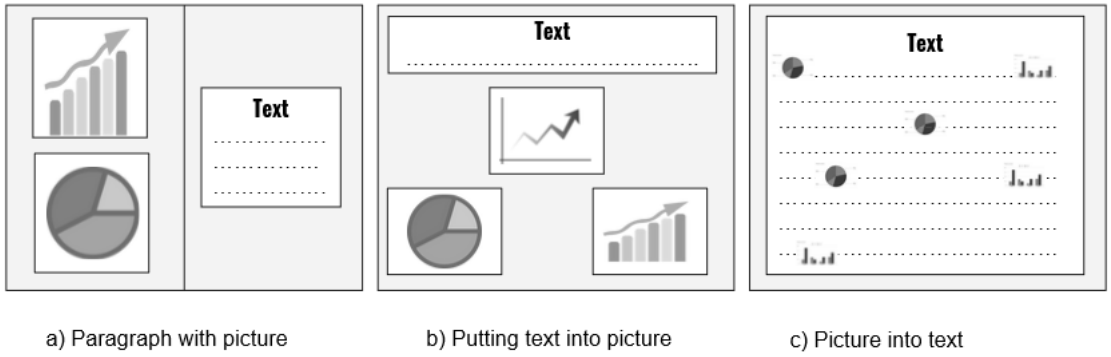


Figure 2.7: Combination of words and pictures (Hasan *et al.*, 2022)

2.4 Usability/UCD and Agile

Agile is lightweight and often avoids upfront planning; moreover, usability activities are often ignored in Agile processes. The involvement of the end user is always encouraged in Agile; however, in practice, it is not a mandatory part of the process (Anwar et al., 2014; Blomkvist, 2005). A Delphi study by Schön (Schön et al., 2017) identified key challenges in Agile requirements engineering. The result revealed six challenges for industries that practice agile processes; two of these challenges are related to users and stakeholders. These challenges can be summarized as a i) lack of involvement of direct end users and ii) difficulty in involving stakeholders in the development process. Similarly, a literature review study revealed that "lack of user participation" is a common issue in Agile (Almeyda, Zapata Del Río, and Cohn, 2021). In their study, Inayat et al. (2015) identified eight challenges in Agile requirements engineering, one of which is

related to usability. Since Scrum is an Agile approach, it faces a few of the challenges faced by Agile when put into practice (Alhazmi and Huang, 2020). Sprint review takes place at the end of a sprint when the Scrum team shows the outcome of the sprint to potential stakeholders to collect feedback (Schwaber and Sutherland, 2020). Normally, end users are not involved in this review process. Their feedback is collected when they use the product. By this time, change is still possible, but rework is costly (Rahayu et al., 2016).

2.4.1 Challenges in integrating usability/UCD in Agile

A recent literature review concluded that in Agile, there is essentially no established approach for considering UX in a product backlog item before its development. But it is necessary. Only then UX can be measured after development (Hinderks *et al.*, 2022). Another systematic literature review investigating the challenges of integrating Agile and UX/UCD identified six challenges. The foremost among these were time constraints, work imbalance between developers and designers, and the failure to prioritize UX and usability. The review also highlighted the specific difficulty of conducting user testing within an Agile environment (Al-Razgan *et al.*, 2022). Curcio et al. (2019) reached a similar conclusion in their tertiary study, confirming that time, work balance, prioritization, and testing are challenges in the integration of usability and Agile software development. These findings emphasize the importance of focusing more on usability when implementing Agile methodologies.

The sprint retrospective is used by Scrum teams to improve their efficiency in subsequent sprints. However, a literature review revealed that none of the studied research utilized this event to improve UX practices (Hinderks *et al.*, 2022). Even while developing a product, Scrum practitioners often neglect creating user experiences based on user needs (Almughram and Alyahya, 2017; Al-Razgan *et al.*, 2022). The Scrum team mainly concentrates on implementing a working function rather than the usability or usefulness of that function, as usability is not a high-priority task and can be improved upon later (Singh, 2008; Al-Razgan *et al.*, 2022). Thus, usability requirements and their implications are often ignored, as they are a low-priority task in the Scrum process (Rafi *et al.*, 2015). In Jia, Larusdottir, and Cajander's study (2012), researchers identified that formal usability techniques are not commonly employed by Scrum practitioners; instead, they tend to rely on informal usability techniques as they are fast.

Overall, in practice, there's no systematic plan or approach to involve end users and no specific person is responsible for ensuring usability. End-user aspects are only considered if the product owner or team has an interest in usability (Cajander, Larusdottir, and Gulliksen, 2013).

2.5 Related work

Usability challenges in Scrum have been widely explored. Extensive research has been conducted to integrate usability/UX/UCD activities into Scrum.

A model was proposed by Yuwei Xiong and Ansheng Wang to reduce the delivery cost and improve the relationship between UCD and software development. Although Agile avoids upfront design, this study proposed that having an upfront design by utilizing UCD can enable developers to have a better understanding of the software (Xiong and Wang, 2010). A similar approach was suggested in other studies (e.g., Ferreira, Noble, and Biddle, 2007); the approach proposes the implementation of an additional UCD team that works one iteration ahead of the development team. This approach is referred to as "upfront UCD design." Argumanis, Moquillaza, and Paz (2021) proposed a UCD framework that can incorporate selected UCD techniques into Scrum.

In another action research approach (Teka, Dittrich, and Kifle, 2018), UCD techniques—such as personas, prototypes, user-pair testing, and heuristic evaluation—were applied in a lightweight manner within Scrum, which enabled the development team to obtain early feedback. Prasad *et al*. (2018) proposed a framework to improve customer satisfaction using five design thinking practices. According to the authors, the framework can be used in projects based on Agile practices, such as Scrum. They further listed customer journey, story mapping, prototypes, POC, and customer profiling as the most suitable methods to identify end-user needs (Prasad *et al*., 2018). Rahayu *et al*. (2016) mentioned three challenges of Scrum: user feedback, testing, and software quality. To overcome these challenges, the researchers integrated usability testing with end users in the sprint review process. They tested their hypothesis in a case study and concluded that integrating usability testing can enhance the effectiveness of the sprint review process (Rahayu *et al.*, 2016).

Numerous studies have proposed new models and different variants of the Scrum framework. Fox, Sillito, and Maurer (2008) conducted a qualitative study by interviewing practitioners who have combined Agile and UCD methods. The results revealed that none of the approaches employed by the practitioners were identical, but there were a few commonalities. All the approaches had an upfront design stage in which contextual inquiry was utilized to collect requirements and perform low-fi prototyping. Aranda Muñoz et al. (2016, pp. 97–115) similarly recommend conducting user field studies in the first sprint.

U-scrum, one of the most frequently cited models for integrating usability into Scrum, was proposed by Singh (2008). The main idea of U-Scrum is that instead of one there will be two product owners; one will focus on traditional functions, and the other will focus on usability/UX. US-scrum is another variant of Scrum proposed by Rafi *et al.* (2015), which was developed by combining feature-driven development (FDD) and Scrum. This model not only addresses the usability issue but also focuses on the aspects of security and accuracy.

A different solution was proposed by Øvad *et al*. (2015) and Øvad and Larsen (2016). They believe that the solution to the problem is to educate developers with certain usability and UX methods (AB usability tests) to apply them to their development process. Developers often lack adequate skills to perform usability activities. But Øvad *et al*. (2015) believe that to be truly Agile, everyone should be able to perform every task. Thus, training developers in conducting certain usability/UX tasks appears to be the rational choice.

Previous research has explored integrating UX and usability into Scrum; however, most proposed solutions either modify the framework itself (Singh, 2008; Rafi *et al.,* 2015) or require a dedicated team of UX experts to manage these activities (Xiong and Wang, 2010; Argumanis, Moquillaza, and Paz, 2021). Typically, UX responsibility falls to a design team, but not all organizations (especially smaller ones) can afford such a dedicated resource. Often, designers are shared across multiple teams or work only part-time (Salah, Paige, and Cairns, 2014). Furthermore, Agile organizations rarely emphasize the explicit inclusion of a UX role within their teams, which frequently leads to communication gaps between designers and developers (Kuusinen and Väänänen-Vainio-Mattila, 2012; Bruun *et al.,* 2018; Lárusdóttir et al., 2016).

While one study (Teka, Dittrich, and Kifle, 2018) have explained how to perform UX activities without a dedicated design team, their suggestions primarily involve adding supplementary roles. Other research (Øvad *et al*., 2015; Øvad and Larsen, 2016) has focused on training developers in UX methods, but this training is often limited to usability testing.

User-Centered Design (UCD) is a broad discipline that is generally not considered lightweight, encompassing numerous techniques for gathering user data, representing findings, and testing outcomes. Although one study discussed the best UCD techniques and practices (Prasad *et al*., 2018), it didn't specify how to generate the necessary artifacts or who is responsible for them in the absence of a UX expert. One study mentioned using these techniques in a "lightweight" manner but did not explicitly detail how (Teka, Dittrich, and Kifle, 2018).

3 Methodology

This study explores how usability activities can be integrated into Scrum-driven organizations to ensure that the user experience of a product is considered during software development. It aims to develop a lightweight approach that fits into the Scrum process without disrupting its workflow. The research is guided by one principal research question: *How can UCD be integrated into Scrum to ensure that the end-user's perspective is considered?* This involves first understanding the role of end users in Scrum, how usability issues are handled, and common practices in organizations related to end-user involvement and usability. These aspects are addressed through RQ1: *What are the main barriers to integrating usability activities in Scrum?* Based on the answer to this question, different solutions to address the challenges were explored; this led to RQ2: *How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?* The result was a canvas solution for incorporating usability activities into Scrum.

3.1 Research philosophy and paradigms

This section provides an overview of the three most popular philosophical perspectives (Easterbrook *et al.*, 2008) in research: positivism, interpretivism, and pragmatism. Each of these provides a distinct approach and methods for conducting research and influencing methodological choices.

3.1.1 Positivism

Positivism is a research philosophy that states that all knowledge should be based on observable facts that can be verified (Easterbrook *et al.*, 2008). Positivists assume that scientific knowledge is built up through verified observation and conclusions. However, positivism has been criticized due to unreliable observation data and the issue that scientific facts can be proven wrong (Easterbrook *et al.*, 2008). In software engineering, positivist research involves testing theories in isolation, which involves laboratory experiments, surveys, and case studies.

3.1.2 Constructivism

Also known as interpretivism (Klein and Myers, 1999), this research philosophy believes that scientific knowledge is deeply attached to the human context and is shaped by society (Easterbrook *et al.*, 2008). Thus, it is necessary to understand how people interpret these theories. Instead of proving theories, constructivism aims to understand how people make sense of the world and how they give meaning to their actions (Easterbrook *et al.*, 2008). Theories developed from this process are always tied to the context being studied. Moreover, it is a common approach in social sciences, where approaches such as positivism do not capture the complexity of social interaction. Constructivism involves methods such as ethnographies, exploratory case studies, and surveys that collect rich qualitative data regarding human actions (Easterbrook *et al.*, 2008).

3.1.3 Pragmatism

While positivism and interpretivism represent two opposite sides, pragmatism sits somewhere in the middle. Pragmatism is an approach that sees all knowledge as incomplete and holds that the value of knowledge depends on how it was obtained (Menand, 1997). Pragmatists adopt a practical approach to solve real-world problems. In addition, pragmatism is somewhat relative (Easterbrook *et al.*, 2008). This implies that the truth can vary depending on the person. Unlike positivism and constructivism, pragmatism is flexible and values practical knowledge over abstract ideas (Easterbrook *et al.*, 2008). According to pragmatism, researchers should be free to employ any methods that help them to answer their questions (Easterbrook *et al.*, 2008).

3.1.4 Philosophy selected in this thesis

This thesis adopts a constructivist philosophical stance. Constructivism emphasizes that knowledge is deeply attached to human context and shaped by society (Easterbrook *et al.*, 2008). A Constructivist stance guided the data collection in all four studies, emphasizing participants' perspectives and context-specific understanding.

In publication I, semi-structured interviews were conducted to explore how Scrum practitioners apply the framework in practice and what usability/UX-related challenges they encounter. A constructivist approach was particularly suitable for this, as it allowed practitioners to reflect on their lived experiences through open discussion. This helped identify practical issues that are best described as Scrum application challenges rather than Scrum challenges. Publications II and III adopted a case study approach to explore how individuals with diverse backgrounds, including both domain experts and non-experts, utilize UCD techniques and comic visualizations. In publication II, participants used personas and storyboards to design solutions based on user needs. In publication III, participants employed comic visualizations to make sense of complex issues and predict future scenarios. Through these studies, we analyzed participant activities, gathered feedback and came to conclusion, a process that aligns with a constructivist stance (Easterbrook *et al.*, 2008). Finally, in publication IV, the lightweight UCD canvas was developed and evaluated with practitioners through hands-on exercises. Methods such as semi-structured interviews, open-ended surveys, and the think-aloud protocol were used. This approach aligns with the constructivist paradigm, as it allowed the author to observe how practitioners engaged with the artifact in practice and interpreted its instructions.

Positivism is not well suited for this study, as it emphasizes the testing of hypotheses, objective measurements, and quantifiable variables—approaches that do not align with the exploratory nature of this research. While a pragmatic stance might have also been appropriate for this thesis, given its flexibility to mix qualitative and quantitative methods based on practical considerations, this study relies mostly on qualitative data and does not incorporate a mixed-methods design. Therefore, constructivism was selected because it supports the aim of understanding usability practices within the organizational context of Scrum practitioners and utilizes the knowledge to inform solutions.

3.2 Research methods

Guided by the constructivism stance, this thesis adopts a qualitative approach, including a selection of qualitative data collection methods, such as semi-structured interviews (Qu and Dumay, 2011), open-ended surveys (Scheuren, 1948), think-aloud protocols (Van Den Haak, De Jong, and Schellens, 2003), and a case study (Runeson and Höst, 2009) to collect rich qualitative data. In this section, the research methods employed in this thesis are introduced.

3.2.1 Semi-structured interviews

An interview is one of the most popular qualitative research data collection methods. It serves as an effective tool for researchers to learn about other's perspectives and experiences. When conducted properly, an interview can yield a rich dataset (Qu and Dumay, 2011). Interviews can be conducted in different forms, including focus group and in-depth interviewing (Gubrium and Holstein, 2001).

The questions asked in an interview can be structured in three different ways—structured, semi-structured, and unstructured (Qu and Dumay, 2011). In a structured interview, an interviewer asks all interviewees the same set of pre-defined questions in the same order. The primary concern in a structured interview is to minimize research bias and increase the generalizability of the data. In contrast, an unstructured interview is informal and adaptable. To make an interviewee feel relaxed, an unstructured interview is shaped according to the interviewee's context and situation (Qu and Dumay, 2011). Sitting between these two extremes is the semi-structured interview, which is perhaps the most common of all qualitative methods (Alvesson and Deetz, 2000, p. 194). Semi-structured interviews rely on a set of pre-defined open-ended questions, which are asked in an organized manner while also including follow-up questions to encourage detailed answers (Qu and Dumay, 2011). It is flexible, accessible, capable of finding hidden facts of human and organizational behavior, and, often, the most effective means of collecting information (Kvale and Brinkmann, 2009).

In this thesis, the author mainly conducted semi-structured interviews with Scrum professionals from different organizations (Publications I & IV). All individuals selected for the interview had practical experience working in Scrum or an Agile environment. Three interviews were conducted with waste management personnel at a Finnish university to explore waste sorting practices and challenges (Publication II).

3.2.2 Survey

A survey refers to a method of collecting data from a sample of individuals who represent a subset of a target population (Scheuren, 1948). A survey serves different purposes and can be conducted through multiple channels, such as emails, phone calls, or in person. Moreover, a survey aims to focus on a segment of a target population, not to study every individual (Scheuren, 1948). An open-ended survey enables individuals to freely share

their thoughts and feelings in their own words. This approach is valuable to researchers, as it captures detailed insights and is cost-effective. Unlike interviews, a survey provides respondents with anonymity, which encourages honest responses (Erickson and Kaplan, 2000).

In this study, survey was utilized as a method for identifying challenges and collecting feedback at four distinct stages. In the first stage, a survey was conducted to assess public awareness of solid waste management and to gain detailed insights into common individual waste management behaviors (Publication II). In the second stage, another survey was carried out for the same case study to evaluate the solutions generated during a co-design workshop (Publication II). Third, after conducting another workshop, an open-ended survey was administered to collect feedback from the participants and facilitators on a particular technique that was used in the workshop; the participants were mainly teenagers (Publication III). In the fourth stage, to evaluate a canvas, an open-ended survey was administered among practitioners to collect information on their experience with the canvas (Publication IV).

In all cases, the surveys were designed to enable participants to openly share their feedback. The surveys provided both quantitative and rich qualitative data to i) understand individual behaviors related to waste management ii) to validate the outcomes from the co-design workshop iii) analyze the effectiveness of comic visualization and help to refine the lean UCD canvas design for further evaluation.

3.2.3 **Think-aloud protocol**

Think-aloud protocols are a predominant method for the usability testing of a software, interfaces, websites, and documents. In this method, users are asked to complete a series of tasks using the tested artifact while verbalizing their thoughts. The collected data is highly valid, as it reflects actual user interaction rather than subjective opinions (Van Den Haak, De Jong, and Schellens, 2003). The method has two variations: concurrent think-aloud protocol and retrospective think-aloud protocol. The key distinction is that in retrospective think-aloud protocols, participants first silently complete the task and then verbalize their thoughts thereafter, based on the recording of the session. There are both advantages and disadvantages of using retrospective think-aloud protocols instead of concurrent think-aloud protocols. One advantage is that a retrospective think-aloud protocol does not interrupt participants while they are performing the task, thereby enabling them to complete it at their own pace. Additionally, it enables accurate recording of the time taken to complete the task. However, a major disadvantage is the overall duration of the sessions, as participants must retrospectively review their actions after completing the task. Another limitation is that participants may forget specific details of what occurred during the task, which can result in incomplete or inaccurate feedback (Van Den Haak, De Jong, and Schellens, 2003).

A concurrent think-aloud protocol has been used in this study with Scrum practitioners to evaluate the usability of the canvas (Publication IV). Participants were given a brief

introduction of the method at the beginning of the session and instructed to verbalize their thoughts and confusions aloud as they interacted with the canvas. All sessions were recorded for further analysis. The data collected from this method helped understand how participants navigated the canvas, the type of challenges they encountered, and areas in which they encountered trouble. Based on the findings, the author was able to make adjustments to the design and concept of the canvas.

3.2.4 Case study

In software engineering, the primary focus is to understand how the software is developed, operated, and maintained by engineers and stakeholders under different circumstances. Due to its nature, software engineering fits well with case study research, as, by definition, case studies are conducted in real-world settings (Runeson and Höst, 2009). A case study is a method that explores phenomena in their natural settings. It enables researchers to develop a deep understanding of how the phenomena interact with the study context (Runeson and Höst, 2009). Often, case studies are based on qualitative data, as such data provide in-depth understanding (Runeson and Höst, 2009).

In this thesis, two case studies were employed to validate a framework and to explore how data in comic visualization can enhance sensemaking in a co-design process (Publications II & III). Both studies involved organizing a co-design workshop.

In the first workshop, participants used personas and storyboards to understand the individual challenges identified from a survey, step into the personas' shoes, and generate solutions (Publication II).

In the second workshop, participants engaged with curated data presented in a comic-based approach. The results revealed that the comic visualization was interesting and an easy way of visualization that helped in data sensemaking (Publication III).

3.3 Data collection and analysis

This chapter provides a detailed description of the research process, data collection, and analysis in each stage.

3.3.1 Data collection

Semi-structured interviews: To better understand the role of end users, how usability issues are handled in Scrum processes in practice, to explore waste management and to evaluate the lean UCD canvas, the author conducted semi-structured interviews (Publications I, II & IV) with 23 Scrum practitioners and 3 individuals responsible for overseeing university's waste management. The interviews were conducted in 2020, 2024 and 2025. Interviewees were first selected through personal networks and recommendations. Subsequently, the snowball recruitment technique was utilized,

leveraging referrals from initial interviewees to expand the pool of participants and increase the number of interviews conducted. Different approaches were employed to contact the interviewees: sending emails, direct outreach, and posting on social media (Facebook and LinkedIn). All Scrum practitioners selected for this study had practical experience in Scrum-driven organizations, playing roles such as Scrum master (SM), product owner (PO), developers, IT professionals, project coordinators, and test automation engineers. More information regarding the Scrum practitioners is presented in Table 3.1.

Table 3.1: Overview of interviewees

No. of participants	Role in organization	Experience (years)
9	Product owner (PO) & Scrum master (SM)	2–15
3	Scrum master (SM)	1–2
9	Developer	1–10
1	Test automation engineer	2
1	IT professional	15

The interviews were conducted in two phases: the first phase focused on understanding the problem, and the second phase aimed to evaluate the proposed solution.

In the first phase, semi-structured interviews were conducted with 11 Scrum practitioners from different organizations in the beginning of the year 2024 and with 3 individuals responsible for university's waste management in the year 2020. Each interview session lasted approximately from 45 minutes to 60 minutes. All interviews were conducted online and recorded by the author himself. Each participant was asked to sign a consent form in which they gave permission to record the sessions. The Scrum interview questions mainly focused on two areas: 1) the role of end users and their involvement in the Scrum process, and 2) the importance of usability and usability evaluation techniques used in practice. During these interviews, the practitioners provided the author with valuable insights into their current and past experiences, how they engage end users in the Scrum process, the importance of usability and the usability evaluation techniques they used in practice, common practices and the reasoning for these practices. The findings from these interviews not only enabled the author to identify four Scrum anti-patterns in organizations but also identify the reasoning for why certain practices are performed by a Scrum practitioner. This guided the author to search for solutions to mitigate the identified issues. The interviews related to waste management provided insights into the

university's current waste management practices, identified the main challenges, and gathered suggestions on potential measures to improve waste sorting.

The second phase took place toward the end of 2024 with 12 Scrum practitioners. The interviews ranged from 30 minutes to 45 minutes and included completing tasks using the canvas. As done in the first phase, interviewees had to sign a consent form, and all interviews were conducted online and recorded by the author himself. The interview questions focused on the clarity of the provided instructions, examples, templates, the canvas's effectiveness in supporting the creation of UCD artifacts, and whether it could be a tool for non-designers to produce these artifacts. Feedback was collected on satisfaction. Data from these interviews confirmed that the canvas is well-designed, provided users with all necessary guidelines to create UCD artifacts, the output can be highly useful, and the overall process was enjoyable.

Surveys: The author conducted a survey with 93 individuals in the first case study to identify challenges related to waste sorting. A second survey, involving 36 individuals, was carried out within the same case study to evaluate the solutions generated from the co-design workshop. Both surveys were conducted in 2020.

Open ended surveys were conducted to collect qualitative data with 26 individuals, where 4 of them were Scrum practitioners and 22 of them were from the data comic case study research. In this thesis, open-ended surveys were designed to collect participant and facilitator feedback, first on the comic visualization and then on the UCD canvas. The first survey was conducted in 2021 after performing a workshop, and the second survey was conducted in 2024. The open-ended survey was used as it is less time consuming and provides respondents with anonymity, which encourages honest responses (Erickson and Kaplan, 2000). A few demographic questions were also asked regarding the respondent's role in the company, working experience, and educational background in the UCD canvas evaluation survey. All four individuals who participated in the UCD canvas evaluation survey had practical experience working in Scrum or Agile environments, with three working as a developer and one as a project coordinator. More details can be found in Table 3.2.

Table 3.2: Overview of survey respondents

No. of participants	Role in organization	Experience (years)
3	Developer	2–5
1	Project coordinator	3

Think-aloud protocol: The think-aloud protocol was utilized to evaluate the canvas with 12 Scrum practitioners. The author chose to use a concurrent think-aloud protocol while practitioners interacted with the canvas to capture insights into their cognitive processes

(Robillard *et al.*, 1998) and the challenges they faced. Semi-structured interviews and the think-aloud protocol were conducted together, so this also took place toward the end of 2024. All sessions were conducted online and recorded by the author himself. Microsoft Teams and Miro board were used as the online tools for conducting the meetings and facilitating the think-aloud protocol. All sessions were facilitated by the author himself.

During the study, the researcher began the session with a brief background overview and guided participants through the task. All participants were assigned the same scenario and set of tasks. We provided a scenario in which participants were asked to create a user persona for the Facebook app. After creating the persona, participants were tasked with creating the journey map for logging into Facebook. Thus, the first task was to create a persona, and second task was to create a journey map for that specific persona. Participants were instructed to verbalize the thoughts, confusions, and difficulties they encountered while performing the tasks. All sessions were conducted with minimal interference from the author. Little guidance was provided and only when there was a question or confusion. Semi-structured interviews were conducted right after the session.

The think-aloud protocol provided valuable data on how participants engaged with the canvas, thus enabling the author to identify areas in which participants encountered difficulties. A few participants faced confusion primarily related to the use of Miro board rather than the canvas itself. Only issues relevant to the canvas were considered for potential improvements.

Case study: The author wanted to evaluate UCD artifacts and the effectiveness of the comic visualization in real-life settings. This is why case study was conducted, as it explores phenomena in their natural settings (Runeson and Höst, 2009).

The first case study included a co-design workshop with 9 participants, primarily students and staff from the university selected for the study. The workshop aimed to generate solutions to improve the university's waste management practices. The session followed the framework process and was structured into the following stages:

- Conversation starters
- Framing the design challenge
- Storyboard presentation
- Tour of the university
- Introducing personas
- Brainstorming
- Peer review and feedback

The workshop began with a brief introduction, followed by a discussion on the intended outcomes. A storyboard was then presented, depicting a student attempting to dispose of waste in a recycling bin, illustrating the challenges faced during waste sorting. The storyboard was developed using insights from the initial survey and interviews. Following

the storyboard presentation, participants were taken on a tour of the university to observe real-life waste sorting challenges. Afterwards, personas were introduced to encourage deeper reflection on the issues. Each participant selected a persona they felt closely represented their own perspective. They were then divided into groups and tasked with generating solutions to the problems identified throughout the session. The workshop concluded with a peer review and feedback discussion.

The outcomes from this case study (Publication II) inspired the decision to use personas and storyboards as effective tools for representing different end-user groups and communicating with stakeholders, particularly in time-constrained scenarios.

The second case study (Publication III) involved a workshop with 10 participants, mostly teenagers. The aim of this study was to understand how comic visualization can be utilized to facilitate engagement with curated data and improve sensemaking of complex issues. The workshop was designed for students from diverse backgrounds, although most of the students who chose to participate in the workshop were from a school that specializes in art and drawing. Recruitment occurred as part of a larger event that was being organized in the city and to which a number of schools had signed up to participate. This workshop was one of six possible workshops that students from participating schools could choose to attend.

During the workshop, participants were introduced to a real-world scenario regarding water pollution. They were familiarized with the history of a local lake, Vesijärvi, near Lahti in southern Finland, which was polluted in the 1960s; a restoration program for the lake was begun in the 1970s. Raw data regarding the lake was collected from water-quality experts who also helped the author to understand the history of the lake and the actions that had affected the water quality. The workshop was based on a concept of data drama . The goal of the workshop was to raise awareness regarding the causes and effects of water pollution. The workshop was conducted over six hours during a weekday. It was not possible for the author to facilitate the workshop independently due to several barriers, such as a language issue. Another researcher (an expert in arts-based methods) along with 12 youth education students from Theatrum Olga agreed to facilitate the entire workshop process. For the 12 youth education students, participation was part of a training program on learning to use arts-based approaches for youthwork through both theory and practical application. Conducting the workshop was part of their demonstration activities for the course. Before the workshop began, all participants and facilitators signed consent forms. All the workshop participants were divided into two groups and engaged in two card games, which were created based on comic visualizations following the concept of data comic. The author used the concept of card games to make the entire process more interactive. The workshop was a part of the drama and at the end of the workshop participants needed to help a mythological character by predicting the future. The prediction was supposed to be made based on what they have learned from the data comic.

The findings from the case study informed the decision to utilize comic visualization and data comic design patterns in the lean UCD canvas, as it helps in sensemaking for non-domain experts and is an interesting and interactive means of sharing information.

3.3.2 Data analysis

This study employs thematic analysis (Braun and Clarke, 2006) to analyze the qualitative data collected from semi-structured interviews, think-aloud protocols, and open-ended surveys. To effectively sort and organize dataset, a software program for thematic analysis called NVivo was used. To analyze the quantitative survey data from the first case study, Webropol's built-in functions were utilized.

Thematic analysis is a widely used method for identifying, analyzing, and reporting patterns (themes) in qualitative data (Braun and Clarke, 2006). We followed the six-phase process specified by Braun and Clarke (Braun and Clarke, 2006) to code the data and generate themes aligned with our research questions. In thematic analysis, themes can be generated via one of two approaches (Braun and Clarke, 2006): inductive or bottom-up approach and deductive or top-down approach. In the inductive approach, the themes are strongly related to the data themselves (Patton, 1990); they should not be driven by the researcher's theoretical interest. In short, the themes must be data-driven. However, it is never possible to completely detach researchers from their theoretical and epistemological commitments (Braun and Clarke, 2006). In the deductive approach, the themes are driven by the researcher's theoretical or analytical interest in the area. This approach does not provide too much description of the overall data. In this thesis, the inductive or bottom-up approach has been utilized in all studies (papers).

The interviews and think-aloud protocol recordings were transcribed and stored for analysis. Answers from open-ended surveys were collected from Webropool. Inductive thematic analysis was performed for all data following the six-phase process. All interviews, the think-aloud protocol, and surveys were conducted, transcribed, stored, and analyzed by the author himself.

The first phase was to become familiarized with the data by reading interview transcripts and survey answers, listening to the audio recordings, and making notes. In case of the interviews, potential transcribed conversations related to the research questions were highlighted, making notes and rephrasing them when necessary.

The second phase was to generate codes from the data. Coding helped the author to generate a summary of the highlighted conversations from the first phase or describing the contents from interview transcripts and survey answers (see Figure 3.1, Figure 3.2, Figure 3.3).

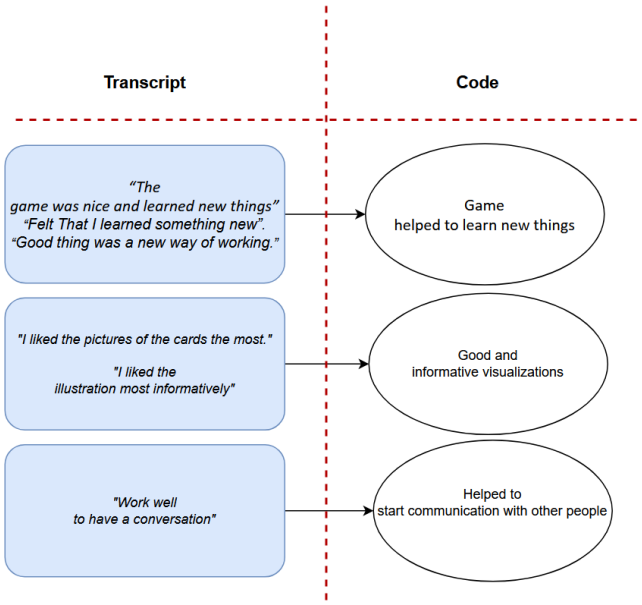


Figure 3.1: Example of generating codes from interview transcripts (Publication III)

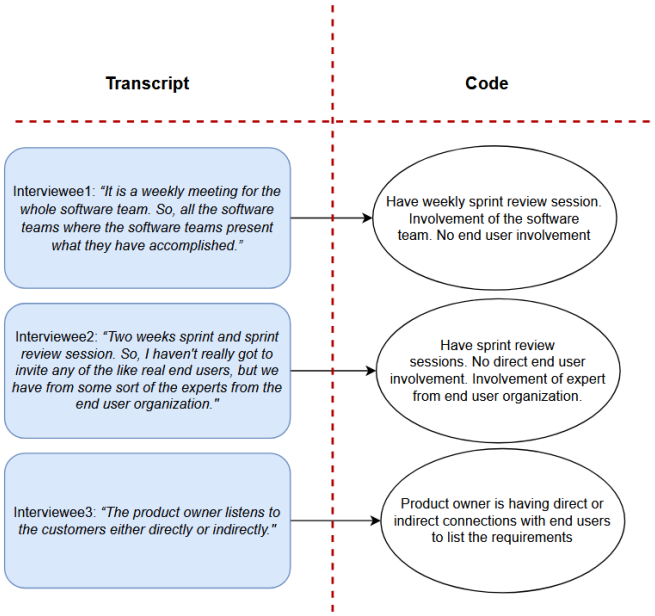


Figure 3.2: Example of generating codes from interview transcripts (Publication I)

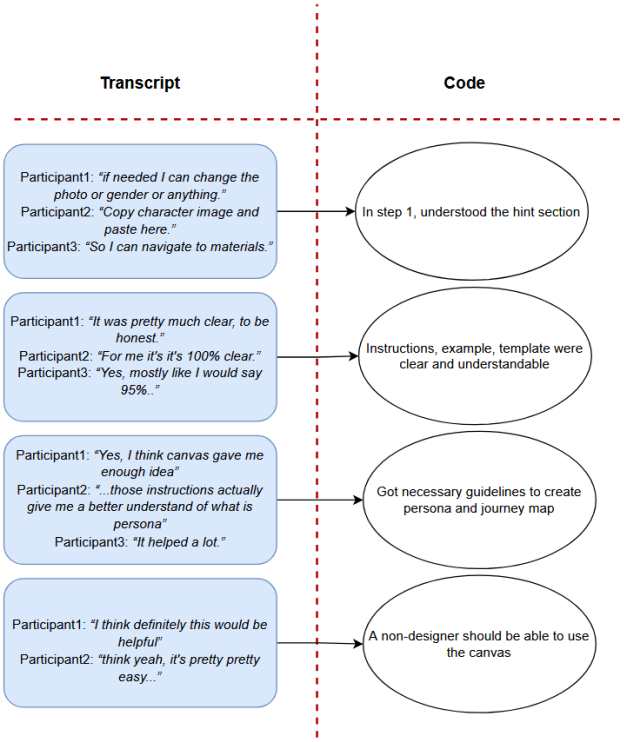


Figure 3.3: Example of generating codes from interview transcripts (Publication IV)

In phases 3 and 4, the author generated potential themes from the codes and reviewed the themes against the extracted data. All coded data from the transcripts were reviewed to produce themes, combining potential themes or dividing them into multiple different themes. For example, in publication I, for the interview question, "What is the role of the end user in your Scrum process?", the author noticed that the codes focused on either involving end users in the Scrum process or no involvement of the end users at all. The author then constructed one theme that contained all codes relating to no end user involvement in the Scrum process and one theme that related to involving end users. The author further divided the theme involving the end user into several different themes: involving end users, involving end users through product owner, involving end users through customer, involving end users through user representative, and involving end users through sales or marketing team. In publication IV, for the interview question, "Were the instructions, examples, and templates provided within the canvas clear and understandable? If not, please specify what was unclear," the responses were divided into two distinct categories: positive responses and negative responses. From the positive responses, we generated the theme "Instruction and Example Clarity"; from the negative responses, we generated the theme "Design Challenges."

In last two phases, the author defined and named the themes that addressed the research questions. It was also noticeable that few identified themes could have been subthemes

of another larger theme. In publication I, involving end users through the product owner, involving end users through the customer, involving end users through user representatives, and involving end users through the service or marketing teams can be described as subthemes of a broader theme called “indirect involvement of end users.” In publication IV, the author explored participants’ experiences by asking them about the complexity of navigating around the canvas, the clarity of instructions, and the artifacts within the canvas. Answers from these two different questions led the author to two distinct sub-themes—“Ease of use” and “Instruction and example clarity.” These sub-themes were subsequently integrated into a broader overarching theme, “Usability,” which reflected the interconnected nature of user experience and interface design. The example process, from generating codes to identifying potential themes, is illustrated in Figure 3.4, Figure 3.5 and Figure 3.6.

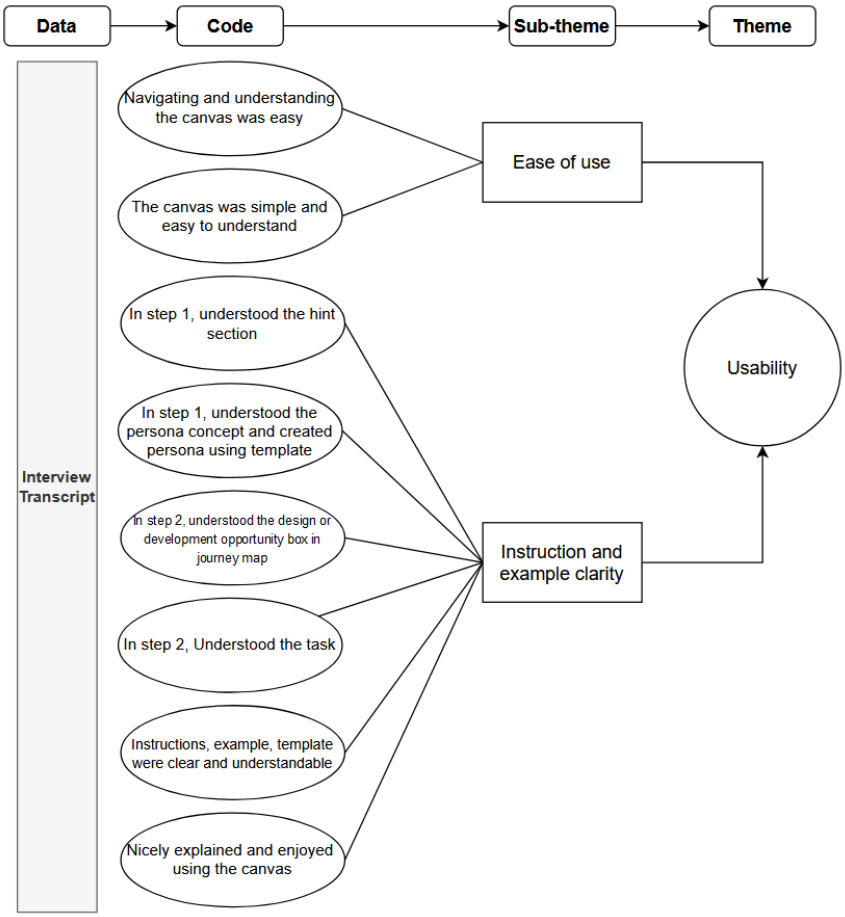


Figure 3.4: Visual representation of the thematic analysis (Publication III)

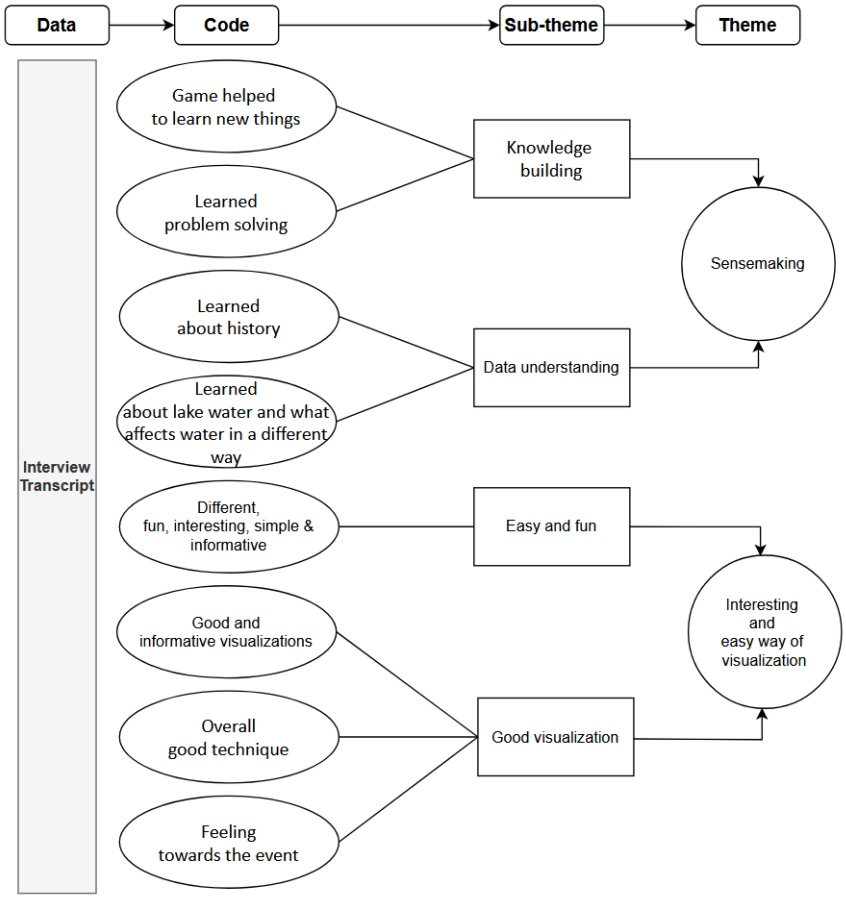


Figure 3.5: Visual representation of the thematic analysis (Publication II)

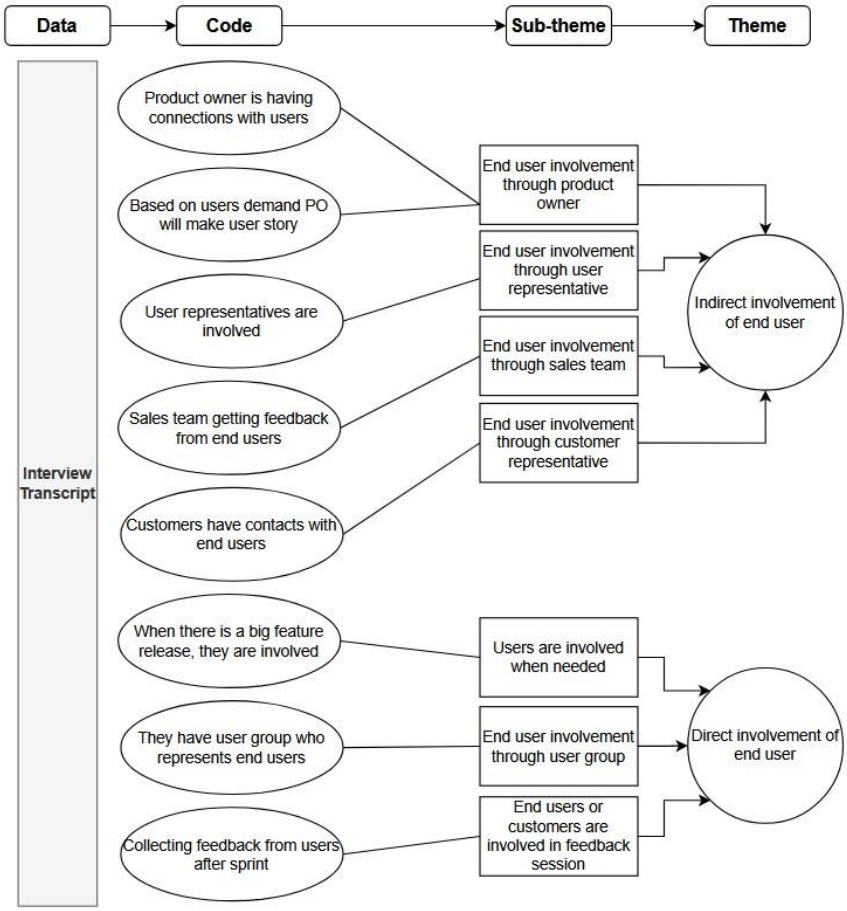


Figure 3.6: Visual representation of the thematic analysis (Publication I)

4 Research findings

This section summarizes the key findings from the publications and discusses their practical implications. The four publications have been attached to this dissertation in the appendix. For more in-depth description, please refer to the attached publications.

4.1 Publication I: Involvement of End Users in Scrum-Driven Organizations: Four Anti-Patterns

4.1.1 Background and research objective

Scrum is a lightweight Agile framework for software development. Scrum is the most widely used Agile approach and focuses on stakeholder involvement and short releases. Numerous studies have identified several challenges of Scrum in organizations, such as ignoring end user's needs and usability requirements as well as lack of techniques for the design process. However, Scrum is a framework, not a methodology. Based on the situation and requirements, different techniques can be used within the framework. The identified challenges are not actually Scrum framework challenges but Scrum application challenges. Therefore, the objective of this publication is to inspect how Scrum practitioners involve end users in their development process and explore how usability issues are handled in practice. Much of the existing research focuses on integrating UCD/UX into Scrum, but often lacks a proper discussion on why usability is often overlooked. While few studies have briefly mentioned the end-user's perspective, the studies do not reveal how they are involved or what roles they play in the Scrum process. This study is guided by following research questions:

In practice, what is the role of end users?

How are they involved in the Scrum process?

How are usability issues handled?

4.1.2 Summary of main results

The objective of this publication was achieved by conducting 11 semi-structured interviews (Qu and Dumay, 2011) with Scrum practitioners from different organizations. Four anti-patterns were identified from the interviews. Anti-patterns are common Scrum practices in organizations that may appear convenient initially but are actually harmful in the long run (Eloranta, Koskimies, and Mikkonen, 2016). The study further discussed the limitations of Scrum practitioners and the underlying rationale for these practices. Four anti-patterns are explained below.

The involvement of the end user is not mandatory, and a customer may be used to representing an end user

From the interview result, it was concluded that a majority of the practitioners do not involve end users in their Scrum process. Few practitioners mentioned that they involve end users but not directly. They include the end-users' perspective through product owners, user representatives, or customers. It is true that the product owner should be the bridge between the end user, customer, and the development team; however, in practice, identifying the end user's need and conveying them to the development team is not only a product owner's responsibility but a team effort. There is no involvement of end users in the sprint review session. Most of the practitioners believe that a sprint review is a session only for the internal team members. Only a few practitioners mentioned involving a few external stakeholders, but they are not end users.

Product demonstration occurs with no end-user involvement

Sprint review is a session in which the Scrum team demonstrates their results before key stakeholders (Schwaber and Sutherland, 2020). The key stakeholders include customers and end users. Product demos are useful after a development iteration to understand the needs of end users; however, in practice, product demonstrations are not regular and, in most cases, they occur without the involvement of end users. According to practitioner's answers, the reasons for this are 1) that the outcome of a sprint is not always sufficiently significant to have a demo session; 2) having the mindset that this is a technical event only for the internal team members; 3) lack of communication with end users.

Low focus on usability and usability is often ignored

The results from the interviews revealed that, even though almost all practitioners agreed that usability/UX is very important, the focus and necessity of usability activities depends on the product and the customer, particularly on the type of customers. There can be two types of customers—internal customers and external customers. For external customers, the priority of usability is rather high. But if the team is working or providing services for internal customers, usability is not a high priority. According to the practitioners, internal customers do not care about usability, they just want something that works. But the main issue is that even if usability is in the product backlog, it is often ignored because functionality is the top priority. A common reason for ignoring usability is because Scrum is fast and lightweight. Scrum sprints are short and focus mainly on delivery. Thus, there is not sufficient time to take usability into consideration. But according to the Scrum guide, output from a sprint should be valuable and useful, and its quality should not be compromised (Schwaber and Sutherland, 2020). Ignoring usability can result in a product that technically works but is not useful.

No formal usability techniques are employed in the Scrum process

The interview results revealed that a majority of the practitioners utilize informal evaluation techniques, such as conversation on Teams or email. Even though formal usability techniques are highly effective, they are not used in practice. Using informal techniques might increase the risk of building a low-quality product.

4.1.3 Relevance to the dissertation

Publication I provided the groundwork for this dissertation and a basis for the research that follows. This publication aligns with RQ1. Addressing RQ1, the publication uncovered four anti-patterns of Scrum-driven organizations that provide clear insights into why Scrum practitioners avoid end-user involvement in their process, overlook usability, and avoid using appropriate usability techniques. Any proposed solution should take into account the anti-patterns and following underlying causes of these anti-patterns:

- User characteristics, expertise, and background
- Scrum's limitation in facilitating optimal end user engagement
- Time limitation

It is important to clarify that the author's focus on user involvement is only on situations where considering the user's perspective is essential, for example, when designing user interfaces or developing new features that directly impact how users interact with the system. This does not apply to tasks such as backend operations or technical configurations, where end-user input is not directly relevant.

It is apparent from publication I that the direct involvement of the end user is often problematic in Scrum due to various constraints and is not always practical. This led the author to explore UCD techniques, where end users are not always directly involved but their perspectives are still considered through experts. However, since a few of the concerns highlighted by the practitioners were valid, this dissertation shifted its focus toward involving end users during the requirement collection and analysis phases rather than emphasizing their involvement solely in subsequent stages. A key challenge here is that Scrum practitioners often ignore formal usability techniques due to time constraints. To address this, the author explored lightweight solutions that could easily fit into the Scrum workflow.

Since the solution and its outcomes will be utilized by a diverse group of stakeholders, they must be presented in a clear and easily understandable manner. This led to the exploration of visual solutions that are easily understandable and facilitate better communication.

Overall, the results from publication I laid the groundwork for investigating solutions that fit within the constraints of Scrum. The identified complexity of employing traditional usability approaches and involving end users within Scrum led to the second phase of the research—exploring alternative lightweight ways to integrate usability activities into Scrum.

4.2 Publication II: Co-designing recycling solutions on campus

4.2.1 Background and research objective

Living lab is a user centric open innovation approach where end users are involved in designing and developing a product. This approach invites end users to actively take part in the design process with the aim of solving problems, generating ideas with users and testing solutions in their real-life environments. Co-design is an integral part of a living lab that entails designing with users through co-design phases. However, to make co-design successful within a living lab approach requires substantial practical knowledge often known as "know-how" that is not always shared with and amongst living lab practitioners and thus, living lab practice mostly depends on practitioners' education and experience. This makes it hard to achieve a successful living lab approach for those who are new and inexperienced. To address this gap, this paper extracted the core principles of living lab and phases of co-design and developed a new framework that combines key principles of these two approaches. The framework was validated through a case study in which a Finnish university campus was transformed into a Living Lab, using a co-design approach to improve the campus's solid waste management practices.

4.2.2 Summary of the main results

The proposed framework maps three living lab principles (openness, empowerment of users, and realism) to four co-design stages: pre-design, generative, evaluative, and post-design. The framework was validated through a case study in Finland, addressing solid waste management on a university campus.

In the pre-design phase, openness was ensured by conducting surveys with students and staff to identify challenges they face in their daily life. Based on the survey result, four personas and a storyboard were created, representing the voices of 93 individuals, to be used in later co-design stages. Three interviews were also conducted with individuals responsible for waste management. In these interviews, personas were used to illustrate the perspectives of students and staff regarding the current situation. This encouraged the interviewees to open up and discuss the challenges they were facing in greater detail.

In the generative phase, empowerment of users was confirmed by involving diverse stakeholders in a co-design session such as students and university staff. Challenges identified in the previous stage were presented using personas and storyboards. These artifacts were chosen for their visual clarity and ability to help participants see problems from someone else's perspective. The personas also encouraged discussing sensitive issues by allowing participants to speak from a persona's viewpoint not from their own. Participants then proposed solutions based on these shared understandings.

The last living lab principle is realism, which was embedded throughout, as all stages engaged real users affected by the issue. In the evaluative stage, we also validated the proposed solutions through a follow-up survey with the students and staff.

4.2.3 **Relevance to the dissertation**

This publication aligns with RQ2 of this dissertation: *How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?*

Although this case study was not conducted within a Scrum or software development context, it provided a suitable environment to test UCD artifacts (persona & journey map) under conditions similar to those found in Scrum such as the need to communicate ideas effectively among diverse stakeholders in time-sensitive scenarios (sprint review, sprint planning) to generate solutions or gather feedback. While aspects such as living lab principles and co-design stages are not directly relevant to this thesis, the application of UCD within the co-design workshop demonstrated that these artifacts could convey information very effectively. They were integral for ensuring all participants understood diverse user needs before beginning their co-design process.

Addressing RQ2, publication II demonstrated how UCD techniques (personas and storyboards) can be applied to address two of the Scrum anti-patterns identified in publication I:

- The involvement of end user is not mandatory, and a customer can represent an end user.
- Product demo occurs with no end user involvement

It is evident from publication I that directly involving end users in the Scrum process is often not practical. Therefore, in this study, personas and storyboards were used to represent the perspectives of different types of end users during a co-design session focused on ideating solutions. However, an initial user study (survey) was conducted to create these two artifacts, aligning with standard UCD practice, where users are viewed as subjects of research (Dubberly & Sanders, 2006). We conducted surveys and interviews as they are the most effective ones in the initial phase (Hasani, Sensuse, and Suryono, 2020). Typically, developing artifacts such as personas require qualitative or quantitative user analysis (Cooper et al., 2014).

Publication II also overcame one of the identified barriers to usability identified in publication I—the challenge of involving end users in Scrum due to their different characteristics, expertise, and backgrounds. It is essential to present challenges or solutions to non-technical stakeholders in a clear and easily understandable way. Findings from this publication demonstrated that both personas and storyboards provided an effective and time-efficient means of representing the challenges identified through surveys and interviews. Personas were also used during interviews to present the challenges identified from the survey. This demonstrated that personas can effectively

communicate end user's perspective in a clear and easily understandable manner. Both persona and story board were utilized in a time-sensitive co-design workshop. The results showed that participants were able to understand the presented information despite the time constraints. This demonstrates that both artifacts are effective in scenarios where time is limited.

The key takeaway from this study was that using UCD artifacts made it possible to represent the perspectives of 93 individuals in the ideation session in a clear and easily understandable way. In line with the co-design approach, diverse stakeholders were invited to the workshop. Reflecting a participatory mindset, this included end-users (Dubberly and Sanders, 2006). However, we are not proposing that end-users should be involved in Scrum events in a similar way; this was solely the focus of the specific case study.

4.3 Publication III: Playing Games Through Interactive Data Comics to Explore Water Quality in a Lake: A Case Study Exploring the Use of a Data-Driven Storytelling Method in Co-Design

4.3.1 Background and research objective

In co-design sessions, data can always provide valuable insights into a problem. But embedding curated data into scenarios such as co-design is problematic as co-design sessions are often limited by time and participants may lack adequate data literacy skills. Data should be presented in a clear and easy manner, thereby ensuring that it can be understood by anyone, regardless of their skill and expertise. A data comic combines the visual storytelling techniques of traditional comics with data visualization. It simplifies the process of sensemaking by presenting data in an interesting and easy-to-follow format. Previous research has already revealed that comic visualization is useful within the co-design process. Thus, data comics can enable the overcoming of the barriers of embedding data into co-design. In publication III, water data was curated and visualized using comic visualization to make sense of water pollution in a Finnish lake and raise awareness of environmental pollution. To make it more collaborative, the data comic was introduced in the form of card games. This publication was guided by the following research question:

What role do data curation and data visualization play to support sensemaking within a time-limited situation, such as co-design?

4.3.2 Summary of the main results

The design pattern of Benjamin Bach was followed to create the comic strip used in this publication. To make it more interactive, comic strips were presented through two card games.

A co-design workshop was conducted with 10 participants and 12 students who acted as the facilitators. The objective was to investigate 1) the usefulness of the card game, 2) whether the design of the card game helps to understand the topic, and 3) whether a data comic makes the learning process easier. We conducted a post-workshop survey with both the participants and the facilitators. The analyzed results from the open-ended survey are presented below.

Sensemaking

The objective of incorporating data comic as a visualization technique was to make sense of the curated data. The initiative was successful. The results revealed that almost all participants understood the concept. All facilitators agreed that data comic card games helped students understand the topic.

Interesting and easy means of visualization

The purpose of comic visualization is to make the data sensemaking process fun and simple. Based on the participants' responses, the entire process was fun, simple, and something that they have never experienced before. All facilitators agreed that the visualizations were an easy way of learning, and 8 out of 10 facilitators agreed that the process enabled the participants to understand the history of the lake.

Interactive

Two card games were used to present the data comic to improve interaction among participants and ensure active participation. Participants mentioned that the card games helped them to initiate conversations and teamwork.

4.3.3 Relevance to the dissertation

This publication aligns with RQ2 of this dissertation: *How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?*

This paper focuses on co-design activities. While co-design is not directly related to Scrum, there are notable similarities between the two, particularly the need to communicate ideas effectively among technical and non-technical stakeholders within time-constrained settings to present issues, brainstorm ideas or gather feedback. Therefore, addressing challenges within a co-design context can provide valuable insights into how similar approaches can be applied within a Scrum environment. The participants in this study were group of teenagers who served as non-domain experts, helping to assess how easily complex information could be understood and interpreted by individuals without specialized knowledge. The findings from publication III demonstrated that data comics are effective for sensemaking—an easy and interesting visualization method.

Addressing RQ2, publication III helps overcome two of the identified barriers to usability identified in publication I:

- Challenge of involving end users in Scrum due to their different characteristics, expertise, and backgrounds
- Time limitation

In the proposed UCD canvas solution, end-user perspectives are represented through UCD artifacts such as personas and journey maps, reducing the need for continuous direct involvement of end users in the Scrum process after the initial stages. However, the extent of user involvement may vary depending on the nature and requirements of each project. Agile is a people-oriented and collaborative approach that emphasizes close cooperation between customers and developers (Miller, 2001; Abrahamsson *et al*., 2017). According to publication I, in practice customers mainly represent end users at Scrum events and in many cases, customers are also end users. Depending on the project, customers may be domain experts or non-technical stakeholders who are not always interested in technical aspects. Therefore, obtaining meaningful feedback from diverse stakeholders requires presenting solutions in a clear, accessible, and easily understandable way. Previous studies have identified that UX or usability experts are often not available during the later stages of Scrum. To ensure the utilization of UCD throughout the process, the responsibility naturally shifts to the Scrum team. But Scrum team, especially developers, lack sufficient knowledge or experience in UX/usability practices. By addressing these issues publication III introduced comic visualization and data comic design patterns as techniques to simplify communication and make complex design ideas more accessible to both customers and Scrum team in time sensitive situations. Comic visualization has already proven successful in many HCI contexts (Sturdee and Lindley, 2019).

The publication focused on exploring how UCD techniques could be adapted to address the barriers identified in publication I. The outcome of this study was the evaluation of comic visualizations and data comic design patterns (Bach *et al.*, 2018) for representing curated data in an accessible and engaging way. These insights later informed the creation of the lightweight UCD canvas, which can be used in Scrum to produce personas and journey maps represented in a storyboard-like, comic format. Since Scrum practitioners often have limited knowledge on UCD, they can be considered similar to non-domain experts in this context. Therefore, the canvas was designed with clear instructions and visual cues to make it easily understandable for all users. While the primary purpose of the canvas is not to visualize curated data, it adopts the same comic-based design principles (Bach *et al.*, 2018) to depict personas and their interactions for effective communication which is one of the characteristics of Agile (Miller, 2001).

4.4 Publication IV: Integrating Lightweight UCD into Scrum: A Canvas Approach for Non-Designers

4.4.1 Background and research objective

Agile methods are lightweight and often avoid excessive processes. These methods either avoid upfront planning or allocate a rather small amount of time for any type of upfront design activities. UCD is a set of techniques or methods that focuses on users' perspectives to ensure that the final product meets user needs. In general, UX professionals/designers take the responsibility of conducting UCD activities. However, in small organizations, it is not possible to hire dedicated UX professionals/design teams. Additionally, Agile organizations often do not prioritize or incorporate a UX role within their team. According to the Scrum framework, every individual in a team should be able to perform every task. Thus, activities related to usability are not limited to only UX professionals; developers should also share the responsibility. But developers often lack an adequate understanding and skills for conducting usability activities. Moreover, conducting these activities while maintaining Scrum's speed is also rather challenging.

Publication IV addresses these challenges by proposing a lightweight UCD canvas designed to be used by anyone in time-constrained scenarios. The canvas helps to produce two key UCD artifacts: persona and journey map. The canvas utilizes the comic visualization technique and represents these artifacts as a storyboard (another UCD artifact). The study is guided by following research question:

How can a lean UCD canvas assist non-designers in effectively adopting and applying UCD techniques in time-constrained scenarios?

4.4.2 Summary of the main results

The main outcome of publication IV is the lean UCD canvas. It is designed to guide users through a series of steps to produce UCD artifacts: persona, journey map, and storyboard. Creating these artifacts individually is a lengthy and time-consuming process. To simplify the process, the canvas provides guidelines that follow the elements of storyboarding, data comic design patterns, persona, and journey maps.

The canvas is rather simple and straightforward. It has the following two steps:

Step 1: Creating the persona. Users become familiar with the concept of personas and their attributes by reviewing the example provided on the canvas. Clear instructions are given to guide users in creating a persona using the predefined template available on the canvas (see Figure 4.1).

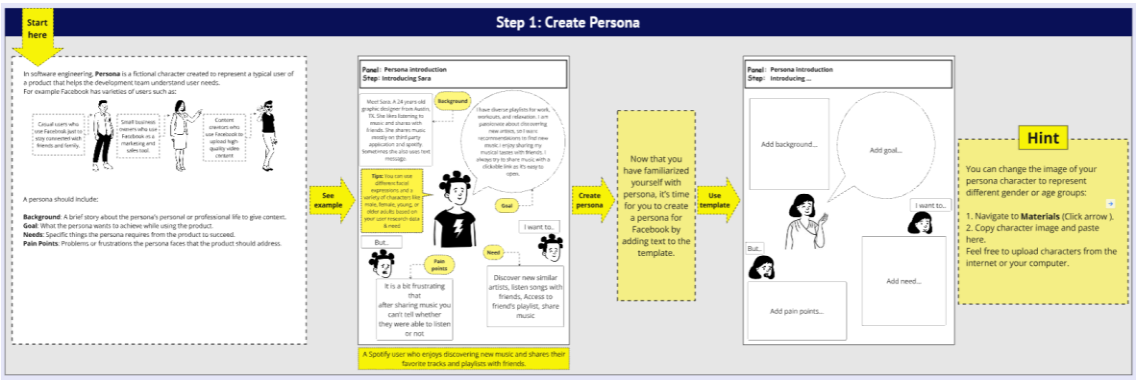


Figure 4.1: UCD canvas (Step 1)

Step 2: Creating the journey map. In the second step, users create a journey map for the persona they have created in step 1. Similar to step 1, users become familiar with the journey map concept by reviewing the example provided in the canvas and then follow the step-by-step instructions to produce the journey map (see Figure 4.2).

The canvas was evaluated by conducting two evaluation sessions that involved 16 Scrum practitioners. The first session was a pilot test with four practitioners to collect initial insights and refine our design decisions for further evaluation. A Miro board was used to share the canvas with practitioners and a survey was administered to collect feedback. Survey feedback from the pilot test helped the author to refine the overall design of the canvas. The second session was conducted with 12 practitioners. In this session, a think-aloud protocol (Van Den Haak, De Jong, and Schellens, 2003) and semi-structured interviews (Qu and Dumay, 2011) were used to obtain deeper insights. The updated canvas was evaluated in the second session and its results were analyzed by conducting an inductive thematic analysis. The results from the thematic analysis were then categorized according to the four common usability attributes identified by Sagar and Saha (2017):

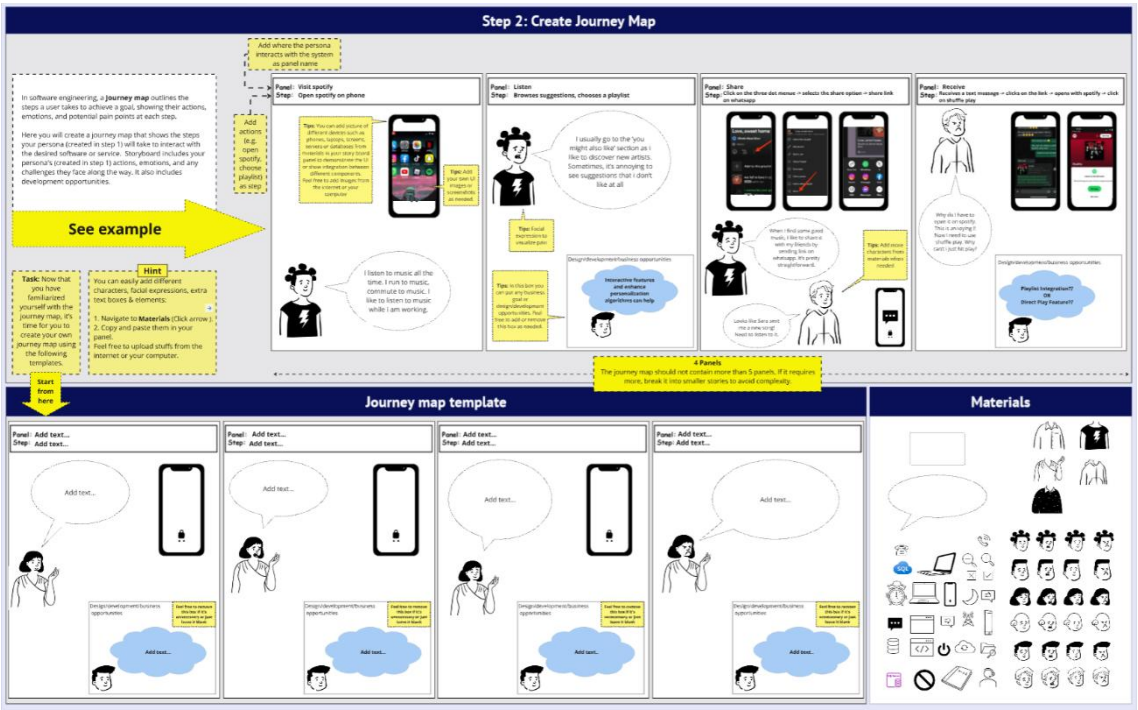


Figure 4.2: UCD canvas (Step 2)

Learnability

The canvas was designed to be used by anyone regardless of their background or design expertise. The relevance of the canvas was assessed by asking practitioners how they believe the canvas could support non-designers. A majority of the practitioners agreed that anyone can use the canvas to produce UCD artifacts. Practitioners also shared positive opinions regarding the usefulness of the canvas output.

During the think-aloud protocol (Van Den Haak, De Jong, and Schellens, 2003), the design and usability of the canvas was assessed by observing how practitioners navigated through the canvas, where they encountered confusion, how often they encountered confusion, and how they interacted with the canvas. Overall, most of the practitioners interacted with the canvas without confusion. The design and usability of the canvas enabled practitioners to complete the tasks without difficulty. A few practitioners encountered challenges, and the analysis revealed common patterns or themes among these issues. Additional changes were made to the design based on these challenges encountered.

Efficiency

On average, practitioners spent 26 minutes working on the canvas to create the artifacts; this is a reasonable amount of time, since the UCD artifacts were a new concept for many

practitioners. The time taken ranged from a minimum of 11 minutes to a maximum of 38 minutes.

Effectiveness

Most of the practitioners responded that they obtained all necessary guidelines to generate the UCD artifacts from the canvas. In particular, the examples provided within the canvas clarified ideas and helped in understanding the concept of persona and journey map.

Positive feedback was received from the practitioners regarding comic visualization. Based on their feedback, comic visualization can enable software development, collection of feedback, and engaging with nontechnical stakeholders.

Satisfaction

It was noticeable from the practitioner's feedback that they found the canvas to not only be effective but also highly satisfying.

4.4.3 Relevance to the dissertation

Publication IV is directly related to RQ2 of this dissertation. One of the usability barriers from publication I was that usability activities are often neglected, and practitioners avoid using formal usability techniques due to time constraints. Publication IV addresses this barrier by developing a lean UCD canvas. The canvas is lightweight and able to integrate UCD into Scrum by enabling Scrum practitioners to create UCD artifacts—such as persona, journey map, and storyboard—in the early development phase or within a sprint without requiring any design expertise. Since the canvas incorporates UCD techniques, the direct involvement of end users is not mandatory in later stages. The UCD artifacts generated using the canvas can be utilized across various Scrum events. For example, during sprint planning, they can help the team brainstorm ideas while considering the user's perspective. In sprint review, the artifacts can be used mainly to communicate solutions effectively with non-technical stakeholders. Finally, in sprint retrospective, they can support discussions on how usability issues were addressed, what worked well, and what could be improved in future sprints.

The results of publication IV revealed that the canvas is user friendly, supports non-designers in understanding and applying UCD techniques, provides all necessary guidance to produce UCD artifacts (persona and journey map), and is time efficient. Designers or UX experts are typically responsible for UCD activities, but using the canvas, anyone can perform basic UCD tasks within a sprint. This not only integrates usability aspects in the Scrum workflow but also helps Scrum practitioners to understand the importance of usability/UX. By doing so, this addresses the third anti-pattern—low focus on usability and usability is often ignored.

4.5 Summary of findings from each publication

The overall summary of the findings is presented in Table 4.1.

Table 4.1: Summary of the findings

Publication		Research Question	Key Findings
I.	The involvement of end users in Scrum-driven organizations: Four anti-patterns	RQ1: What are the main barriers to integrating usability activities in Scrum?	Identified four anti-patterns in Scrum-driven organizations that cause usability to be ignored. Highlighted underlying reasons for certain practices. Provided a foundation for investigating alternative approaches to integrate usability activities in Scrum, such as considering both direct and indirect involvement of end users, shifting focus on the requirement collection and analysis phase, exploring strategies for engaging nontechnical stakeholders, and exploring lightweight solutions.
II.	Co-designing recycling solutions on campus	RQ2: How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?	UCD artifact persona and storyboard can effectively represent diverse end-user perspectives. Both personas and storyboards are efficient

			for communicating with non-technical stakeholders. Both are effective in time-constrained scenarios.
III.	Playing games through interactive data comics to explore water quality in a lake: A case study exploring the use of a data-driven storytelling method in co-design	RQ2: How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?	Comic visualization and data comic design pattern that can be used to effectively communicate with nontechnical stakeholders by making complex issue visually appealing and easier to understand.
IV.	Integrating lightweight UCD into Scrum: A canvas approach for non-designers	RQ2: How can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?	A lightweight UCD canvas to produce UCD artifacts (persona, journey map, and storyboard) in time-constrained scenarios using comic visualization. The results reveal that the canvas is user friendly, effective, and relevant. The canvas focuses on usability even when the direct involvement of end user is not possible. The canvas is designed to be used by non-designers, such as developers. By

		actively engaging in usability activities, developers can understand its importance, thus leading to a stronger focus on usability.

4.6 Implications

The findings from publications make several contributions to enhance usability integration within Scrum. Overall, this thesis makes several practical contributions for Scrum practitioners.

One of the key contributions of this study is the identification of four anti-patterns in organizations practicing Scrum by investigating the role of end users and how usability is being handled in practice. The identified anti-patterns can provide valuable insights to Scrum practitioners in improving their processes. Practitioners can reflect on their current process to identify and address any harmful practices. Table 4.2 illustrates the key insights that practitioners can obtain from the four identified anti-patterns.

Table 4.2: Key insights derived from the anti-patterns (Publication I)

Anti-pattern	Insights for practitioners
The involvement of an end user is not mandatory, and a customer can represent an end user	• Developing a clear understanding of various stakeholders (such as internal and external stakeholders) and recognize their role. While examining the role of end users in the Scrum process, it was found that practitioners mostly involve internal people in sprint review. A few practitioners mentioned involving customers or user representatives. Based on the interview analysis, internal stakeholders refer to people within the organization (such as product owners, Scrum team, individuals from another team) while external stakeholders include customers

	and end users, who might, in certain cases, be the same group of people. • Distinguish between the roles of customers and end users. Customers can have knowledge about the end user, but they cannot always fully represent the needs and preferences of an end user.
Product demo occurs with no end user involvement	• Implementing strategies to collect feedback from end users, such as through frequent product demonstrations. If the outcome of a short sprint is minor, practitioners should establish a milestone where the outcome is sufficiently substantial to gather user feedback. • Turning sprint review into a session where feedback can be collected from diverse stakeholders rather than restricting it to a technical discussion only for internal stakeholders
Low focus on usability, and usability is often ignored	• Focusing on usability regardless of whether the product is for an internal customer or an external customer • Efficiently allocating time for usability activities to ensure that they are not neglected due to time limitations.
No formal usability techniques used in Scrum process	• Relying on formal usability techniques rather than depending on informal evaluation

The major contribution of this dissertation is the UCD canvas, a lightweight canvas approach to integrating UCD activities in Scrum. Typically, UX/usability experts or designers are actively involved at the beginning of the project. Their involvement in the project decreases as development progresses (Bruun *et al.,* 2018). However, in the iterative development environment, new requirements emerge frequently (Beck *et al*., 2001). Very often, a new requirement, such as a new functionality, needs usability consideration. A common issue in such cases is the unavailability of UX/usability experts, thus leading to usability being overlooked. As the lean UCD canvas is designed to be accessible to anyone regardless of their background, it ensures focus on usability even in the absence of a UX/usability expert. Scrum team members, such as developers, can easily utilize the canvas to understand how a new feature will be used by end users.

The canvas can be utilized in a structured way to implement the UCD activities defined in the ISO 13407 standard (Jokela *et al.,* 2003). First, during the requirement collection phase before the actual sprint starts which is often referred to as sprint 0 (Argumanis, Moquillaza, and Paz, 2021), particularly when creating user stories for the product backlog (see Figure 4.3). The dissertation proposes both direct and indirect involvement of end user. Direct involvement occurs only during this initial stage through traditional user research methods such as interviews, focus groups, surveys, and observations (Hasani, Sensuse, and Suryono, 2020). According to ISO 13407 standard, this stage corresponds to understanding and specifying the context of use, which includes the identification of end users (Jokela *et al.,* 2003). The next activity is specifying the user and organizational requirements, which is achieved by creating UCD artifacts by using the canvas (Jokela *et al.,* 2003). UCD artifacts complement user stories by providing visual and structured support for user-centered analysis. It must be noted that personas do not need to be repeatedly created. It is sufficient to generate them at the beginning of the project and then use them as reference points throughout the development process. If new types of users emerge subsequently, additional personas can be created. Indirect involvement continues throughout the whole process by using the UCD artifacts.

The canvas can be an effective tool in sprint planning sessions (see Figure 4.3). During the evaluation of the canvas, a few practitioners highlighted its potential as a valuable tool in this regard. In sprint planning, the Scrum team selects tasks from the product backlog for an upcoming sprint. The journey map should be created for specific product backlog item during sprint planning to produce design solutions by illustrating user interactions and pain points (Jokela *et al.,* 2003). During this event, the canvas can help in better understanding how end users will interact with the function selected from the product backlog as well as explore different types of end users and their interactions with the product, thereby facilitating a user-centered approach in the development process. However, not all product backlog items require a journey map—for example, backend tasks.

The UCD artifacts can be utilized both in sprint review and in sprint retrospective. In sprint review to help the Scrum team evaluate designs against user requirements (Jokela *et al.,* 2003) and communicate solutions with non-technical stakeholders to gather

feedback. In sprint retro to help discuss usability tasks. A previous study found that the Scrum team never discussed usability or UX aspects in their sprint retrospectives (Hinderks *et al.*, 2022). The canvas provides a means to address this gap.

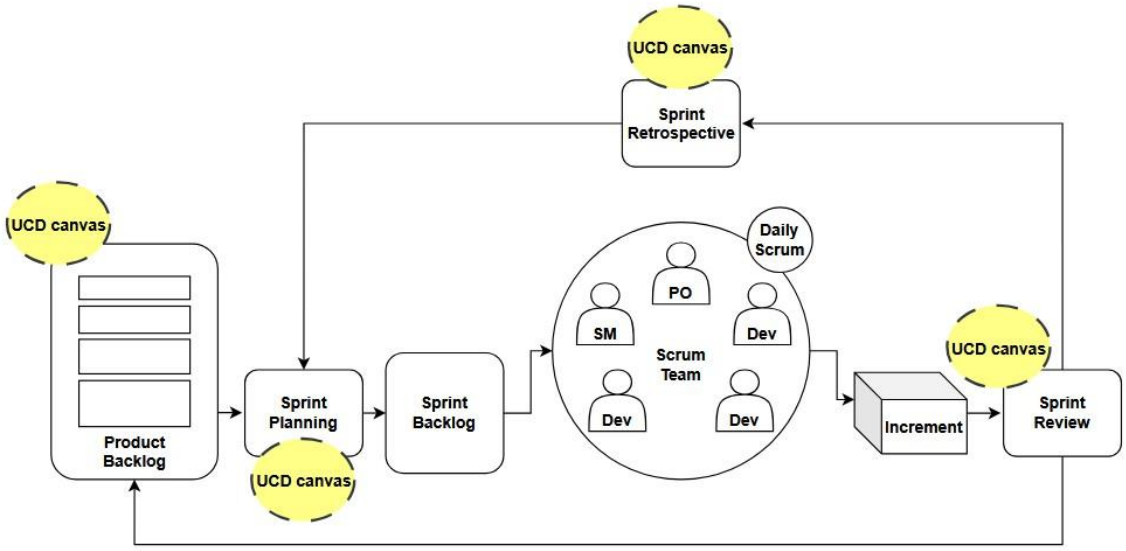


Figure 4.3: Lean UCD canvas in the Scrum process

Further, the canvas was evaluated by Scrum practitioners. The practical implications of the canvas became evident through their comments and discussions. During the evaluation session, Scrum practitioners were asked about the usefulness of the canvas output. There were many positive opinions. The opinions were categorized into the following two themes (see Table 4.3):

- practical application of the canvas output
- practical application of the canvas

Table 4.3: Practical implications of the UCD canvas (Publication IV)

Practical application of the canvas output	Practical application of the canvas
"I can easily create a UML diagram from the journey map, which will help me go into the development process faster."	*"Before design, before implementation, the canvas is a very good way for the developer."*
"Would help to figure out user experience much better."	*"With my work, or if I want to open a business, this can be a good start."*

"It really helps to understand the developer, as the developer thinks—or at least I think—there is only one type of user."	*"I think this kind of canvas will reduce gap between designers and developers."*
"This kind of thing gives you to have a different perspective because the user story is kind of dry."	*"I might start using and implementing it in my personal life."*

Practitioners also shared their insights on the practical applications of comic visualization, which are summarized in Table 4.4.

Table 4.4: Practical applications of the comic visualization (Publication IV)

Advantages of comic visualization	Participant quotes
Software development	*"It (The canvas) can play a vital role in software, like improvements."* *"This kind of output is very effective for development because a developer can get a clear message regarding what actually needs to be built."* *"(The canvas) can be effective in guiding designers and developers to create their product."*
Collecting feedback and engaging with nontechnical stakeholders	*"(Canvas can be used) when you want to actually get feedback or want to get an understanding from the real end users."* *"Sometimes we have to represent our work to nontechnical persons; in such a case, this can be an easier way to do so."*

5 Discussion

This chapter presents a discussion of the key findings from the publications included in this dissertation. Each finding is examined in relation to existing literature and practice. Their limitations and threats to validity are discussed in the latter part of this section.

5.1 Barriers in Scrum

The first sub-research question of this dissertation is, "*what are the main barriers to integrating usability activities in Scrum?"* To address this, four anti-patterns were identified in organizations practicing Scrum from publication I. The anti-patterns revealed underlying reasons for ignoring usability in practice, offering a new means of addressing and understanding the issue.

5.1.1 End-user involvement in Scrum

An observation from this research was that, in practice, Scrum practitioners do not provide a direct role for end users in the development process. Instead, their perspectives are often conveyed through customers or user representatives. While prior studies have considered end-users' perspective in Scrum (Rahayu *et al*., 2016; Almughram and Alyahya, 2017; Bordin and De Angeli, 2016; Rafi *et al*., 2015), they have not thoroughly discussed their role in practice or how organizations involve them. This dissertation not only explored the end-user's role in Scrum but also provided insights into why practitioners often struggle to include the end users. Two main challenges in involving end users were consistently highlighted by practitioners:

- User characteristics, expertise, and background
- Scrum's limitation in facilitating optimal end-user engagement.

Since end users are not always domain experts, it is necessary to train them first in order to obtain valuable feedback, which is both time-consuming and often not practical in Scrum workflows, where speed and iteration are prioritized.

The second challenge was related to the Scrum workflow. Although the sprint review is a session in which all stakeholders should be present to provide feedback, this is often not possible in practice. Many practitioners perceived it as a technical session only for the internal stakeholders. Others believed that it is the product owners' responsibility to translate technical concepts and collect feedback from end users. Additionally, a few practitioners were worried that involving users in sprint review could lead to unstructured discussion, thereby creating a chaotic environment and delay the entire process. The short duration of sprints is also an issue, as incremental outcomes may not be sufficiently large. The results from publication I revealed that involving end users in sprint review sessions is often impractical due to the short sprint cycles, as not every sprint produces a significant outcome that requires feedback. This is also identified in a previous study (Lárusdóttir,

Cajander, and Gulliksen, 2014). This implies that there's no product demonstration with end users, which corresponds to the second anti-pattern identified in publication I. These findings align with prior research in which Scrum practitioners admitted they could not recall the last time a product demonstration was held with end users (Flouri and Berger, 2010).

The findings from publication I revealed that customers often represent end users in practice. However, even customers are not regularly involved in sprint review sessions. This is because, findings from publication I indicated that in practice, sprint reviews are technical sessions and customers tend to have limited interest in technical aspects. This practice has also been identified as a Scrum anti-pattern in prior research (Eloranta, Koskimies, and Mikkonen, 2016). The primary reason for involving end users or customers in a sprint review session is to obtain their feedback. But it is visible from this thesis that integrating this into Scrum practice presents challenges. Given these limitations, a suggestion was made to establish a milestone at which the Scrum increment would be worth testing with end users.

5.1.2 The direct and indirect involvement of end users

A key insight from this study was that end-user involvement is not a straightforward issue but rather a complex decision. There were several factors that influenced this decision. One practitioner mentioned that the nature of a project plays a crucial role in end user involvement. Not all types of software projects require active participation from users.

Since numerous challenges were associated with direct user involvement, this dissertation considered both direct and indirect approaches to user involvement:

- Direct involvement occurs at the beginning of a project through user research (e.g., interviews, surveys, observations) to create UCD artifacts. Previous studies refer to this phase as Sprint 0. The canvas allows Scrum team to incorporate the user's perspective without directly involving them in the Scrum process. However, creating artifacts such as personas requires conducting user research, such as interviews, surveys, or observations (Dubberly & Sanders, 2006; Cooper et al., 2014).
- Indirect involvement continues throughout the Scrum process through UCD artifacts (persona & journey map), which represent end-user perspectives without requiring their continuous participation. The indirect approach refers to situations in which user's perspectives are conveyed through experts, customer or user representatives, which has been shown to be a common practice in Scrum, as identified in prior research (Flouri and Berger, 2010; Abras, Maloney-Krichmar, and Preece, 2018).

This does not mean that the study suggests direct involvement of end users should be ignored in later stages. While indirect involvement provides a more flexible approach, direct involvement should be pursued whenever necessary. A few practitioners in publication I mentioned involving users indirectly through customers, user representatives, and product owners. However, these individuals are also not always domain experts, which makes it difficult to effectively communicate technical solutions.

5.1.3 Usability in Scrum

As mentioned earlier, Scrum sprints are short. Previous studies have revealed that, Scrum practitioners often struggle to find time to include usability activities within the sprint (Cajander, Larusdottir, and Gulliksen, 2013; Singh, 2008; Law & Lárusdóttir, 2015). The results from Publication I reflected similar findings. Scrum practitioners believe that the Scrum framework does not support appropriate feedback sessions with end users, as conducting feedback sessions with users are time consuming and difficult to align with Scrum's basic cycle: *"It might take a bit more time, so it doesn't always fit into the basic cycle of two or three weeks."* A few practitioners also felt that the overall Scrum framework places limited focus on usability aspects: *"Well, the Scrum process isn't really it…it doesn't really take usability aspects or UX into account much."* One practitioner suggested that introducing a lightweight user involvement method could make it easier to integrate usability into the Scrum process: *"That could be then a nice thing that maybe we could have some lightweight user involvement methods that we could use in the Scrum process"*

While the time constraint is the most cited issue related to neglecting usability techniques (Flouri and Berger, 2010; Jia, Larusdottir, and Cajander, 2012; Lárusdóttir, Cajander, and Gulliksen, 2014; Cajander, Larusdottir, and Gulliksen, 2013), this dissertation identified another factor: the type of customer. Publication I presented two categories of stakeholders based on insights from practitioners (see Table 5.1).

Table 5.1: Type of stakeholder (Publication I)

Stakeholder type	Description
External stakeholders	Customers who finance the project and end users who use the product or benefit from the product. In many cases customers and end users can be the same group of people.
Internal stakeholders	Management of the company, Scrum team, product owners, program manager, functional manager, supervisors, customer support people, and UX designer. In many

	cases, an internal stakeholder can also be a customer or end users.

The customer can be external or internal. The findings suggested that usability activities depend on whether the customer is internal or external. If the customer is an external stakeholder, then the priority of usability is very high; however, if the customer is an internal stakeholder, then usability is not a top priority.

This dissertation argues that usability should not be treated as such. Usability is not only related to interface design; it has a deep connection with HCI. The first two factors of usability are effectiveness and efficiency: Can users solve their tasks using the software? Can users solve their tasks without major problems? Thus, usability should be focused upon regardless of the type of stakeholder. This suggests that Scrum practitioners may not always have a good understanding of usability, which explains why usability is often being overlooked within Scrum. Although, the results of publication I revealed that Scrum practitioners agreed that usability is very important, it is often ignored.

5.2 Addressing the barriers

The second sub-research question of the dissertation, *"how can UCD techniques be applied or adapted to overcome the identified barriers in Scrum?"* focused on adapting UCD techniques to overcome the barriers identified in publication I. To address the second research question, a canvas solution was proposed through publications II, III and IV.

5.2.1 End user representation through persona and storyboard

Since direct user involvement is often problematic and not feasible throughout the Scrum process, publication II explored how UCD artifacts—personas and journey maps—could be utilized to represent user perspectives without engaging users directly. A key point is that, while direct involvement may not be required in later stages, it remains necessary to involve users initially through user research to create these artifacts (Dubberly & Sanders, 2006). A significant issue is that developers often lack the specialized skills to analyze qualitative data from methods like interviews, observations, and think-aloud sessions (Øvad *et al.*, 2015). In publication II, this was addressed by primarily using surveys and analyzing the quantitative data with built-in platform tools, a process less complex than qualitative analysis.

However, the choice of research method should be based on project requirements. UX/ usability experts are typically available in initial phases if more in-depth qualitative

methods are required (Bruun *et al.,* 2018). In such case, conducting user research is not the responsibility of developers but falls to these specialists.

5.2.2 Comic visualization for usability integration

One of the key contributions of this dissertation was the use of comic visualization techniques to enhance usability integration. The solution proposed in publication III included the introduction of a comic style visualization technique that transformed complex data into understandable visuals. The case study results from publication III revealed that this approach was interesting, an easy means of visualization, and helped in sensemaking. In the case study, to make the comic strips more interactive, they were presented in the form of a card game.

Building upon these insights, publication IV introduced the lightweight UCD canvas for Scrum practitioners that incorporated the patterns and visual storytelling approach of a data comic. However, the goal of the canvas was not to present complex data, but to create a persona and illustrate scenarios from a journey map. This integration ensured that the canvas itself is easy to use, giving non-designers a basic understanding of the UCD artifacts and clear instructions on how to create them, and that its output is more visually accessible to both technical and non-technical stakeholders. By utilizing elements of comic visualization and data comic design patterns, the canvas enhances communication and shared understanding of the needs of end users across Scrum team and other stakeholders.

5.2.3 Lean UCD canvas

UCD was selected for several reasons. First, given the identified issues, in many cases, direct involvement in Scrum process is not practical, thus making UCD an appropriate approach as it considers both direct and indirect involvement of end users. Results from publication II showed that UCD artifacts (persona and story board) are suitable for both the direct and indirect involvement of end users. Also effective in time constraint scenarios.

Second, existing research revealed that UCD techniques are frequently used in Agile software development and have proven to be effective in many cases. The objective of this thesis was to propose a solution that would not disrupt Scrum's process as much as possible and could be easily integrated.

Additionally, the results of publication I indicated that time constraints were not the only reason usability is often ignored; Scrum practitioners also have an unclear understanding of usability. If we can get Scrum practitioners to participate in basic usability activities, it will not only enhance their understanding of usability but also enable them to develop features that are more user-centric. Additionally, this will ensure user-centered design even when a dedicated UX/usability expert is unavailable. However, developers often lack formal UX training or design expertise. Thus, the aim was also to develop a solution

that could be easily used by anyone, thereby ensuring that usability activities do not fall solely on UX experts/designers. This can lead to involve other team members, particularly the developers, in basic usability activities. This is further supported by previous research that revealed that Scrum teams believe usability is the responsibility of either the Scrum manager or the entire team (Cajander, Larusdottir, and Gulliksen, 2013).

As the focus shifted to the requirement collection phase, UCD techniques that are particularly effective during this stage were selected for the canvas. Results from publication II already demonstrated the effectiveness of using personas and storyboards. Based on existing research, the most commonly used UCD techniques among Agile practitioners are personas, journey maps, and storyboards (Schön, Thomaschewski, and Escalona, 2017; Øvad and Larsen, 2015; Schön, Thomaschewski, and Escalona, 2017). The canvas was intended to provide all necessary information—guidelines to first learn about these UCD artifacts by referring to examples and then creating the artifacts using the predefined template in the canvas.

To ensure the canvas was lightweight, only the necessary content was included. The attempt was to keep it simple and easy to use because if practitioners are unable to understand the canvas, it loses its value. To validate this, a pilot test was conducted with four Scrum practitioners to collect initial feedback on the canvas and refine the canvas design if needed. The results revealed several flaws in the initial design. Practitioners reported that the canvas contained excessive guidelines text, thus making it confusing and difficult to follow. This caused them to lose interest and become distracted from the primary goal. Practitioners also noted that the canvas instructions lacked consistency and was confusing. Additionally, it was observed from the output generated by the practitioners that they struggled to effectively produce the artifacts. There were signs of misunderstanding the key concepts. To address these issues, several improvements were made to the canvas design:

- Long textual guidelines were replaced with real-life visual examples that integrate guidelines and explanations.
- Sequential steps were introduced to structure the entire process, thus ensuring that participants could follow a logical flow.

Following these refinements, the second evaluation of the canvas was conducted. The results from the second evaluation were mostly positive. Practitioners were able to easily navigate the canvas following the step-by-step guidelines and successfully created the artifacts. In addition, Scrum practitioners who participated in the second evaluation session were from different backgrounds and non-designers. The majority of the participants were able to use the canvas. This satisfied the objective of making the canvas accessible to anyone, regardless of their background.

The key challenge identified in integrating usability in Scrum was the time limitation (Flouri and Berger, 2010; Jia, Larusdottir, and Cajander, 2012; Lárusdóttir, Cajander, and Gulliksen, 2014; Cajander, Larusdottir, and Gulliksen, 2013). Since a lightweight UCD

approach was developed here, one of the hypotheses was that it will be less time-consuming. The time taken by practitioners to create the UCD artifacts on the canvas was calculated, and the results revealed that the average time required to create two artifacts was reasonable. While rather simple example scenarios were used to create the artifacts, it is important to note that it was the participant's first attempt at using the canvas and a few of these participants had never had experience with UCD before. Since they were learning the concepts and using the canvas guidelines for the first time, subsequent usage would likely be faster. Overall, the results provided a strong argument that canvas can be efficient and feasible to use as part of the Scrum process.

5.3 Limitations and future research

While this dissertation provides valuable insights into the roles of end-users, the reasons why end-users are not involved, and why usability is often ignored in Scrum-driven organizations along with proposing a solution to integrate usability addressing the identified challenges, it is also important to recognize the limitations of the thesis to identify opportunities for future research. This section outlines these limitations and provides a few directions for future research.

5.3.1 Limitations

The first two limitations in this dissertation are related to the anti-patterns identified in Scrum-driven organizations from publication I. The first limitation concerns the sample size used for identifying the anti-patterns. Eleven semi-structured interviews (Qu and Dumay, 2011) were conducted with Scrum practitioners and stopped once a saturation point was reached where no new insights were emerging. While this is valid in qualitative research, the relatively small sample size may limit the generalizability of the findings to a larger population of Scrum practitioners or organizations.

The second limitation is related to the type of project. Scrum practitioners were not selected based on what type of projects they were working in nor were they asked specific questions regarding the nature of their projects. However, several practitioners mentioned that they were working on projects such as infrastructure development, backend systems, or supporting internal teams where end-user involvement or usability is not considered a priority. One practitioner noted that projects such as app development, which are typically aimed for external stakeholders, tend to prioritize end-user involvement and usability.

For future research, it would be valuable to engage a larger and more diverse group of practitioners working on different types of projects. This would help to validate whether the identified anti-patterns are common across all types of projects or if project type plays a role in the relevance of these anti-patterns.

Another major limitation of this dissertation concerns the applicability of the lean UCD canvas in real-life projects. While the canvas was evaluated with Scrum practitioners in a controlled setting, it has not yet been tested in actual development environments.

Although this dissertation presents a rationale based on practitioner's feedback and suggests that the canvas is well suited for Scrum environments, how practitioners would adopt and use it in real-world settings remains unexplored.

Further, the canvas itself is built upon existing UCD artifacts—such as personas, journey maps, and storyboards. However, the canvas proposes a new means of creating and presenting these artifacts using comic visualization and data comic design pattern. Although publication III demonstrated the potential of comic visualization, its effectiveness in representing UCD techniques in real projects has not been assessed or compared to traditional methods in terms of time efficiency. Future research could investigate the canvas' use in real-life Scrum projects. Such an investigation could analyze practitioners' adoption of the canvas, identify the stage/Scrum event in which the canvas is useful, assess whether the generated artifacts help developers better consider end-users' needs, evaluate if the overall process leads to software with improved usability/UX, and ascertain its effectiveness in integrating usability in a more lightweight and accessible manner than conventional approaches.

5.4 Threats to validity

To evaluate the validity of this study, Wohlin's guidelines were followed (Wohlin *et al.,* 2012).

5.4.1 Construct validity

In publication IV, the construct validity of the UCD canvas evaluation was strengthened by utilizing multiple methods (think-aloud protocols, semi-structured interviews, and an open-ended survey). This approach allowed for the triangulation of findings from different data sources. Furthermore, across all four publications, the questions used in interviews and surveys were reviewed by two other researchers to ensure they were aligned with the research objectives.

Specific measures were also taken in individual publications to address particular risks. In publication I, a primary risk was that participants might misinterpret the questions. To mitigate this, all questions were sent to the interviewees a few days in advance, allowing them to review the content and seek clarification if needed. For publication II, an expert researcher was invited to help ensure the workshop met its intended objectives. Similarly, in publication III, volunteer facilitators were recruited to help guide the workshop and keep it focused on its goals.

5.4.2 Internal validity

As all the thematic analysis was performed by one primary researcher, there is an inherent risk that the identified themes might have been influenced by researcher's assumptions,

biases, and perspectives. To mitigate this all generated themes were reviewed and discussed with two other researchers.

5.4.3 External validity

Data for publication I was obtained from 11 Scrum practitioners working in different organizations and on diverse projects. Although the sample size was relatively small, participants were selected from various companies and held different roles within their organizations, which addressed threats to external validity by capturing a range of perspectives. In publication IV 16 Scrum practitioners were involved with diverse roles including developers, testers, IT professionals, and Scrum Masters. This enhances the richness of the findings related to the evaluation of the UCD Canvas. However, recruitment relied primarily on personal and professional networks, which may have introduced sampling bias. In publications II and III, the workshops were conducted with relatively small but heterogeneous groups of participants. The inclusion of individuals from different backgrounds and experiences enriched the data and provided valuable insights, partially compensating for the limited sample size.

5.4.4 Conclusion validity

The conclusions of this study are supported by empirical evidence collected through multiple studies that employed diverse research methods, including semi-structured interviews (Qu and Dumay, 2011), open-ended and closed-ended surveys (Scheuren, 1948), case studies (Runeson and Höst, 2009), and think-aloud protocols (Van Den Haak, De Jong, and Schellens, 2003). Triangulation strengthens the credibility of the findings by overcoming biases. One of the major outcomes of this dissertation, the UCD canvas, was evaluated using a combination of semi-structured interviews, open-ended surveys, and think-aloud protocols which enhanced the validity and reliability of the canvas evaluation. The thematic analysis of qualitative data was explicitly described, and coding transparency was maintained by including direct quotes from participants across interviews, surveys, and think-aloud protocols in publications I, III, and IV.

6 Conclusion

This dissertation investigated usability practices in Scrum-driven organizations, explored how UCD techniques can be integrated into Scrum, and proposed a lightweight canvas-based solution to support the creation of UCD artifacts. The proposed canvas can be used by anyone regardless of their background, and its output is designed to be visually clear and easily understandable. The study addressed one principal research question by answering two sub-research questions: RQ1 and RQ2.

According to the Scrum Guide, Scrum is a flexible framework that can accommodate various methods. However, in practice, integrating usability and end-user considerations remains difficult. To understand why, this research first identified the practical issues that Scrum practitioners face in addressing usability and involving end users. Publication I addressed this by answering RQ1 using semi-structured interviews with Scrum professionals and identified four anti-patterns that explain organizational practices and the barriers to integrating usability into Scrum. The limitation of this study was that the findings cannot be fully generalized due to the small sample size and the fact that interviewees worked on different projects where end user's perspective is not important. However, many of the identified issues were consistent with findings from prior research, strengthening the credibility of the results.

The anti-patterns identified in publication I provided the foundation for addressing RQ2. One of the key insights from publication I was that involving end users during sprint review session, is often impractical. Therefore, this study focused on considering end users perspective in the requirement gathering and analysis phase—when user stories are developed. As in practice, direct participation of end users is not always feasible, this dissertation proposes both direct and indirect involvement of end users.

Publication II demonstrated how personas and storyboards could be created from quantitative survey data. The case study successfully demonstrated the value of UCD artifacts in a time-constrained, collaborative environment. Although the context was not Scrum or software engineering, it involved similarly diverse stakeholders (domain and non-domain experts), engaging in a process that mirrored Scrum's stakeholder gatherings (sprint review or sprint planning) for ideation and feedback.

Publication III also addressed RQ2 by focusing on how UCD techniques can be adapted to improve communication among both technical and non-technical stakeholders. Based on the outputs of publication I, customers represent end users and in many cases customers are also end users. Customers may not always be domain experts or interested in technical details. Therefore, presenting solutions in a clear and accessible manner is essential in events such as sprint review. Also, since developers often lack knowledge of UX/usability practices, the instructions for the UCD canvas and the process of generating these UCD artifacts must be presented in an easy and understandable form. Publication III directly addressed this issue by exploring comic visualization. The insights from this study informed the design of the Lean UCD Canvas, where comic visualization was integrated to simplify the presentation of personas and journey maps. The goal was to

create a tool that anyone—regardless of background—could understand and use effectively.

Publication IV directly addressed RQ2 by developing and evaluating the Lightweight UCD Canvas—that addresses barriers from publication I. As agile development is an iterative process, the UCD canvas remains valuable beyond the initial development stage. For instance, after creating personas, the journey map can be utilized to design functionality in subsequent sprints without requiring direct end-user involvement. During sprint planning, the Scrum team can leverage the journey map to brainstorm how a specific persona would interact with a proposed feature. This practice ensures a user-centric perspective throughout the entire Scrum process. A key limitation here is that the canvas was not validated in real-world projects. Therefore, future work should investigate the canvas's effectiveness in real Scrum projects.

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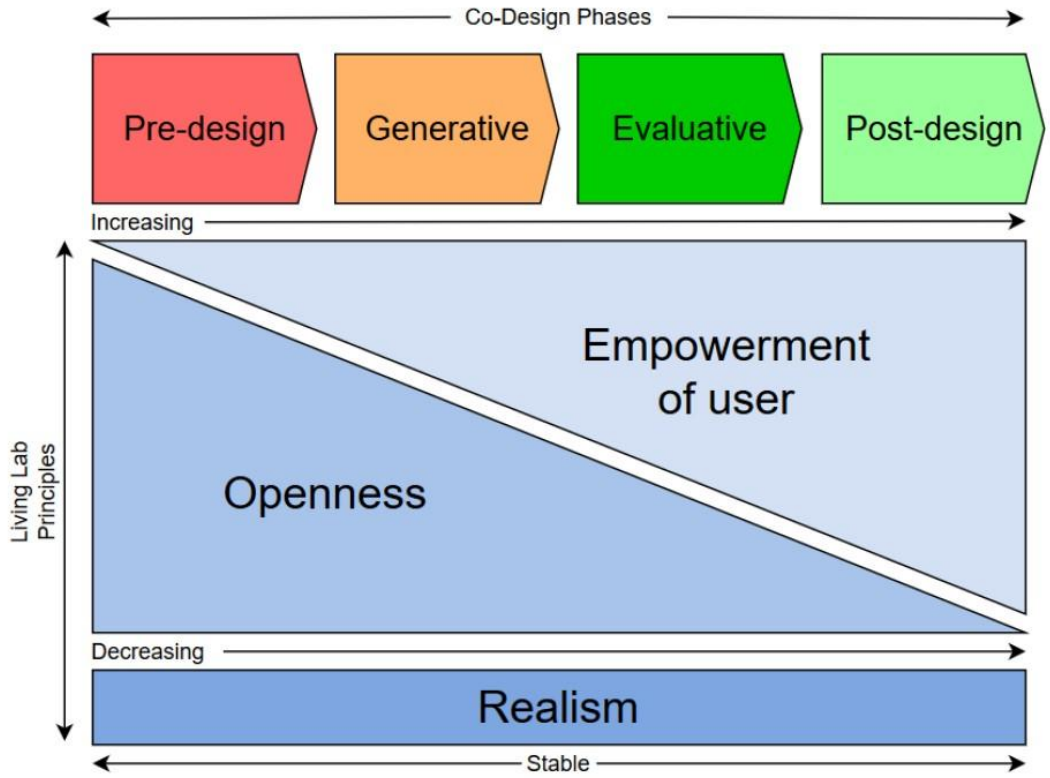
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Appendix A: Interview questions from Publication I

1. What kind of software development process do you use?
2. How often do you have a review session after a sprint?
   a) In general, who is invited to this review session?
   b) How often do you involve stakeholders in these review sessions? (stakeholders includes anyone who's going to be affected by the product or software; both internal and external)
3. Who are these stakeholders?
   a) How do you define the term "user" or "end user" in your own words?
   b) How do you define the term "customer" in your own words? (determining the difference between customer and end user)
4. What is the role of end user in your Scrum process?
   a) Are they usually involved in any feedback session?
   b) How?
5. Can you provide an example of a project where end-user involvement was particularly important? What challenges have you faced?
6. How do you ensure that the end-users' feedback is heard and addressed during the development process?
7. Have you noticed any benefits of involving end-users in the Scrum process? If so, what were they?
   a) What are the benefits do you think there could be?
8. What techniques or methods do you follow to conduct feedback sessions or evaluate the usability of the product with users?
9. Did you conduct a product demonstration with the user?
10. How important is usability or user experience in your Scrum process? Where does it stand in the priority level and how is it handled?
11. In your opinion, what do you think is the main challenge of involving end users in SCRUM?

Appendix B: The framework and Survey questions from Publication II



1. Participation in the study

- I understand the purpose of the research and volunteer
- I don't want to participate in the research

2. Have you ever heard of waste sorting, especially solid waste?

- Yes
- No

3. If Yes, write two activities which you think are related to waste sorting.

4. As an individual how much do you care about waste sorting?

- Strongly
- Neutral

- Not interested
- I don't know

5. I use different recycle bins –

- Every time I need to dump waste
- If the bin is near to me
- Depends on the situation
- I don't have time to recycle

6. On a scale of 1-5 how important is waste sorting to you?

- 1 Not at all important
- 2
- 3
- 4
- 5 very important

7. How many types of waste container are available in university campus? (e.g Plastic containers for plastic bottles). Name all you can think of.

8. In the past two months did you dispose trash into the appropriate bin inside university campus? (e.g. plastic bottles in plastic bins, papers in paper bin etc.)

- Every time
- Sometime
- Never
- Didn't notice

9. In the past two months if you sorted, was waste sorting difficult? (e.g. You have to think carefully about which bin the waste belongs to?)

- Every time
- Sometime
- Never
- Didn't notice

10. On scale of 1-5 how confident are you in your ability to select appropriate bin?

- 1 not at all confident
- 2
- 3
- 4

- 5 very confident

11. According to you which statements are true for not using appropriate bins? (You can choose multiple answers)

- Time consuming
- Visibility problem
- Lack of awareness
- Not interested
- Others

12. In your opinion, what are the barriers against using proper recycle bins? Mention at least three problems.

13. Do you think the recycle procedure (waste bins) of the university should change? Why? How?

14. In your opinion how can people on campus be more motivated to use separate containers for separate waste?

15. Thinking about the questions you have answered, is there anything else you would like to add?

16. Age?

- 17-24
- 25-34
- 35-44
- 45-64
- 65 +
- 77

17. Gender?

- Male
- Female
- Wish not to answer

18. What is your main role at the university?

- Staff
- Bachelors student
- Masters student
- Docoral students

- Other

19. My Finnish language skill is_

- None
- Poor
- Average
- Excellent
- Native

20. What is your mother tongue?

Appendix C: Feedback survey questions from Publication II

1. Which of the following system is the best to improve waste recycling situation inside the campus and why?

A. Recycle points will have a system which will read student's magnetic key and they need to type which kind of waste they have. After determining the waste only, the lid of proper bin will open. The system will count the points of the students for using the proper bin. Based on that there will be three top selected persons for the hall of fame each month. The names will advertise everywhere, and they will be given complementary lunch as a reward from any of the university restaurants.

B. A smart system which will give feedback whenever someone dumps a waste. So, if someone is dumping a waste into a wrong bin then it will show a sad emoji which means the bin is wrong. Over the time the person will realize and change his behavior.

C. Implementing a KIOSK(Digital display) inside the university. Basic design of the KIOSK is given below. On the left is the display and options. On the right side is the description of the options. There will be a survey to find the most popular waste.

- A
- B
- C

2. Which of the following changes can improve the waste sorting situation inside the campus and why?

A. Introduce biodegradable coffee cups instead of paper cups and mugs. This will reduce waste generation.

B. In every place if there are more than one bins then the sequence should be the same. For example: first bin is bio waste, second bin is metal waste, third bin is paper waste. This sequence should be maintained in every recycling points inside the campus. Benefit is soon people will learn about the bins and they will know for example the 3 bin is metal which is correct for him.

C. Perform some research on which place needs how many bins. For example: Building no 6 needs three bins. Then we don't need to put random bins here and there. We can just place those three bins. This will force the people not to put wrong waste in random bins because there are only three bins in that building.

D. A map indicating bin location. Don't need to be a touch screen map. It can be a road map. Just like the instruction we have for the room numbers right now.

E. Implement footsteps direction like in the schools. For example, for the bio waste you have to follow the yellow footsteps, for the metal you have to follow blue footsteps.

F. Instead of paper cups or the suggested biodegradable coffee cups we can use re-usable cups which will have LUT logo on it and also some words regarding sustainability or waste sorting which can raise awareness among the people.

- A
- B
- C
- D
- E
- F

3. Which of the following attachment to the current system can be useful and why?

A. Visual images of waste on the recycle bins. Put images of commonly used waste in recycle bins. For example: A Coca-Cola can and beer can image in metal container, Water bottle image in plastic container.

B. Symbol and different color in waste and waste bin. For example: Every paper waste will be equipped with a black star symbol on it. The container for the paper waste will be also equipped with black star symbol. So, there will be no confusion and people don't have to think for sorting. They will just dump waste based on the color and symbol.

C. It is simple that the university doesn't need every bin inside the campus. A simple research can say which are the mostly used or commonly used bins. University can just increase the amount of those bins.

- A
- B
- C

4. Which idea is the best to raise awareness among the students and staffs and why?

A. Tutor plays an important role for every student. They teach students about various things. Showing bin location can also be a part of tutor's task. Tutors can guide students about sorting and different bin locations inside the university.

B. Students visit moodle or university website very often. We can put the bin information there. For example: the location of different types of bins around the campus and some general knowledge about sorting.

C. Introduce some short courses. For example: courses of one or two credit on waste sorting. This course will give them proper knowledge and ideas about waste bins and sorting.

D. For the new students we can arrange some tasks in the orientation weeks. As we all know in the orientation weeks, we have many games and tasks from the guilds. We can also add something which will teach new students about recycling and sorting.

E. Most of the international students don't know about waste sorting. So, workshop or teaching event can help them to learn about it.

- A
- B
- C
- D
- E

DEMOGRAPHIC QUESTIONS

5. Age?

- 17-24
- 25-34
- 35-44
- 45-64
- 65+

6. Gender?

- Male
- Female
- Wish not to answer

7. What is your main role at the university? Staff

- Bachelor Student
- Masters Student
- Doctoral Student
- Other

8. My Finnish language skill is

- None
- Poor

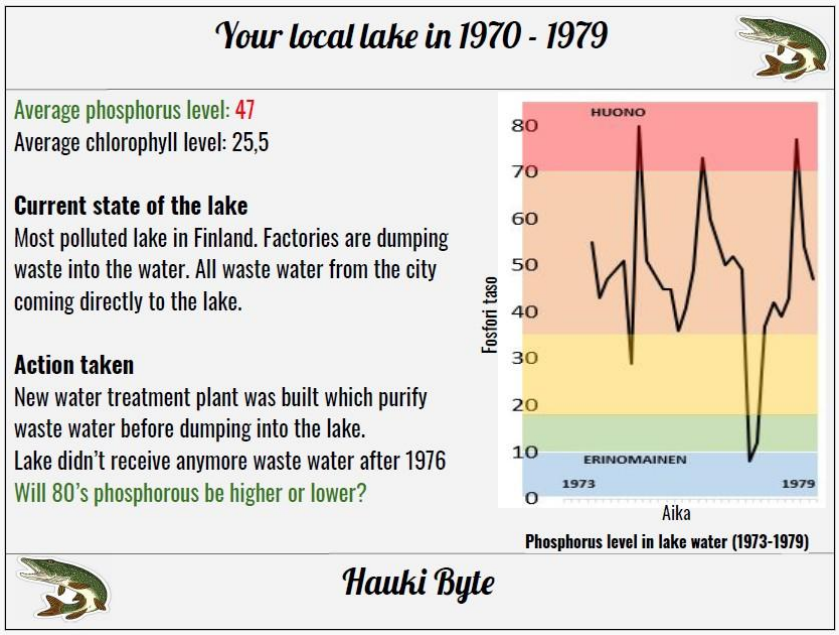
- Average
- Excellent
- Native

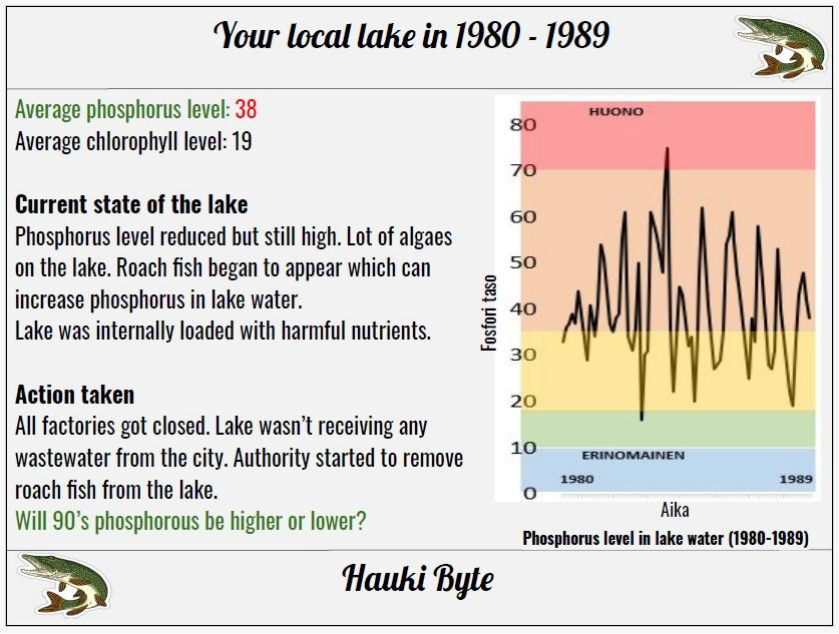
Appendix D: Survey questions from Publication III

1. What did you like most about the card games (Speed data-ing and Hauki bytes)?
2. What did you dislike about the card games (Speed data-ing and Hauki bytes)?
3. Card games helped students understand the topic.
    a) Strongly agree
    b) Agree
    c) Disagree
    d) Strongly disagree
    e) Neutral
4. It was easier to understand the impact of phosphorous and chlorophyll in water through the card game
    a) Strongly agree
    b) Agree
    c) Disagree
    d) Strongly disagree
    e) Neutral
5. It was easier to understand the history of the lake through the card game
    a) Strongly agree
    b) Agree
    c) Disagree
    d) Strongly disagree
    e) Neutral
6. Card games helped students come up with ideas on how to solve water-quality problems
    a) Strongly agree
    b) Agree
    c) Disagree
    d) Strongly disagree
    e) Neutral
7. If you facilitated your group, how did it go? What was difficult and what was easy? Are there any tips you would like to share with the game designers to make facilitation better?
8. Overall, share your thoughts about the game. For example, was it useful? Was it difficult? Was it informative? Something else?

Appendix E: Data comic card games from Publication III

1. Hauki Byte



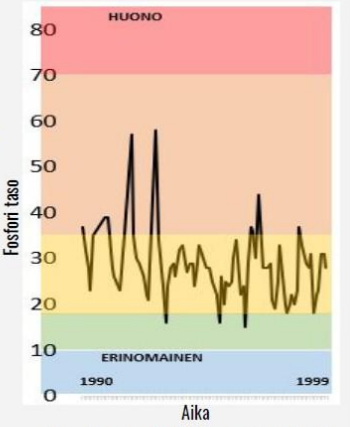


Your local lake in 1990 - 1999

Average phosphorus level: 28
Average chlorophyll level: 9,2

Current state of the lake
Level of phosphorus reduced. Phosphorus affects algae growth. With less algae people are able to swim.

Action taken
Authority continued removing roach fish. 1000 tons of fish removed and the use of the lake for leisure is promoted
Will 2000's phosphorous be higher or lower?



Phosphorus level in lake water (1990-1999)

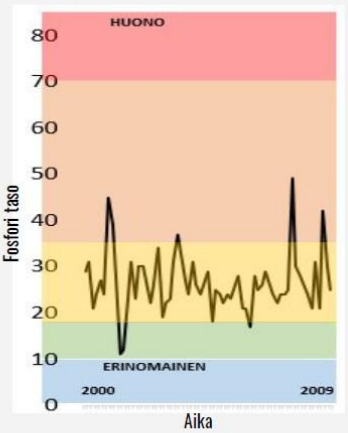
Hauki Byte

Your local lake in 2000 - 2009

Average phosphorus level: 25
Average chlorophyll level: 9,6

Current state of the lake
Phosphorus level still fluctuates, but variability is less. (When lots of nutrients in lake the variations are much larger)

Action taken
Necessary steps were taken to improve water quality even more. For example: Water treatment (oxidation)
Will 2010's phosphorus be higher or lower?



Phosphorus level in lake water (2000-2009)

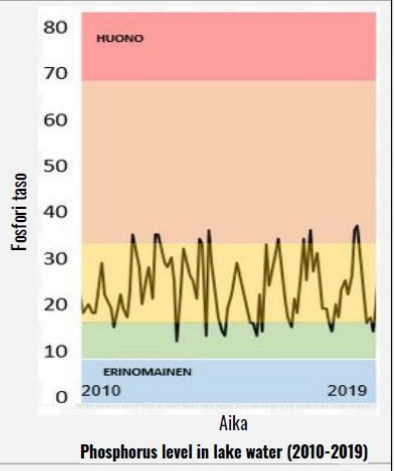
Hauki Byte

Your local lake in 2010 - 2019

Average phosphorus level: 24
Average chlorophyll level: 8,9

Current state of the lake
It was possible to see improvement effects of water treatment at deep levels but not at surface. However, overall water quality was ok and appeared to be stable.

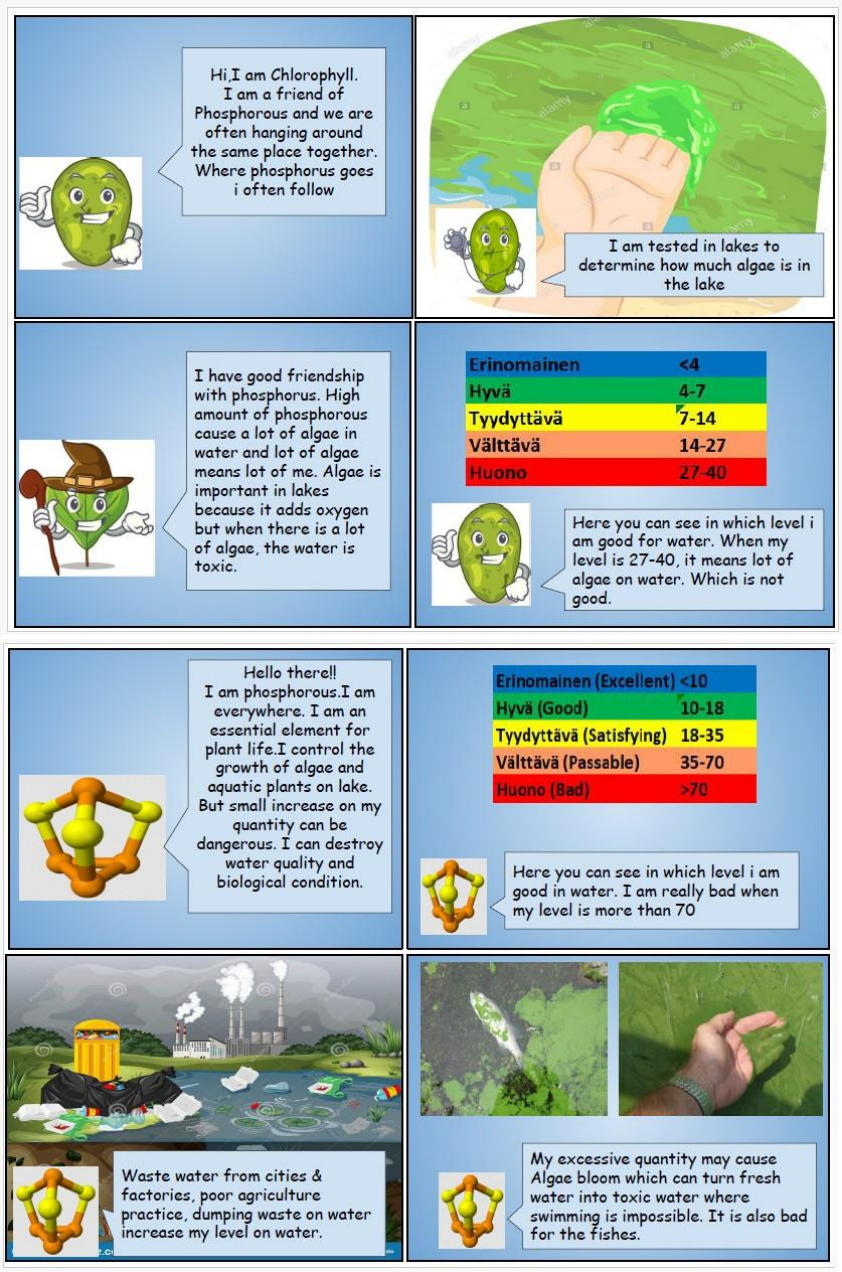
Guess what happened between now and the future??



Phosphorus level in lake water (2010-2019)

Hauki Byte

2. Speed Data-ing



Appendix F: Survey and interview questions from Publication IV

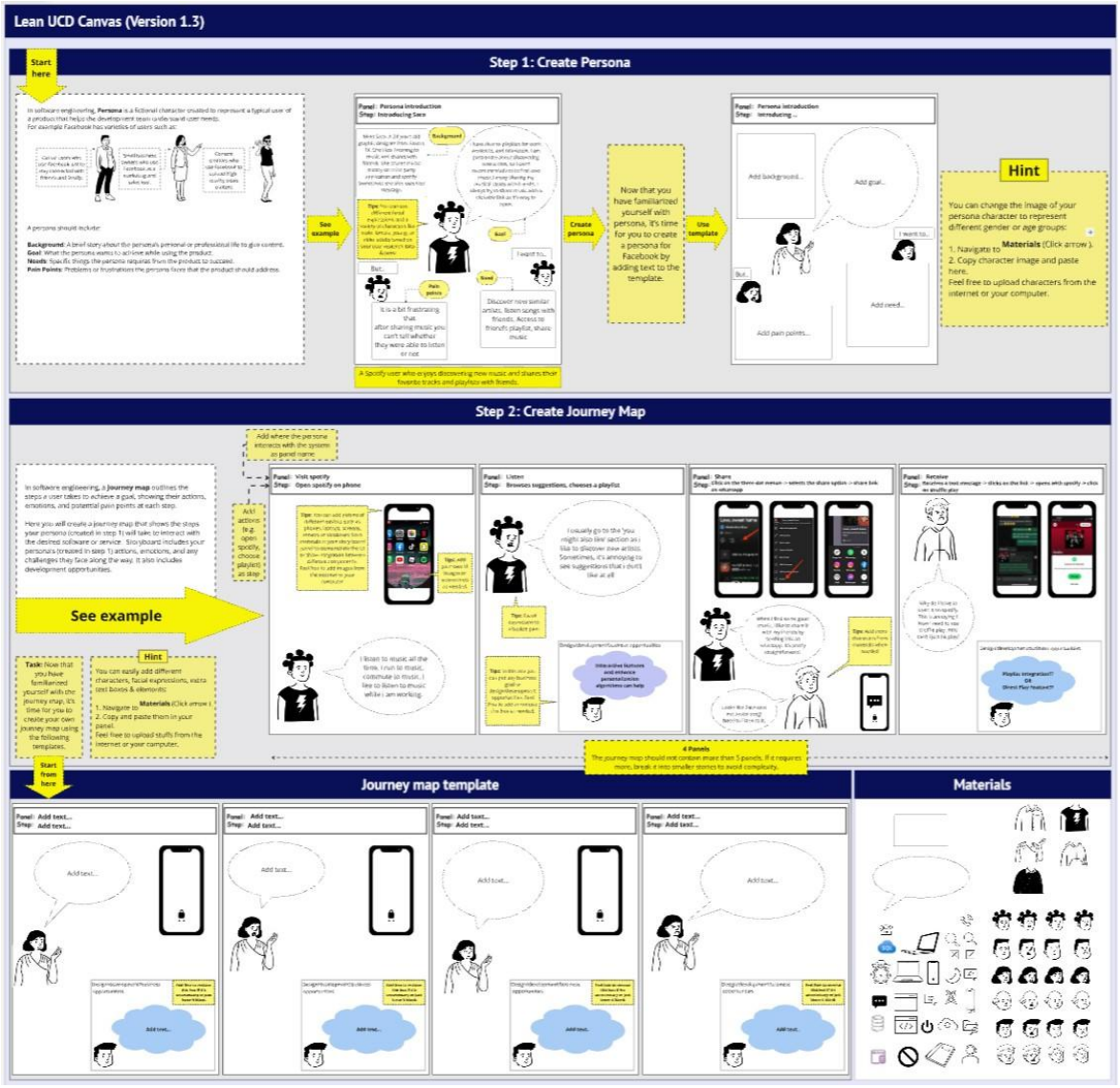
Survey questions from the first evaluation

a. Were the instructions provided with the canvas clear and easy to understand? Please mention any specific points that were confusing.
b. Overall, did you find the steps outlined in the canvas easy to follow? If not, what were the challenges?
c. Did you find the guidelines in "B. Storyboard Guidelines" easy to follow? If not, what were the challenges?
d. Did the canvas provide adequate support for creating personas, journey maps, and storyboards? If not, what were the challenges? What would have been helpful?
e. Write down one aspect of the canvas that you found most beneficial and one aspect that you felt needs improvement.

Interview questions from the second evaluation

a. How easy or difficult was it to understand and navigate through the canvas during this session?
b. Do you think that instructions, examples, and the templates that were provided on the canvas were clear and understandable?
c. Did the canvas provide you with sufficient guidance on what a persona and journey map are and how to create them?
d. Do you believe comic visualization can be effective and actionable for guiding product development?
e. Do you have any suggestions to improve canvas usability, functionality, or design?

Appendix G: The lean UCD canvas from Publication IV



Publication I

Hasan, M.T., Wolff, A., and Knutas, A.
Involvement of end user in Scrum-driven organizations: Four anti-patterns

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Involvement of end user in Scrum-driven organizations: Four anti-patterns

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Abstract

Agile software development processes, especially Scrum, have become more common in the software industry due to the fundamental principles that focus on speed and communication. However, it is noteworthy that even though the Scrum framework emphasizes frequent stakeholder involvement, the involvement of end users is often not a primary focus within organizations using Scrum, nor is usability a high-priority task. In the worst cases, the result is a low-quality product that is unable to meet user needs. To better understand the role of end users and how usability issues are handled in Scrum processes in practice, we performed interviews with professionals practicing Scrum, focusing on two areas: 1) the role of end users and their involvement in the Scrum process, 2) the importance of usability and usability evaluation techniques used in practice. Our results identify four novel anti-patterns in organizations using Scrum: 1) Involvement of end user is not mandatory, and a customer maybe used to represent an end user; 2) Product demonstration occurs with no end user involvement; 3) Low focus on usability and usability is often ignored; 4) No formal usability techniques used in Scrum process. Based on our interviews we discuss the reasoning of *why* certain practices are performed by a Scrum practitioner. Finally, the study is subject to several limitations that have also been addressed.

Keywords

Scrum Challenges, End User Involvement, Agile Software Development Process, Usability

1. Introduction

Agile is a popular approach within software development processes [31]. However, requirements engineering (RE) challenges in agile have been identified by many studies. Agile software development is a lightweight method, that focuses on delivering working software, organizing, and making coding more efficient. As such, issues related to usability are often neglected, and end user involvement – while advocated for is not mandatory in the process [3, 5]. A Delphi study was conducted by Schön et al. [20] with 26 experts in agile to identify the key challenges in agile requirement engineering. The result showed six challenges for industry practicing agile processes and two of these challenges are related to stakeholders and users. A summary of these challenges is i) lack of involvement of direct end users ii) difficulty in involving stakeholders in the development process. A systematic literature review on "Obstacles Faced by Requirements Engineering Professionals in an Agile Context by Applying User Experience" also showed that "lack of user participation" is the common problem mentioned in most of the studies [1]. More specifically, to identify agile requirements engineering practices and challenges a systematic literature review was conducted by Inayat et al. [10]. The study identified eight challenges posed by agile requirements engineering and one of them is neglecting nonfunctional requirements such as testability and external quality usability. According to Helmy et al. [9] the issue related to nonfunctional requirements should be taken care of to improve the product quality. The aforementioned factors demand the necessity for engaging end users and focus more on usability when enacting agile methodologies.

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In this paper, we focus on inspecting the most popular agile approach, Scrum [22], and how Scrum is utilized in industry. Scrum is a lightweight agile project management framework mostly used for software development. Scrum focuses on frequent stakeholder involvement and short releases, which is why Scrum is broadly adopted by industries. Since Scrum is an agile approach, it also faces some of these same requirements engineering challenges when put into practice [23]. Specific challenges include lack of time to focus on the end user's needs [2], lack of techniques for the design process [21], and usability requirements being ignored [29]. However, proper utilization of Scrum depends on professionals. The identified challenges are not a Scrum framework challenge but a Scrum application challenge. According to the Scrum guide the Scrum framework is purposely incomplete. Different types of techniques and methods can be used within the framework in different situations and environments [8]. So, our goal is to focus on how practitioners are involving end users in Scrum processes and how they are ensuring usability of the product. The paper aims to answer the following research questions:

In practice what is the role of end users?
How are they involved in the Scrum process?
How are usability issues handled?

Our results identify four novel anti patterns in organizations using Scrum. Here we refer to anti patterns as the common Scrum practices that appear convenient in the beginning but harmful in the long run and therefore must be avoided [28]. The results also provide holistic understanding of the challenges faced by software companies related to end user involvement when employing Scrum.

The structure of the paper organized as follows. Section 2 provides related work. Section 3 discusses the research methodology and data analysis. The results of the study are presented on Section 4 and discussed on Section 5. Section 6 and Section 7 respectively address the limitations and conclusions of the study.

2. Related work

2.1. Involvement of end user in Scrum

Normally in Scrum, most frequent review activities such as end of sprint review and testing involve the product owner and the development team. End users are not commonly involved in these activities, but instead may be involved in feedback sessions when they are using the product. By this stage, change is still possible, but requires costly rework [15]. Even though agile processes such as Scrum are iterative approaches that conduct user testing in the process, when put into practice Scrum often does not focus on creating user experience based on user needs [2]. Most of the time, the Scrum team is having communication only with the customer. It is possible that the customer who is representing the end user is inexperienced, not efficient in providing feedback and not having enough knowledge. In that case the product may not work well with the end user [27]. There are exceptions where a sample of real users are also involved, but this happens only occasionally. Usually, it is the customer who represents user needs, requirements and gives feedback [24].

Even the role of end users versus customers is still unclear in many software development companies [18]. Whilst doing an analysis on the fitness of Scrum and Kanban towards user experience, researchers [25] noticed that interviewees were using the terms "customer" and "user" as synonyms which motivated them to do additional research on the issue. The result showed that whilst many respondents did understand the difference between "user" and "customer" there were still a significant number who were unclear about the differences and couldn't answer properly or answered incorrectly.

One of the core values of agile development is that working software should be demonstrated to the end users throughout short development iterations. The reason behind this value is that working software elicits more detailed information toward understanding end user needs than written words. However, the situation is different in practical life. In one organization, team members were interviewed after a transformation to Scrum from a more traditional waterfall process. In their interview, they specified that they never did a full product demonstration. In other words, there wasn't any demonstration after each sprint where they could gather feedback. Instead, they delivered the software when all the required functions were working. So essentially, they ended up doing a waterfall approach every three weeks [4].

2.2. Usability techniques used in Scrum

In many Scrum projects the Scrum team (product owners & developers) concentrate more on the goal of implementing a working function rather than the usability or usefulness of that function. A common argument here is that the usability can be improved later [17]. In practice, the overall result is a product that may obtain all the functional requirements but still fails in the market due to poor user satisfaction [14]. In [11] authors performed a survey study to find what usability techniques are used in Scrum processes and how often they are used. They listed 13 usability techniques used by the practitioners. Later, they performed another survey to find the usefulness and usage of these 13 techniques. Results showed that according to 75% of the respondents, formal usability evaluation with users is a very good technique but only 30% of participants used

formal usability in their project. The author mentioned one possible explanation and that is that Scrum is all about speed. Scrum sprints are short (2-4 weeks). It takes a longer time to use a specific usability technique which may slow the whole Scrum process down. Also, the output of such sprints is minor, and evaluating them in a quantitative way is not useful as changes are small. Therefore, Scrum professionals tend to rely on informal usability techniques such as lo-fi prototyping, and meetings with users instead of formal techniques such as heuristic evaluation, questionnaires, field studies, scenarios, or digital prototyping. One advantage of informal usability techniques is they can be performed quickly without prior preparation. In some cases, informal usability techniques are helpful, but the main drawback of this approach is the number of users involved. Only a few users are taking part in informal usability evaluation, so evaluators are not sure if they can rely on the result or not [12]. In agile, especially within Scrum, usability requirements and their implementations are ignored most of the time. Usability or usefulness is often ignored or a less prioritized task in a Scrum process. Usability features are often absent in product backlog [29]. Another study result, on the user perspective in Scrum practice, showed that the involvement of a user in the process is informal, meaning that user involvement is done based on an individual's own initiative, preferences and knowledge. There's no systematic plan or approach within the development process and often there's no specific person responsible for usability issues, nor any clear usability goals. As such, the user perspective is only considered if the product owner or team has an interest in usability [6]. One survey study was conducted on user involvement methods in Icelandic software industry [13]. 37% of the respondents were using Scrum as the software development process and rest of the respondents were using other methods including organization's own process models, waterfall, and XP. The result shows that developers who were using Scrum are the most skeptical compared to others when asked about whether usability is important or not. The definition of usability within that study described it as "Usability is a qualitative attribute that assesses how easy user interfaces are to use. Usability is mainly made up of three factors: Effectiveness – Can the users solve their tasks with the software? Efficiency – Can the user solve their tasks without major problems? Satisfaction – How satisfied are the users?

Most of the literature on Agile or Scrum mainly focused on integrating UCD, UX [2, 17, 5, 3, 1, 15, 25] or requirement engineering challenges [4, 31, 7, 20, 10, 9]. Usability issues are considered in several studies [11, 12, 29, 13, 6] but with no proper discussion on the *reasons* for ignoring usability. While several researchers have been interested in end user's perspectives in Scrum, the research does not reveal what is the role of end users in practice, or how practitioners are (or are not) involving them. Therefore, our research will focus on understanding not just what gaps exist in end user involvement in practice but why the involvement of end users and usability is often ignored.

3. Method

To familiarize ourselves with the issues related to end user involvement and usability within companies' agile processes, we conducted semi-structured interviews with professionals having experience working with Scrum or working as a product owner (PO), scrum master (SM) or developers in a Scrum driven software organizations. We continued conducting interviews until a saturation point was reached in which no new information was emerging, which happened after 11 interviews had been conducted. We observed consistency among the interviewees as they articulated recurring themes and issues. All interviews were conducted online (MS Teams) and recorded by one researcher. The interviews lasted from 45 minutes to an hour.

3.1. Participants

The 11 individuals were selected randomly through personal networks, recommendations from people who had already been interviewed and via a post on social media (Facebook, LinkedIn). All individuals who participated in the research had practical experience working in Scrum or Agile as SM, PO and developer. There were in total 9 males and 2 females. More information can be found in Table 1.

Table 1
Description of interviewees

Number of interviewees	Role	Experience
6	SM, PO	4-15 years
2	SM	1 year
2	SM, PO	2-5 years
1	SM, PO, Developer	5 years

When participants were asked about what type of software development process they were using, most of them didn't answer directly Scrum. Only two participants replied they were using Scrum. The Scrum process that others were using had been modified to adapt to the needs, requirements, and context of the organization. Similar to the outcome presented in [25],

most of the organizations were using features of Scrum and combined Scrum with another agile development process, Kanban. A mix of Scrum and Kanban. Participants answered the questions based on the project they were working on.

Table 2
Example of coded transcripts

Transcript	Codes
Interviewee1: *"It is a weekly meeting for the whole software team. So, all the software teams where the software teams present what they have accomplished."*	Have weekly sprint review session. Involvement of the software team No end user involvement
Interviewee2: *"Two weeks sprint and sprint review session. So, I haven't really got to invite any of the like real end users, but we have from some sort of the experts from the end user organization."*	Have sprint review sessions. No direct end user involvement Involvement of expert from end user organization
Interviewee3: *"The product owner listens to the customers either directly or indirectly."*	Product owner is having direct or indirect connections with end users to list the requirements

They were also asked to answer from one previous project where they had to involve end users, or the involvement of end users was mandatory. The interview questions were designed to be adaptable to the level of experience of the individuals and in accordance with the situation. The interview questions focused on two areas: 1) end user and 2) usability in the Scrum process. To collect demographic, background information (role, education, experience) and sign the ethical form (voluntary participation & data privacy), we performed a survey, using the GDPR-compliant survey tool Webropol, prior to the interview.

3.2. Data analysis

Interview recordings were transcribed and analyzed with inductive thematic analysis [30]. The first phase was to get familiarized with the data by reading transcripts, listening to the audio recordings, and making notes. In our case we highlighted all potential transcribed conversations related to our research question, putting notes and rephrased when necessary. The second phase was to generate initial codes which provided us with a summary of the highlighted conversations or describing the contents (see Table 2).

In phases 3 and 4 we constructed potential themes from codes and reviewed the themes against the extracted data. We reviewed all coded data from interviews for each question to generate themes, collapsing potential themes together or splitting them into multiple different themes. Example: For question, *"What is the role of end user in your Scrum process?"*, we noticed that the codes focused on either involving end users in the Scrum process or no involvement of the end users at all. We then constructed one theme containing all codes relating to no end user involvement in the Scrum process and one theme relating to involving end users. We further divided the theme involving end user into several different themes: involving end users, involving end users through product owner, involving end users through customer, involving end users through user representative, involving end users through sales or marketing team.

In phase 5 we defined and named the themes addressing our research questions. We also noticed some identified themes can be subthemes of another theme. Example: involving end users through the product owner, involving end users through the customer, involving end users through user representative, and involving end users through the service or marketing team can be described as subthemes of a broader theme "indirect involvement of end users".

4. Results

Here the interview results will be presented focusing on two major areas: 1) the role of end users in Scrum, 2) the importance of usability and usability evaluation techniques used in practice.

4.1. Involvement of end users in sprint review session

According to the Scrum guide, a sprint review is one of the mandatory events to gather feedback from key stakeholders (which includes end users) but in practice, involving end users is not necessary. In sprint reviews, feedback sessions and testing are often done without any involvement of end user. Instead, domain experts are playing the role of an end user. In almost all cases, major participants in sprint review are the Scrum team and internal stakeholders or customer representatives who play the role of an end user. In this case, it is hard to determine how relevant the feedback is with respect to the needs of real end users [26]. Almost half of the interviewees (five out of eleven) answered that they are only involving internal stakeholders in the sprint review session. Three interviewees had two different sessions but only on demand. One is for the internal stakeholders and another session is for the customers or end users. Three interviewees informed us that they are involving external stakeholders in the sprint review session. Stakeholders are those groups of people who are impacted by the product outcome. Based on our analysis, we identified the following categories of stakeholders from interviews. The categorization is necessary to understand different types of stakeholder interviewees addressed during the interview:

1. External stakeholders: Customers who finance the project and end users who use the product or benefit from the product. In many cases customers and end users can be the same group of people.
2. Internal stakeholders: Management of the company, Scrum team, product owners, program manager, functional manager, supervisors, customer support people, UX designer. In many cases an internal stakeholder can also be a customer.

4.2. Defining the term “end user” and “customer”

Customer and end user are different terms, and both have different roles in the development process. Scrum teams and literature sometimes do not specify the differences between a customer and an end user in a project. From the previous research [25, 18] we identified that agile practitioners are not always clear on the role of end user and customer. The purpose of defining “end user” and “customer” was to determine the participant’s understanding of these terms. Based on the answers almost all participants had a proper understanding and clear distinction between the two terms “customer” and “end user”.

4.3. Role of end user in the Scrum process

After defining end user and customer, all interviewees were asked about the role of the end user in their Scrum process. Analysis of the answers resulted in three different themes: a) Indirect involvement of end users b) No involvement of end users c) Direct involvement of end users. Four out of eleven interviewees mentioned that they are indirectly involving end users through a middle person such as a product owner, sales team, UX designers, customers, or user representatives. However, one interviewee also mentioned that even though end users are not involved in the Scrum process, they have close contact with them. Almost half of the interviewees (5 out of 11) replied they are having no end user involvement in their Scrum process. Some are involved, but only in special cases: *“I would say that no, not in daily, like all the sprints that we have. But it will be more of a special case when we have a special need. So, it is more of this special occasion when we decide that, hey, OK, now, we want to talk with them.”*
“*But in the scrum process we don't involve the end user actually”.*
Only 2 interviewees were directly involving end users in their process however one of them stated that they are involving end users only when they need, not on a regular basis: *“We haven't gotten them involved in the like every Sprint, so to speak, but there when we do something some kind of feature a little bit bigger thing we usually uh set up some kind of pilot group and we talk to them over teams and sort of gather their feedback from that”*

4.4. Ensuring end users' feedback during the development process

We already noticed from the results that most of the interviewees are not involving end users in their Scrum process, so how do they ensure their feedback is heard? Everyone described their own different approaches. We categorized them into two sections: 1) getting feedback from end users; 2) getting feedback from user representatives: *“but their customer make decision should we do the changes or not? We have these different persons who are basically listening the users.”* 8 participants were collecting feedback from user representatives. Only 3 participants mentioned that feedback is coming directly from end users.

4.5. Importance of usability, user experience and benefits of involving end users in the Scrum process

Even though from the responses it is apparent that end users are seldomly involved in the Scrum process,

almost all of the participants (8 out of 11) agreed that usability or UX is very important, and their priority is either high or very high: *"it is always about providing the user with the best experience. So, it's definitely the most important thing."*

"I would say that it should not be the highest but one of the highest. If it doesn't satisfy a customer, then what's the point anyway."

However, it was also pointed out from the conversation that the priority of usability or UX also varies from case to case: *"let's say if the customer or end user is some external customer or external client, then I would say the user experience should come first but for internal customers the situation is a bit different."* Furthermore, participants were asked about their opinions on the benefits of involving end users and two themes emerged from the analysis: 1) Enhanced User Understanding and 2) Enhanced Product Quality.

4.6. Techniques or methods used to collect feedback or evaluate the usability of the product

When asked about the methods used for usability evaluation and collecting feedback, we got quite diverse answers. We tried to sort the answers into two categories: 1) formal usability evaluation; 2) informal usability evaluation (no systematic plan or approach sometimes based on individuals' own initiative). Only 3 participants mentioned that they are using some type of process although two of these participants are not using any techniques in the current project. They referred to a past project where they have used formal usability techniques such as questionnaires, or feedback tools. Only one participant was using formal methods for usability testing. The rest of the participants didn't have any systematic plan or approach to collect feedback or evaluate usability. Mostly they were using team conversations and sending emails: *"Uh, yeah, just uh, we are sometimes requesting this information in teams."* Some also mentioned initiating meetings when needed: *"Uh is the informal techniques, so if we need it then we initiate a meeting request and then book a calendar and then get the feedback from them".*

4.7. Product demonstration with end user

Product demonstration is one of the core values of Agile to understand end user needs. Almost all interviewed individuals informed us that they are having either regular or irregular product demonstrations. But our goal was to find out if they are having these demonstrations with end users or not. Analysis showed that only three interviewees were involving end users in product demonstrations, where one company was having regular product demos and another two had irregular product demos. The rest of the interviewees had product demonstrations with no end user involvement.

4.8. Main challenges of involving end users in Scrum process

We asked participants to point out the challenges of involving end users in Scrum from their own experiences. It was interesting to see that there wasn't that much diversity in the answers. Almost everyone had similar opinions. There were mainly two types of answers: 1) User characteristics, expertise, and background; 2) Scrum's limitation in facilitating optimal end user engagement. According to the interviewees, choosing the right end user is very important and a big challenge:

"I would say it depends on the product and what you consider as end user and it it's also very important to choose the right end users."

"You need to get involved the right people on the right time and on the right place."

Some interviewees also pointed out one issue and that is lack of knowledge and experience. Sometimes you need to train the general end users to get meaningful feedback. Also, the feedback should not come from just one individual's opinion. Feedback should come from the whole user group. So, it is always easier to involve someone knowledgeable and experienced who's able to represent end user groups such as super users or user representatives. The other challenges mentioned by interviewees are all related to Scrum's limitation in facilitating optimal end user engagement. The most common challenge was not being able to include end users in the sprint review process. According to the interviewees, the Scrum sprint review is intended to be only for the internal team members. Involving general users is often not possible as these sessions are mostly technical: *"they're talking sort of user language and we're talking developer language. So, if you involve them in the same kind of in this well in the Scrum, scrum day-to-day or Sprint week to week thing, then that could be a little bit challenge."* One interviewee mentioned that it is the role of the product owner to translate the technical language and have conversations with end users. Involving end users in sprint review just will create a chaotic environment and also might delay the whole process. Furthermore, one interviewee added that Scrum focuses more on delivery so fitting user's perspective in two to three weeks sprints is hard. End user involvement needs different planning.

5. Discussion

5.1. Involvement of end user is not mandatory, and a customer maybe used to represent an end user

There are several implications for Scrum due to low user involvement, these are (1) software testing: In Scrum, testing is done by the product owner and development team instead of the user, (2) Poor software quality: The user might not be satisfied from the software as no testing has been performed, (3) No feedback: There are no feedback sessions after sprints. A feedback session takes place after the software has been created. As a result, the Scrum team has little idea if they are going in the right direction or not [15]. In our interviews we asked interviewees about the role of end users in their Scrum process and whether they are involved in sprint review sessions. The results show that the majority of the interviewees are not involving end users in their Scrum process. Interviewees who agreed that they are involving end users mostly having indirect communication with the end user through product owners, user representatives, or customers. From studies, it is indicated that customer representatives often have daily contact with the developers which can make them too familiar with the inner software working process. So, they cannot truly represent an actual end user during the feedback session [21]. According to the Scrum guide it is true that the product owner should be the link between end user and development team. But in practice this is not always true. One study [19] showed that some product owners think that identifying end users need and communicating those needs with development team is not product owner's responsibility but a team effort. Also, product owners in Scrum are mostly overwhelmed with marketing and sales concerns and often lack formal education and skills to design effective user experiences [17]. Sprint review sessions are mainly for the internal Scrum team or internal stakeholders. Only a few mentioned they involve external stakeholders, but external stakeholders are not always end users. According to one of these interviewees, their external stakeholders are experts from an end user organization. So, our study results stay partially in line with [2, 26, 28, 15] that product owners, user representatives, or customers play the role of end users and participate in the Scrum sprint review. End user involvement is not mandatory in the Scrum process. We pointed out from our interviews that some of the interviewees are developing products for internal customers who are also end users. When end users are internal customers then the focus is more on functionality rather than usability. Moreover, some interviewees were working in sectors where UCD is not important, such as the telecommunication sector where the priority is to have a system that works as a mediator to other systems. Optimizing something for human use is not always a key requirement. So, the necessity of involving end users in the Scrum process varies case by case.

5.2. Product demonstration occurs with no end user involvement

A product demonstration can be useful after short development iterations to better understand user needs. However, even though product demonstration is one of the core values of agile, in practice it is not happening. We found out from our study that in most cases, product demonstrations happen without any involvement of the end user. In few cases, users are involved but the session is not regular. Only when there's something big to demonstrate. Based on our results reasons behind not having a product demonstration are: 1) the outcome of a 2–4-week sprint lacks substantial significance for demonstration; 2) thinking of the product demo as a technical event only for the internal stakeholders; 3) lack of communication with end users.

5.3. Low focus on usability and usability is often ignored

Usability is not something that is only related to the aesthetic or visual design of the interface. Usability has a deep relationship with Human-Computer Interaction (HCI). There are several human factors in HCI and if they are implemented properly the usability of software could be improved. Studies have shown that even completed software increments have sometimes been rejected by the end user due to poor usability [29]. Our results contradict the previous research outcome [13], [29] on the importance of usability among Scrum practitioners as our study shows that almost all the participants agreed that usability or user experience is very important. But no matter how important usability or user experience is according to the interviewee's answers, their focus and necessity in the Scrum process vary based on the product and customer. As mentioned before, end users and customers can be the same group of people. The customer can be an internal or external stakeholder. The priority of usability depends on the nature of the customer. For external stakeholders, the priority is high: *"let's say if the customer or end user is some external customer or external client, then I would say the user experience should come first. But for internal customers the situation is a bit different. The expectations are different, so they just want something that works for them, so they don't care about the usability or such."* The same goes for the product a Scrum team is developing or the service they are providing. For the team who are

developing products to support internal stakeholders, usability is not the top priority. But if the team is developing something for an external stakeholder such as a mobile app for a bank's general users, then the priority is high. Based on the interviewee's answers, UX or usability is not important or a top priority task in the case of internal stakeholders who are also the customer and end users. According to [13], the first two factors of usability are: Effectiveness – Can the users solve their tasks with the software? Efficiency – Can the user solve their tasks without major problems? - means the product a team is developing should be accessible to the end user regardless of whether the end user is an internal or external stakeholder. This leads us to a new argument *that Scrum practitioners do not always have a very good understanding of UX and usability*.

A common reason for neglecting usability issues in Scrum is having a short time frame. Scrum sprints are very short, and the focus is on the delivery. According to [6, 25, 17], Scrum practitioners do not have enough time to take usability activities into account. Even if usability is in the backlog, it is not on the top and is often ignored as functionality is the top priority. Now it can be asserted that it's the product owner's responsibility to enhance product quality by focusing on usability aspects. It is indeed factual that the product owner is an indirect leader and has some authority to make decisions. However, a product owner rarely wants to be the only person making decisions for the whole team. This was identified as one of the challenges of a product owner. A product owner is often guided by the development team in what is technically right. Another challenge is to act as a communication link between end users and developers. Because there are development teams who do not know who their end users are [19]. It is challenging to prioritize usability when there's no clear understanding of the end users.

The output from a 2–4-week sprint is minor. So, there's no point in evaluating in a quantitative way. We found the same from our study by showing one of the main challenges of involving end users in Scrum by professionals is its own limitations such as not having any proper feedback session with end users: *"that could be then a nice thing that maybe we could have some lightweight user involvement methods that we could use inside the scrum process"*, short sprint cycles (usually 2-3 weeks): *"It might take a bit more time, so it doesn't always fit into the basic cycle of two or three weeks"*, focusing more on output not usability: *"well, scrum process isn't really it...it doesn't really take usability aspects or UX into account much."*

5.4. No formal usability techniques used in Scrum process

Professionals practicing Scrum often do not follow any systematic approach to usability. Scrum practitioners rely on informal qualitative evaluation. There's a difference between the top usability techniques and usability techniques commonly used by Scrum practitioners. Even though some techniques are effective, they are not being used in practical situations. Our results show that the commonly used usability technique by the participants is informal usability evaluation. Only one participant was using a specific user testing method (formal usability evaluation) in their ongoing project. Our result stays consistent with practitioners' top and frequently used usability techniques [11] which discussed that formal usability techniques are not used by practitioners even though practitioners think formal usability techniques are highly effective [18].

6. Limitations

The first limitation of this study is the sample size of the data. 11 interviews might be considered relatively small to generalize the findings to a larger population of Scrum practitioners or organizations. Also, the perspective of 11 individuals may not cover the diversity within the Scrum community. Secondly, the data was analyzed by one researcher and reviewed by two other researchers. Many aspects of the result might have been influenced by the researcher's assumptions, biases, and perspectives. Lastly, project type can influence the need to involve end user in the Scrum process. The findings of the study may be limited to projects that may not prioritize end user involvement such as infrastructure development or internal process improvement.

7. Conclusion

This study investigates the role of end users and usability in Scrum processes by conducting interviews with Scrum practitioners. The study addresses our research question by identifying four novel Scrum anti patterns. Anti-patterns identified in our research can provide insights to the Scrum practitioners in improving their Scrum process by 1) reflecting on their process and becoming aware of any harmful practices; 2) having a clear understanding of different stakeholders (internal, external) of the product, what are their roles and value; 3) reconsidering the role of a customer and end user in their Scrum process. Engage the right stakeholder at the right time; 4) seek ways to ensure end user's feedback such as by introducing frequent product demonstrations with stakeholders. Sprint review is an event to engage with different stakeholders and gather feedback. Our

recommendation is to avoid turning sprint review into a technical event only for the internal stakeholders. The outcome of a short sprint may be a minor update, in that case, practitioners should set a milestone where the increment would be worth testing. Involve end users from the beginning of the process when necessary to ensure objectives aligned with end user needs; 5) focusing on usability regardless of whether the product is for an internal stakeholder or external stakeholder; 6) using formal usability techniques (such as formal usability evaluation, prototyping, workshops, interviews, surveys) instead of getting feedback from online conversations; 7) managing time effectively so that usability activities cannot be ignored due to time limitation. The study contributes to the literature regarding usability in Scrum and extends the line of research by highlighting a significant gap: how end users are involved in the Scrum process, why end users are often neglected and what are the challenges of involving end users? Furthermore, we discussed certain limitations experienced by Scrum practitioners and the rationale behind specific practices. The outcome we derived from the interviews gives a good picture of why end user involvement and usability are often ignored and difficult in many organizations. The field of research is building a deeper understanding of the challenges and as future research, specific strategies such as lightweight methods for integrating usability aspects and end user evaluation within the Scrum framework, to mitigate the identified challenges are needed.

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Publication II

Hasan, M.T., Wolff, A. and Porras, J

Co-designing recycling solutions on campus:
A case study exploring openness, realism and empowerment of users in a Living Lab

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Co-designing recycling solutions on campus

A case study exploring openness, realism and empowerment of users in a Living Lab

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The Living Lab concept is often distinguished from other user-centered and co-innovation approaches by an aspiration to adhere to principles of openness and empowerment of users, with a focus on innovating ideas that are realistic to the real needs of end-users. However, when it comes to implementing a living lab approach it is sometimes unclear how to put these principles into practice. In this case study, we introduce a framework for mapping three living lab principles to co-design stages within a single co-design framework and describe its use in innovating solutions for solid waste management on a university campus as a pre-cursor to scaling up the approach across the city. We present an evaluation of the framework in use against evidence of adherence to the three living lab principles of openness, empowerment of users, and realism. This work may act as a reference point for researchers to start exploring which concrete co-design activities have better impact in achieving living lab aims.

CCS CONCEPTS

Additional Keywords and Phrases: Living Lab, co-design, solid waste management, recycling, university campus, openness, realism, empowerment

1 Introduction

A living lab is a user-centric open innovation approach where users are involved in designing and developing a product. Instead of setting limitations on people's involvement, this approach invites them to actively take part, as equals in different stages of the design process. Participation may take place over a while and through different methods. The goal is to solve issues, generate ideas along with users, and test the generated ideas in their real life, thus being more responsive to users' real needs than more top-down design processes. In essence, the practices of a living lab are related to direct and indirect collaboration with users [1]. To make co-creation successful within a living lab approach designer need to utilize a wide range of knowledge. This includes knowledge related to:

- Facilitating long-term co-creation with users" [7], such as: setting up environments to facilitate and support collaborative innovation, often in the form of workshops; interacting and collaborating with a wide range of stakeholders; and motivating participation to innovation activities.

- How to co-design whilst practicing adherence to core values of living labs, such as realism based on the open innovation approach and the empowerment of users within the co-creation activities.

These types of knowledge are predominantly procedural knowledge often known as “know-how”. However, this “know-how” is not always shared with and amongst living lab practitioners and thus, living lab practice mostly depends on practitioners' education and experience. Yet there is little guidance on how to realize these principles in practice through co-design processes. This may make it hard to achieve a successful living lab approach for those who have not built up this knowledge over a while and through practice [6].

In this study we have extracted the core principles of living lab [5] and phases of co-design [8] via a literature review and developed a new framework that combines key principles of these two approaches. Furthermore, we validate the framework through a case study where a university in Finland was turned into a living lab and a co-design approach was used to improve the solid waste management situation of the University. Solving waste management on a large campus with many different types of people who use the facilities with differing motivations/language/cultural backgrounds is a wicked problem and co-design and living lab approaches are considered appropriate for tackling such problems. The original context for the living lab was to design a new smart waste management system, for example using IoT. However, in our case study we describe a design process that was intentionally agnostic to the type of solution being developed and whilst inviting technology-based solutions allowed all ideas to be proposed.

2 Development of a conceptual framework based on the principles of Living lab & Co-design

Nowadays designers are facing complex design problems. Co-design is a design process where participants from different mindsets and backgrounds are involved. It is distinguished from other types of participatory design process through the act of designing with users, rather than for them. Co-design typically progresses through a number of stages which [8] define as:

- **Pre-design phase:** exploring and understanding the problem, often by learning current experiences of users.
- **Generative phase:** a co-design phase of ideating solutions, often with the support of toolkits.
- **Evaluative phase:** evaluating proposed solutions.
- **Post design phase:** completion activities, which can include understanding how people use a finished product, sometimes leading to new design activities.

Co-design as a concept is supported by a wide array of validated practical tools and methods that support co-design practitioners in different stages of design.

Living labs are defined as “open innovation environments in real-life settings, in which user-driven innovation is fully integrated within the co-creation process of new services, products and societal infrastructures” [4]. All over Europe, living lab is now an umbrella concept for multiple sets of innovation that aim to produce solutions that are responsive to and closely fit to the needs of the users [3]. Co-design is an integral part of a living lab, but whereas the core of the co-design philosophy is based around the principle of designing with users, the living lab approach integrates three further key principles [5]. These are:

- **Openness:** the innovation process should be as open as possible, bringing a diversity of perspectives and enough power to achieve progress quickly. People should be able to join, no matter what their backgrounds are.
- **Realism:** the results should be compatible for the real markets, so it is mandatory to encourage genuine situations and behavior as much as possible. Realism is important also because focusing on actual users in real life situations distinguishes living lab from other co-creation environments.
- **Empowerment of users:** Engaging users in the design process is fundamental for innovating ideas in a preferred direction depending on the desires and needs of the users. Effectiveness of Living Labs depends on participants/users’ creative ideas. So, this is very important to increase the motivation of the users so that they engage in the process.

Of these, the principle that most distinguishes living lab from other co-creation approaches is that of realism, which [5] identify as also being the hardest to achieve. They point to the need to gather more knowledge in this area as well as to understand better how the three principles are related.
One issue is that living lab is a relatively new research area. Therefore, unlike co-design the number of theories that support understanding the concept is very low [2]. The situation is the same when we are talking about methodologies, methods, and tools. This points to a need to create a wider range of concrete tools and methods from which to draw when trying to implement living labs and when aiming to meet these core principles.
In this case study paper, we introduce a framework that integrates the living lab principles of [5] to different co-design stages of [8] to demonstrate how existing co-design methods and tools may be used to support living lab principles. We then describe one instantiation of this framework and evaluate it against the core living lab principles of openness, empowerment of users and realism.

2.1 A conceptual framework based on the principles of Living lab & Co-design

The proposed framework is illustrated in Figure 1. All three principles are present in four co-design phases. The framework indicates that openness should already be considered when defining activities for the pre-design stage. As such, it is important to choose activities at this stage that are already inviting multiple perspectives to the problem and which will inform the design of actual co-design activities in the generative phase. Next, it is during this generative phase that empowerment of users starts to occur, as different stakeholders are invited into the creative process. From the framework we can see that openness is decreasing from pre-design phase to post design phase and empowerment of user is increasing from pre-design to post design phase. Finally, although it is intended that the entire co-design process itself will produce realistic outputs. realism can best be validated by designing with a view to evaluating outcomes in the wild. All of these considerations will have an impact on which co-design methods are chosen and how the process is structured.

3 Case study

In our case study we turned LUT university in Finland into a Living lab and performed a co-design approach to improve the solid waste management situation inside the university. Solid waste management in this scenario refers to the recycling of non-food waste on the university campus including paper waste, food packaging and

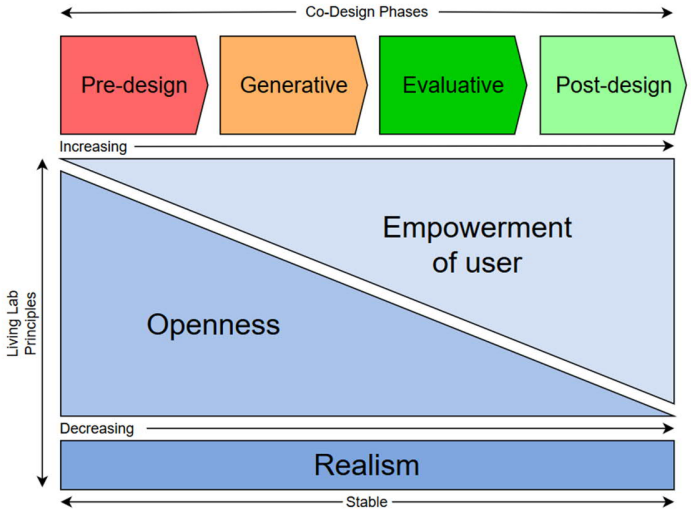


Figure 1: Conceptual framework based on the principles of Living lab & Co-design

coffee cups, and drink cans. We have different waste bins for different kinds of waste in our campus but not everyone is using the proper bin. So, the focus of this study was selective sorting and how it can be improved. We followed the new framework throughout the whole process. We designed our research steps (see Figure 2) by reflecting on the co-design approaches - which comprised the four phases of pre-design generative, evaluative phase and post design – in the context of the three living lab principles. All these phases are interdependent. The study lasted for six months.

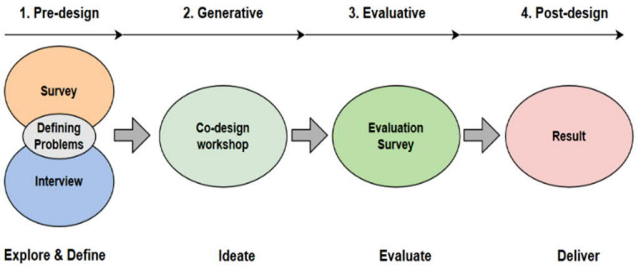


Figure 2: Research steps

3.1 Survey

To learn public awareness about solid waste management and get precise knowledge on common waste management behavior of an individual, we conducted an online survey regarding the issues related to waste recycling inside the university campus. As we were using the LUT campus as a living lab, all the participants were LUT students and staff. The survey occurred in March 2020. Several methods were used to approach the participants. Firstly, the link of the survey was sent through personal email to all known persons who study or work at LUT, with proper introduction and details of the survey. Secondly, the link to the survey and proper description was published in the university intranet. Lastly, the invitation was posted into the international group of LUT on social media. The survey mainly targeted people who are, or were, related to the university. In total 93 participants including students and staff took part in the survey. The age of the respondents was 17-24 (26%), 25-34 (42%), 35-44 (18%), 45-64 (14%). By analyzing participant's responses, it was confirmed that the current waste sorting methods need to be upgraded because a significant number of participants answered that they are not using proper bins. A new design may help to overcome the current situation. Responses from each participant were different. By manually clustering different responses we identified four unique user groups that we represented as user personas. Each of these user groups has some different perceptions, motivations, and frustrations.

3.2 Interview

We conducted three interviews with people from LUT based on a common set of questions that were designed and agreed amongst the research team beforehand. Participants were a newly appointed sustainability manager of the university, a person responsible for university cleaning and the university's head of development program. A single researcher conducted all the interviews over a conference call, each of which was recorded for later analysis. Through the expert interviews we gained information about the current waste management situation of the university, the biggest challenges and what they thought could be effective to change the current waste sorting situation. To show them what students and staff were thinking about the current condition we also presented the four user group personas. This helped them to open up a bit more and discuss the issues they were facing.

3.3 Co-design workshop

The co-design workshop was set up to explore, find and create possible solutions to improve the current waste management situation. From the interviews and survey, we already identified some common issues regarding the waste

management of the university. The aim of this workshop was to find solutions which can minimize those issues. Because of COVID-19 we conducted the workshop with only 9 participants. We invited five master's student, three university stuff (Junior researcher) and one restaurant owner of the university. We also planned to invite the three participants from our interview session, but it wasn't possible due to restrictions. We conducted the co-design workshop in June 2020 at LUT university. Because of COVID-19 we had to follow safety procedures and social distancing, which influenced partly the activities that were possible. The whole session was divided into the following parts:

- Conversation starters (20 mins)
- Framing the design challenge (15 mins)
- Storyboard (15 mins)
- Tour inside the university (30 mins)
- Introducing the user groups (15 mins)
- Brainstorming (30 mins)
- Peer review and feedback (30 mins)

Conversation starters. We started the event with some introduction and greetings to make the atmosphere comfortable and relaxed. We then started explaining our topic, our goal and why we were doing this workshop.

Framing the design challenge. The second activity was to frame our design challenge. We presented all issues as design questions, described what impacts we were trying to achieve and also presented some possible solutions. These solutions were based on the survey results and interviews. Framing the design challenge helped us to get off on the right foot. We also had some discussions with the participants here. Participants asked questions about the problems and our suggested solutions. This session made them open up their thoughts about the topic a little bit. We also introduced examples of existing smart solutions to support participants who may not know much about technologies for smart systems, or how to design them.

Storyboard. The third activity was the storyboard. Participants encountered a short story of a student who's trying to dump his waste in the recycle bins. The scenario showed what type of difficulties the student faces while selecting the correct bin for a waste.

Tour inside the university. This session was basically a competition among the participants. There were nine participants and we divided them into three groups. Groups were formed beforehand according to participant's education background. For example, we put a participant whose background is IT in every group, other participants were from a different background. This type of diverse group can help to develop different ideas. This grouping was maintained throughout the rest of the co-design session. The objective of the game was to find different bin location inside the campus, identify some problems with that recycling point and at finally take a selfie at that recycling point with all the group members. The main idea was to let the participants find different recycling points and identify the issues of that point by themselves. We also assigned points for different tasks. Finding a bin location is 3 points, each identified problem gave them 1 point and taking a selfie with all group members is 1 point. At the end, the team with maximum points won the competition. This competition was a part of problem identification process.

Introducing the user group. To help the participants think more deeply about the problems, we presented the personas of the four user groups that we created based on our survey results. We asked all the participants to study the personas and find which persona was closest to their personality (Figure 3, left).

Brainstorming. This was the most important part of the workshop. In this session participants worked as a group and ideated some solutions (Figure 3, middle). We maintained the same groups as during the competition. Each group was given a task to think about the problems they had encountered so far and create a design of a solution which could help. Related to the previous task, each of the participants had also chosen one persona which reflected their own behavior and tried to make solutions for that persona's frustration. As all the personas had different and multiple frustrations one solution couldn't help everyone. That's why participants came with multiple solutions.

Peer review and feedback. This was the last session of the workshop. After finishing the brainstorming session one member from each group came and gave a short speech about their project. Participants used papers, pencils, colorful pen, and other necessary stuffs to sketch and write down their ideas (Figure 3, right). They explained why they choose that solution and what kind of benefits were possible from that solution. After presenting the solution, members from other groups delivered their comments and feedback about the solutions.

4 Results

As we found multiple different solutions from the workshop participants, selecting just one idea from them was not a good decision. Also just implementing one change will not solve the present waste sorting situation. According to the conversations we had with the interviewees we found that several things could potentially



Figure 3: **Co-design activities**

improve the solid waste management situation. Finding a completely new solution is not the only option. Implementing a small change in the current system could also lead to big improvements on the current situation. For example: adding different types of recycle bins, proper labeling on the bins, changing the location and placements of the bins. Taking these issues into account we decided to categorize the solutions into different solution types and collect feedback on them through a follow up evaluation survey that went out to staff and students in the university. The survey respondents were asked to choose both their favorite type of solution as well as the best idea within that solution type. 36 people responded to this second survey, which was performed in July 2020 which is a common vacation month and therefore the number of respondents was lower. Among the participants 44% were staff members, 3% were bachelor students, 33% were master's students, 14% were doctoral students and 6% were other people related to the university. The solution types and different solutions are shown in Table 1.

Table 1: Proposed solutions from the workshop

Solution Type	Proposed solutions
Implementing a new system	KIOSK (Digital display)
	Smart recycling points
	Smart recycle bins which will give feedback
Implementing a change	Biodegradable coffee cups
	Bin sequence
	Change bin location
	Every building should have every type of bin
	A map
	Footstep direction
	Reusable cups with LUT logo
Attachment to the current system	Put popular waste image in recycle bins
	Different symbols and colors in waste and waste bins
	Increase the amount of commonly used bins
Raising awareness	Tutors can show students different recycling points
	Introducing short courses
	Task in orientation week regarding waste sorting
	Organize workshop or teaching events for new students

The favorite solutions based on the result of the evaluation survey were:

- **Implementing a new system:** A smart waste bin which will give feedback. Students will know if they are making mistakes. Sooner or later they will learn from feedback.
- **Implementing a change:** Introduce biodegradable coffee cups to reduce paper and plastic waste.
- **Attachment to the current system:** Visual images of the most common wastes on the recycle bins to identify what kind of waste they have and which bin they should use without confusion.
- **Raising awareness:** Arranging programs for the new students where they will learn about waste problem and how to sort it properly before dumping.

5 Reflection on key principles of the framework in case study

In this section we will explain how the new framework helped in our research process. As mentioned earlier the framework is based on three principles of Living lab (Openness, Empowerment of User, Realism) and four stages of Co-design (Pre-design phase, Generative phase, Evaluative phase & Post-design phase).

5.1 Openness

The first principle, openness, in a living lab highlights that the innovation process should be as open as possible. It's a foundation of the process user-driven innovation. Openness invites multiple perspectives to bring power in the development process. The pre-design phase of co-design is all about people's experiences. Researchers during this phase can gain a clear view of public experience. Reason why openness is in the peak level during this phase. In our case, we performed a survey among staff and students to find out their experiences, thoughts, frustrations, and expectations towards the current waste management system. The survey was made to be as open as possible and invite different perspectives. These were then curated into four user personas that represented this data and essentially allowed the voices of 93 others to be heard in later stages of co-design, even when they could not be present. This was especially important as COVID restricted numbers allowed in co-design. This approach demonstrates one way to achieve a measure of openness in a living lab.

5.2 Empowerment of users

The second phase of co-design is the generative phase. In this phase, new design opportunities are ideated. In our case, we performed a co-design workshop inside the campus with students and staff from different backgrounds. Together they generated several solution ideas about the current waste management situation by participating in games and different events that were arranged during the workshop. The second principle of living lab starts to reflect mostly from this phase. Empowerment is defined as a person's capacity to make effective choices; that is, as the capacity to transform choices into desired actions and outcomes. The extent or degree to which a person is empowered is influenced by personal agency (the capacity to make purposeful choice) and opportunity structure (the institutional context in which the choice is made) [9]. Generative phase reflects both on openness and empowerment of user. In the co-design session, participants had freedom of making choices. That's why the result was many different types of solutions, not just some stereotypical solutions, or even technology-based solutions which were the original intention. The key point of the principal of empowerment of users and openness in living lab is to make inventions based on human needs and to use the creativity of users, which was perfectly visible in this phase. The third phase of co-design is the evaluative phase. At this point, design decisions are already made and now it's time to verify the ideas by using prototypes or evaluation. The main idea of the evaluative phase is to evaluate the ideas with the help of real people and then implement the idea in post design phase to gather public reaction. Here user has the power to choose what type of solution they would like to see. Empowerment of user is in the peak level in these two phases. Because of the COVID-19 situation, making prototypes and monitoring user reactions was not possible, so instead we conducted a further survey to evaluate our findings from the generative phase.

5.3 Realism

Realism is the principle which separates Living lab from other systems. It highlights the importance of using real situations and behaviors to generate results which are valid for real markets. As the goal of co-design process is to make a realistic solution, realism is stable in all the co-design phases since all the steps we took was with real users. We did survey with people who were going to be affected by the result, interviewed people who are responsible and managing the issue, did co-design with people from different roles. But in terms of assessing realism it is important to note that many of the proposed solutions could have been implemented and tested quite easily in the post design phase: they were not outside the bounds of possibility. Also, through the evaluation survey it was clear that many of the solutions were popular even with those outside the initial co-design team of 9. The full process of our research is illustrated in Figure 4.

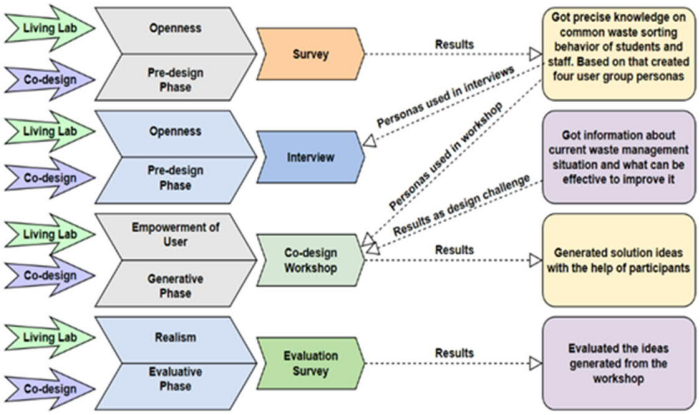


Figure 4: Research steps by merging living lab and co-design

6 Limitations

Our case study was related to the university and our goal was to improve the waste management situation inside the university. During the time of the research, the university was closed due to Covid-19 restrictions and only small number of people could be recruited which limited the data we could collect and the conclusions which could be drawn from them. The post design phase where prototype ideas should be implemented for monitoring user reactions was not accomplished through this case study, also due to COVID. It was therefore not possible to fully assess the outcomes in terms of realism, nor was it possible to have a fully open design process.

7 Conclusion

In this work we presented our experience from a living lab and co-design study with university students and staff in the context of finding solutions of a wicked problem which is solving solid waste management issue of a university. We have demonstrated that giving clearer consideration to how living lab principles and co-design activities work together may lead to better possibilities for evaluating whether living lab principles have been achieved. Based on literature there is no general methodology on how to involve users in living lab. Openness is one of the core principles of living lab, but very few studies succeed to reflect on it. Co-creation or empowerment of users is another essential characteristic of living lab but most of the time living labs do not support true user innovation [6]. Our aim was to show how living lab and co-design can be achieved at the same time and reflect the principles of living lab in each co-design stages by using the framework. We identified different phenomena and challenges from our experience of dealing with users in a real world context and offer some ideas for how to measure such successes, using our solid waste management living lab as a case study. We tried to keep the design process as open as possible and ensured empowerment of users. From our experience we noticed that it created a major impact on our outcome. The initial goal of this case study was to innovate an IoT-based design solution. In choosing to be agnostic to the type of solution being sought, we got a mix of both technology and non-technology-based solutions and this demonstrates that it cannot always be assumed that tech will be the answer.

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Publication III

Hasan, M.T., Wolff, A., Knutas, A., Pässilä, A. and Kantola, L.
Playing games through interactive data comics to explore water quality in a lake: A case study exploring the use of a data-driven storytelling method in co-design

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Playing games through interactive data comics to explore water quality in a lake

A case study exploring the use of a data-driven storytelling method in Co-design

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ABSTRACT

This case study explores the use of an existing data-driven storytelling method, called data comic, within a co-design process, to improve the sense-making of data. Data can often support a co-design process by providing additional insight towards a problem that is being solved. A large number of methods are available to facilitate different aspects of a co-design process, but when it comes to embedding curated data, alternatives are limited. Not everyone has the expertise to understand raw data and co-design scenarios are typically time-limited meaning that learning new data skills is not the focus. Therefore, appropriate data curation can help to bring data into a design process while reducing time spent on data manipulation or upskilling. At the same time, it is important that participants are encouraged to think critically about the data and not simply accept a pre-determined viewpoint. We have adapted an existing technique called Data Comic for curating data for use in a co-design situation by turning the comic panels into a card game so that they can be used interactively and collaboratively. In this paper, we describe and reflect on its use within a workshop with participants who were mostly teenagers. We led them through a process of engaging with curated data to make sense of water pollution in a Finnish lake since the 1970s and raise awareness towards environmental pollution. We present our results from the participant's feedback and workshop to reflect on how data comics played a role in sense-making.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Collaborative interaction.

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KEYWORDS

Data Comic, Co-design, Data driven storytelling, Data visualization, Data curation

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1 INTRODUCTION

A comic is an appealing medium of sequential art. While often considered as lightweight entertainment, comic forms have also been used to tackle serious subjects. For example, documentary graphic novels are, according to [19], "examples of the utilization of the comics medium to create documentary works that deal head-on with serious issues such as war, culture, and identity". Visualization of a subject matter into a comic strip form can help readers to transport into the story and travel through a fictional world. For this reason, HCI has accepted comics in various research contexts [16] such as explaining ongoing work with technology and research through design [8], data-driven storytelling [3], and provocation [15]. But the area is still underexplored. Comic visualization has also been used in co-design in many different aspects. For example, [2] used comic visualization as a fiction approach where they used comic and character cutouts for co-designing fiction. A participatory design technique comic boarding has been demonstrated as a productive approach in co-designing with children [12]. This technique provides comic structure and incomplete content to support individuals in coming up with ideas. Comic boarding has also been used as a participatory design method for productive brainstorming sessions with children [14].

There is no shortage of approaches for using comic visualization in co-design but when it comes to embedding curated data, options are limited. Data collected through activities such as citizen science,

or open data sets, can act as a useful resource in co-design to support problem-solving. The interest in involving general people in scientific research activities is increasing day by day in citizen science projects. Extensive research has been done to find the outcomes of citizen involvement but the road to ensure adequate participant engagement is still narrow. Human computer interaction (HCI) research can facilitate citizen participation and how it is done as the principles of HCI highlight design elements to attract and engage users [1]. But most often this data is difficult to understand for people who lack data literacy skills. This process of engaging with and making sense of the data should be fun, interesting, and straightforward in order to be engaging and to lower barriers to participation. The goal is not just to learn about the data but it's essential to perform further workshop activities.

One possible solution is to make this data simpler so that people can learn and interact with it. HCI has developed and designed techniques for visualizing data for behavior change such as encouraging physical activity [6], monitoring health issues such as diabetes [13], and in sustainability where sensors can monitor water activity [11]. Recently the HCI community got interested in the human experience of data [9] considering how this data reflects on regular social encounters and characterizes the past and future in new ways [10].

Data comics combine the visual language and storytelling concept from traditional comics with data visualization. They are intended to help to explain the insight about the data and make the visualization process simple [18]. Given that the comic form has been demonstrated to be useful within co-design, we propose that Data Comics may provide an answer to overcome barriers to embedding data into co-design. However, in its existing static form, a data comic may not lend itself very well to collaborative activities. We, therefore, propose that an interactive form of a data comic may lend itself better to this scenario and have designed such a comic in the form of card games which we have tested in a co-design workshop performed with teenagers.

In this paper, we focus on one research question:

What role do data curation and data visualization play to support sense-making within a time-limited situation, such as co-design?

2 CARD GAME

Two interactive card games were designed and used in this research a) Speed data-ing and b) Hauki Byte. Both games were designed based on the concept of data comics to help participants to understand two different elements in water: phosphorous and chlorophyll. In Speed data-ing key information on the cards were a) basic introduction of the element as a comic character; b) their levels and impacts in lake water; and c) causes of fluctuations in their level. Hauki Byte is based on a US television show card game where they start playing the game with a random card and the player has to guess if the next card (facing downwards) will be higher or lower. We used the same strategy here focusing on water data about a Finnish lake and using the phosphorous level as the value to guess. The Hauki is a freshwater pike that is common in Finnish lakes.

Both games have different rules. Speed data-ing has two cards containing data comics regarding two water elements (Phosphorous and Chlorophyll) (see Figure 1). Participants get a certain amount of time to read the data comic and learn about the elements. In Hauki Byte there are 5 cards in total. The timeline of each card is a decade, meaning that each card contains information of 10 years. We had data about the lake water from 1970-2020. So, five card timelines were divided as 1970-1979, 1980-1989, 1990-1999, 2000-2009, 2010-2019. Each card contains the level of Phosphorous and Chlorophyll in the water, the state of the lake during that decade, actions taken by the authority to improve the condition, and a visual representation of the phosphorous level (Graph). (see Figure 1). By this time students would already have ideas about Phosphorous and Chlorophyll from the first game (Speed data-ing). The game rule was to look at the first card (1970-1979) and guess the level of Phosphorous in the next card (1980-1989) based on the action taken by the authority. They could do that by looking at the information in the card and also using knowledge from the previous game, or even by looking at external information sources. All visualizations had been collected from websites or created from the raw water data. Information was provided by an expert in water quality who helped to understand the history of the lake and the actions that had affected the water quality.

3 DESIGN PATTERN FOR THE COMIC

The data comic strip used in this research was designed based on the design pattern of Benjamin Bach. In his paper, he facilitated comic creation from data by introducing some design patterns. The narrative power of comics comes from the combination of pictures and words. In data comic, pictures are the visual evidence of the data. Data comic combines both word and picture for better understanding. This combination can be a) paragraph with the picture, b) putting text into the picture or c) picture into text [3] (see Figure 2). We used the second combination, putting text into the picture for our data comics. According to [4] we selected the temporal pattern for the Hauki bytes comic panels, with each panel representing a decade's worth of data. This pattern is designed to highlight a temporal change in the data. We combined this with a narrative pattern where we explained the history, made a connection between the reader and the data, and lastly ended up with a question & answer for the subsequent card which involves the reader with the story (see Figure 3). Speed data-ing was intended to give background information that is needed to interpret the data sets in hauki bytes. The comic panels in these cards reflect the expose pattern under narrative pattern which introduces data context, problems, questions, demonstrating importance (see Figure 3).

To create a meaningful story, the sequence and connection between comic panels are very important. In the original data comic technique, the panels are generally ordered on a 2D canvas which can be a paper, screen, slide, etc. [4]. However, we made a key change in order to use these within co-design. We prepared each panel as a separate card. We introduced a simple game mechanic and flow of activities to encourage viewing the cards in an intended, coherent, linear order (in temporal order by decade) but this also meant that participants could pick up cards and look at them in any order or compare different decade against each other more easily (for example the first and last). There were two reasons: first, having separate cards made it easier for several people to engage with the comic at the same time. Second, we wanted to encourage critical thinking

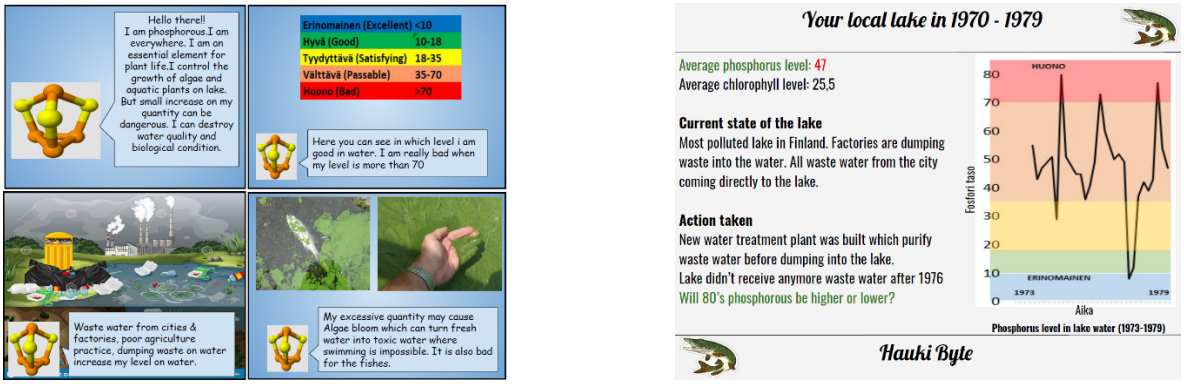


Figure 1: Speed data-ing(left), Hauki byte(right)

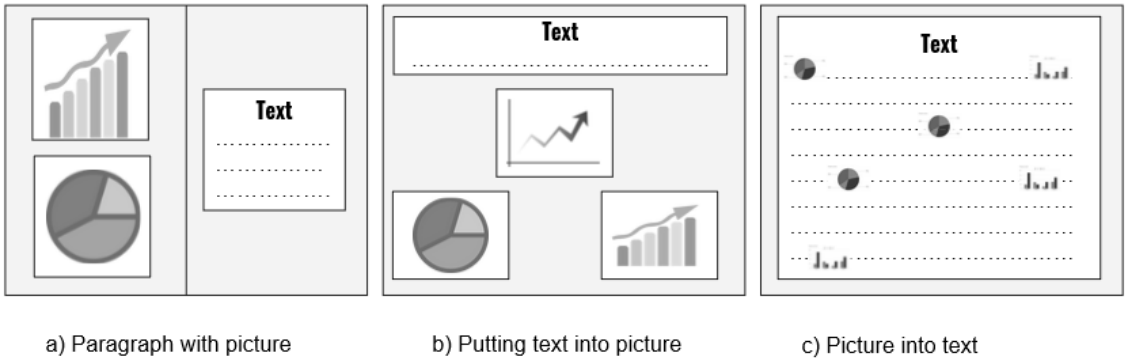


Figure 2: Combination of words and pictures (reproduced from [3])



Figure 3: Data comic patterns. Narrative pattern (left), Temporal pattern (right). (reproduced from [4])

and to give the impression that the data and interpretations were somehow not fixed and did not necessarily tell the whole story.

4 WORKSHOP

We ran a workshop to investigate 1) the usefulness of the card game; 2) whether the design of the card game helps to understand the topic; 3) whether data comic makes the learning process easier



Figure 4: Co-design workshop. Top left card game Hauki Byte, top right card game Speed data-ing, below left participants playing the card games, below right the mythological character Näkkitär (by Pihla Karhu, costume design by Mira Silvennoinen, props maker by Aino Kallioinen, hair and makeup by Janina Kuparinen, photo by drama educator and director of Theatrum Olga Lasse Kantola)

The workshop was based on a concept of data drama and included some arts-based methods that supported the process of interpreting data. Arts-based methods are based on specific form of arts. We can classify them into different categories such as visual art, moving images, performing art, live art, literary art, multiple method approach. In our research we used data drama which lies under performing art category [7]. This paper focuses primarily on the card games rather than the entire data drama experience, so we only introduce the aspects of the arts-based approaches that are needed to explain what happened. In the data drama participants were tasked with helping a mythological character called Näkkitär, who came from the future with a puzzle (see Figure 4). Some of the water in the future lake was good quality and some were not. Näkkitär thought that the answer to why this happened could be found from history, but all she had were some fragments of data and historical facts in the form of the data games and she could not read the data herself. She, therefore, asked for help from the participants, to analyze the historical water data through card games. Participants were given two physical sets of card games "Hauki byte" and "Speed data-ing" (see Figure 4). Then they were asked to play the games in two groups and ideate solutions for the mythological character. The groups were led not by researchers but instead by 12 older students who were part of a training program learning to use using arts-based approaches for youthwork through both theory and practical application. Conducting the workshop was part of their demonstration activities for the course and as such an important part of the process was in preparing these youth workers to lead the school students in the activities, which meant i) learning to make sense of how to read and interpret data comics ii) to co-create space for their use iii) find a way to use them as tools for facilitating discussions
around water issues iv) rehearsing using data comics: ideating, testing and rehearsing what drama convention to use, how and why.

4.1 Participants

The workshop was tailored for students from different backgrounds, although most of the students who chose to participate in the workshop were from a school specializing in art and drawing. Recruitment occurred as part of a larger event that was being organized in the city and to which a number of schools had signed up to participate. This workshop was one of six possible workshops that students from participating schools could choose to attend. There weren't any specific skill requirements and due to the COVID-19 situation the number of students was limited. Participants worked in groups and also individually.

4.2 Procedure

The goal of the workshop was to teach teenagers about lake water pollution and familiarize them with the history of a local lake, Vesijärvi, near Lahti in southern Finland which got polluted in the 1960s, and a restoration program that started in the 1970s. The goal was to tell the story in an interactive way with games based on data comics and raise awareness about water pollution's causes and effects. The main workshop was carried out over 6 hours during a weekday. The program started with an introduction to the workshop and a warm welcome from the organizers. Some introductory games were played to break the ice and get to know each other. After the introduction session participants were divided into two groups. Groups were formed based on a match-making game which was played in the introduction session. Each group had one or two student facilitators who facilitated the games and other activities. Groups were given two sets of card games (the first game was Speed data-ing and the second game was Hauki Byte). Participants played both of the games and had short discussions with the group facilitators regarding the contents of the games. As the workshop was formed as a part of the drama, in the end, they needed to help the mythological character who came from the future by predicting which of the two futures (good water

versus bad water) they imagine would happen based on the trends identified from the hauki byte cards (which had shown a steady improvement in phosphorous and chlorophyll) and then what could have happened to cause water pollution between now and the future, in the case where the water had got worse. Participants made different kinds of sketches based on the history of the lake which they learned from the card games.

5 RESULTS AND IMPLICATIONS

In total 10 students participated in the workshop and 12 youth education students from Theatrum Olga agreed to facilitate the whole workshop process. We succeeded to perform a face-to-face workshop but due to the covid restrictions, it was not possible to increase the number of participants. Yet it was enough to yield rich qualitative data to inform our research question.

5.1 Assessing the impact of comic visualization and card game through online surveys

To understand how the data comic and card games affected participants' output we performed surveys both with the participants and facilitators. The open-ended survey responses were analyzed with inductive thematic analysis [5]. Following their six-phase process, we coded the answers and generated themes informed by our research question. with the analysis, the process resulted in four different themes. They are summarized as follows. We highlight the main themes in bold and include some selected quotes for illustration. We also demonstrate some close-ended survey answers to support the themes.

Sensemaking. The main objective of including data comics as a visualization technique in the workshop was to make sense of the data that we curated. It happens to be a successful initiative because almost all participants understood the concept that we were trying to explain. 10 out of 10 facilitators agreed that the card games were simple and helped students understand the topic. Our goal of engaging citizens with the physical environment through relevant data was accomplished.

> "My opinion of the game surprised me positively, as it allowed me to learn and bring out the interest of the group and it was not very difficult for anything other than the initial awkwardness."

> "I learned a lot about the water lake and what things affect our water and how they relate to each other."

> "I learned different ways to express information, which is for example but diagrams, how to make it clearer and can express it in different ways and practical terms. I also learned something about the health of the lake and the dangers to its health."

Interesting and easy way of visualization. The process of visualizing the data should be simple and fun. Instead of reading some plain texts or looking at complex datasheets, comic visualization of data is interesting and easy to use. According to the responses it was fun, simple and something different which they have never experienced before. 10 out of 10 facilitators agreed that the game and visualization were simple ways of learning about water elements and 8 out of 10 facilitators agreed that it was easier to understand the history of the lake through this process.

> "The game was certainly a fun and different way for participants to learn and internalize things. I also learned a lot about this format and how it is utilized in teaching. Also, as a facilitator, I learned a lot about the topic and things stayed in my mind, I think the participants did too."

> "I liked the simplicity of card games the most. For simplicity, the cards are easy to use in different ways"

> "I liked the pictures of the cards the most."

> "I liked the illustration most informative"

Interactive. The role of the card games was to improve interaction and ensure active participation in the co-design process. Participants were given different tasks along with the card games which involved them in discussion and ideating solutions as a team. Participants mentioned that the games were effective to start a conversation and motivating teamwork.

> "About teamwork and how everyone got involved and participated well"

> "[the card games] work well to prompt a conversation"

Design issues. We also found some drawbacks in our design. Some of the participants quoted that it was difficult for them to understand the concept and gathering informative data to come up with a solution. Based on the result analysis the overall concept of understanding the data we curated and guessing the future based on data was complex for teenagers.

> "Evaluating what will happen in the future was difficult"

> "We could have reviewed more in-depth information about what phosphorous and chlorophyll are, where else you can find them etc. Now they remained only as chemical names. Lake history cards were all similar to each other, which makes sense, but made them boring in the long run. "

5.2 Assessing game activities through analyzing workshop video

The results of the survey mainly focused on demonstrating the effectiveness of comic visualization in card games. To understand how participants were engaging with the game activities within the broader context of the arts-based approach and data drama, we analyzed the video captured during the workshop activities. For analysis, we used a three-stage qualitative analysis process with an aim of generating an evidence-based description of the workshop process and guidelines for coding video material [17]. For initial codes, we used codes for the process [20], data comic literature, and teamwork codes as interpreted from Dickinson & McIntyre [21] by [22]. Based on the analysis, the game process had three distinct stages (see Figure 5). The first stage was understanding the concepts of water elements by playing speed data-ing. In this stage, the framing allowed participants to turn from students into communicators where they needed to explain the quantitative data to

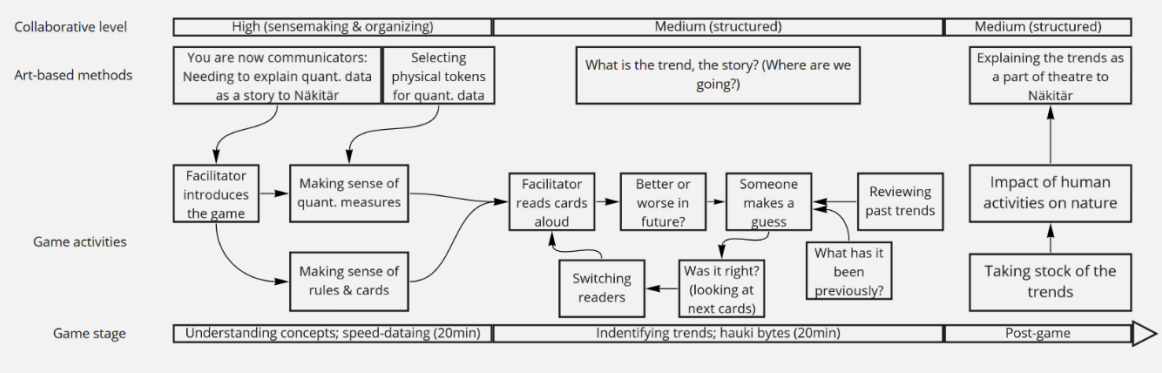


Figure 5: Video analysis of the workshop

the mythological character (Näkkitär) by selecting physical tokens (cards of speed data-ing).

Facilitators played their part by guiding students through the games. After making sense of the first data comic card game, participants went to the second stage. According to the analysis of the video, the collaboration level was high in this stage. In the second stage, participants played Hauki bytes. The facilitators helped participants to read the cards and based on this they needed to guess the future (in other words, the next decade). The process went on until they finished playing all the cards. We label this stage as 'identifying trends' as that was the game purpose and collaboration level was medium. Finally, in the post-game stage participants used their knowledge gathered through the games and explained to the mythological character (Näkkitär) what they had discovered, as a part of the ongoing drama. In the last stage of analysis, analytical coding, we distill the key characteristics and advantages of the data comics workshop through the lens of our research question. They are listed as follows.

Art-based methods added real stakes to game activities. Because of the story and art-based approach, participants engaged themselves more with the game.

Art-based method framing turned participants from students to communicators. Participants were tasked to explain the data we curated to a mythological character. It was a part of the learning process where participants turned themselves into communicators, as opposed to being passive recipients of data.

In the end, students identified trends and placed them in the context of human activities. At the end of the game, students understood the history of the lake and related their learning to a real-life context.

Student-driven setup, facilitation, and preparation were important parts of the process. The first stage with speed dataing was more collaborative and student-driven. Once the groups felt that they were ready and had understood the concepts, the sensemaking and game activities in the middle were quicker and more mechanical. However, the game and the story context added stakes to the process: Had there not been the cards and the narrative, the students might not have taken similar efforts to prepare.

6 LESSONS LEARNED AND INSIGHTS FOR PRACTITIONERS

Our initial research question was 'what role do data curation and data visualization play to support sense-making within a time-limited situation, such as co-design?'. Based on our experience in conducting this workshop and analyzing the process, we identify the following points:

- The combination of arts-based framing alongside card games was of the most importance. Neither approach would work well on its own. The card games supported participants to quickly understand the data, but the arts-based approach was crucial to support collaborative aspects of sensemaking, relating the data to own experience and coming up with ideas.
- A tool such as a data comic can be used to increase engagement and to quickly convey information, but the work has to be done somewhere and by someone - in this case originally by the water expert, then by the researchers in creating the games, and subsequently by the facilitators in learning how to use the cards with the final users, the school students. In each stage, the level of expertise and time required to engage with and understand the information was reduced. This implies that in a co-design scenario where many are already data and/or domain experts some of this effort may be unnecessary but if participants are neither data experts nor domain experts it is highly effective to convey information efficiently to inspire collaborative ideation.

7 CONCLUSION

Data can be a useful resource within a co-design process for understanding problems and ideating solutions. However, the complexity lies in making sense of this data. Especially for people with limited data literacy skills. Typically, co-design workshops are time-limited, and it is not possible to teach data literacy skills during the workshop. Data comic is one approach that can bring data closer to people in a short time. Existing comics have been framed as static artifacts which limits them to use within a participatory process. In our approach, we created interactive data comics in the form of a card game. Cards are quite typical tools used in co-design scenarios. Cards have the benefit that they can be picked up and passed

around. By curating data on cards, we make the data more tangible and by making the game interactive we can provide a storyline with different ways that people can interact with the story such as by encountering cards in different orders. In the case study, we explored the extent to which these interactive data comics help participants to understand the data quickly and without learning new skills. During the workshop, the participants worked in two separate groups to make sense of the information contained in the interactive data comics. Our findings were that overall the facilitators felt the games worked well and participants picked up the information quickly. The principles that Bach identified lent themselves well to being turned into interactive form, by making each panel into a separate card instead of curated into a fixed sequence. We did not test all the principles but could do so in the future. The game principles were easy to apply. The biggest resource was in having an expert to make sense of the data in the first place and to help to put it into the context, thereby providing a good starting point for others to start making their own sense. The raw data in the form of an excel sheet would have been too difficult for the participants to use without guidance. In the future, we would be interested to see how the cards support the exploration of different parts of the data such as framing new questions from it. E.g., not just predicting the future, but also finding out about different aspects of the lake which were in the dataset but not on the cards.

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Publication IV

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Integrating Lightweight UCD into Scrum: A Canvas Approach for Non-Designers

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Integrating Lightweight UCD into Scrum: A Canvas Approach for Non-Designers*

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***Abstract*— Scrum is the most widely used Agile framework. While it allows flexibility in incorporating various techniques, usability activities are often overlooked by Scrum practitioners. To address this challenge, we developed a Lean UCD Canvas to help practitioners create user centered design (UCD) artifacts such as personas and journey maps represented as storyboards using comic-style visualization. We evaluated the canvas in two sessions with 16 Scrum practitioners. The first session was a pilot test aimed at refining the canvas design and content; based on feedback, we updated the canvas. In the second session, we assessed the usability of the updated canvas via think-aloud protocols and semi-structured interviews. Our findings indicate that the canvas is effective, time-efficient, and can be easily used by anyone without design skills. However, the study has limitations, including the need to evaluate the canvas's practical implementation in real-world projects.**

Keywords—UCD, Scrum, Lean UCD, Agile UCD, Agile Software Development

I. Introduction

In software engineering, development methodologies are critical for successful projects. Unlike plan-driven methods, Agile methods are lightweight, avoiding excessive processes while maintaining structure. Generally, Agile methods either avoid upfront planning or allocate limited time for upfront design activities as the goal is to adapt quickly to changing requirements [23]. User-Centered Design (UCD) is a set of techniques and methods that place end users at the center of the development process to ensure the product meets their needs [24]. In UCD, often end users are not directly involved in the development process. Researchers or experts collect data, interpret findings, and use various techniques to define user requirements [24]. The primary goal of UCD is to enhance usability and improve the overall user experience (UX) by ensuring the product aligns with user needs.

While UX professionals or designers are responsible for UCD activities, not all organizations can afford dedicated UX specialists or design teams. Large organizations may have UX professionals or design teams, but in small companies, UX professionals or designers are often shared among many teams or work part-time [23]. Moreover, successful Agile organizations rarely emphasize the explicit inclusion of a UX role in their teams [25], [26]. Scrum is the most widely used Agile development framework in the software industry [25], [23]. In Scrum-driven organizations, UX experts and design teams often face communication challenges. They may feel like outsiders, rather than part of the development process [29]. According to the Scrum guide [31], a team should be cross-functional, with members capable of contributing to all tasks. Thus, usability/UX related activities should not solely be the responsibility of UX experts or designers. Developers should also acknowledge the importance of usability/UX and consider the end user's perspective during development. However, integrating usability/UX activities into Scrum remains problematic in practice. Conducting these activities while maintaining Scrum's speed and simplicity is challenging. Previous research has identified challenges involving end users and integrating usability/UX practices within Scrum [21]. While Scrum does not restrict usability/UX activities, practitioners often neglect them due to time constraints [21] and a lack of skills [27]. A potential solution is to streamline usability/UX activities for easier integration into Scrum sprints and to train developers in basic usability tasks.

This paper addresses these challenges by proposing a lightweight, time-efficient approach to integrate UCD activities into Scrum. We introduce a Lean UCD Canvas designed to be used by anyone, regardless of their background in time-constrained scenarios to help produce two key UCD artifacts (persona and journey map) represented as storyboards using comic visualization. Comic visualization was chosen for its effectiveness in communicating curated data to non-domain experts [16]. We validated the canvas with Scrum practitioners by answering the following research question: *How can a lean UCD canvas assist non-designers in effectively adopting and applying UCD techniques within time-constrained scenarios?*

The paper is structured as follows. Section II covers the background and related work. Section III discusses research methodology. The results of the study are presented in Section IV and discussed in Section V. Section VI concludes the study.

II. Background and Related Work

Requirements from customers and data from user research are presented and shared with stakeholders to inform and guide design solutions. These can be effectively communicated through visioning, which helps facilitate team discussion and visualize how end users will accomplish their tasks with the new system. UCD techniques have proven effective for visualizing and representing product or software requirements. Some popular UCD techniques among Agile practitioners include [2], [1], [3]: 1) persona; 2) journey map; 3) storyboard. The persona concept originated from Alan Cooper [4] to guide designers in making informed design decisions. Persona is a fictional character that represents end users with similar goals, motivations, and frustrations. It helps to define software requirements based on what an end user wants rather than on what developers think an end user might

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want. Personas help developers to build empathy for the end user. Through personas developers are able to understand the end user's behavior, their needs, and how they will interact with the software. There is no standardized template for personas. A persona can be represented in a textual description or as a visual representation. Narrative personas describe end user needs from the proposed product while visual representation may include expressions [33].

A journey map is a visual representation of the step-by-step experience an end user has with software from initial engagement to completion. It documents the goal of the end user, journey phases, tasks, pain points, and opportunities. A journey map is identified as a pattern that can help to stay focused on the big picture and solve continuous requirements management issues [22].

Storyboards are scenarios expressed through graphical language to increase the ability to recognize and understand. There are different styles and usage of graphical representations, but comic visualizations are identified as the universal graphical depictions for storyboards [5].

A. Related Work

Much research has been conducted to integrate UCD activities into Scrum. One approach is 'Upfront UCD Design', where an additional dedicated UCD team works one iteration ahead of the development team [15], [28]. Another successful approach emphasizes communication and collaboration, with UI designers and developers working closely and exchanging feedback [15]. A different approach was proposed by Øvad et al. [27] and Øvad and Larsen [7]. They focused on teaching developers certain usability methods to apply them to their development process. U-Scrum is another popular approach by Singh [8], recommends including two product owners instead of one. One fulfills the responsibilities outlined in the Scrum guide while the other manages usability/UX tasks. One key observation is that most research on integrating Agile and UCD practices tends to focus on the evaluation phase. There is a gap in research exploring UCD techniques for requirement gathering & analysis.

Personas, journey maps, and storyboards have often been utilized concurrently or in conjunction with one another in numerous research. Integrating personas into storyboards is an effective approach and proposed by many researchers. McQuaid et al. [6] in their research used storyboards and personas with scenarios to create a pleasurable experience for library customers. They concluded that persona with scenario could be better with visualizations rather than just narratives. Sedeño et al. [9] introduced a method where they used personas and visualized their context of use by storyboards to create a new type of user story named Context-based Persona Story (CBPS). Kantola and Jokela [10] proposed a storyboarding approach called SVSb ('simple and visual storyboards') where they used personas as the main building blocks of the storyboard. Personas were also integrated with journey maps to produce a robust storyboard [11], [12]. Personas help to understand what type of users go through specific journeys and sequences of steps [13]. McCarthy et al. [14] developed a journey mapping tool during a healthcare project using a human-centered design research approach where they combined elements from personas, journey maps, and storyboards.

We are going to extend the line of research by introducing a Lean UCD Canvas that utilizes comic visualization to develop personas and journey maps represented as storyboards. By combining these elements, the canvas produces outcomes that leverages the benefits of all three methods: personas, journey maps, and storyboarding. This integrated approach provides a comprehensive and insightful representation of different end user groups, user requirements, pain points, and opportunities. The canvas is designed in such a way that anyone can produce UCD artifacts without any training or prior knowledge and is lightweight enough for use within a Scrum sprint.

III. Methodology

The Lean UCD Canvas is designed to guide users through a series of steps to visualize user requirements through UCD techniques: persona, storyboard, and journey map. Creating these artifacts individually can be a time-consuming process. To minimize effort, we created guidelines that follow the elements of storyboarding, comic design patterns, persona & customer journey maps.

A. Canvas design

We adhered to the guidelines outlined by Truong et al. [17] for producing storyboards and Bach et al. [18] for comic design patterns.

Storyboard panels. A user story needs to be broken into smaller sections which will be visualized in panels. According to Truong et al. [17], three to five frames/panels are optimal for one storyboard in most cases. If the scenario or the application requires more than five frames/panels, then it needs to be broken into small storyboards. We followed the rule of keeping the storyboard under five panels. We used a linear panel layout for comic design, as it is compatible with all comic design patterns and is suitable for visualizing the journey map. We chose to utilize either time-sequence patterns, which depict moments over time, or multiple explanations, which highlight different aspects of the data [18].

Inclusion of actors and emotions. Personas will be used as the actors in the storyboard. The level of information to be integrated into the persona depends on the product type and variety of end user profiles. According to Brangier and Bornet [19], a persona can have three types of data: the persona profile (demographic, introduction), aspects related to attitude and behavior (goals, needs, frustrations), and context of use. But it is not always necessary to have a full description of the persona. In our case, to keep it lightweight practitioners will create a persona in the first step by having them describe their own background, frustrations, and motivations through comic visualization (see Fig.1) and in the second step, practitioners will describe the persona's journey. The inclusion of human characters is necessary for getting feedback on the interaction experience. However, it should be avoided while collecting feedback on usability, technical, or aesthetic features. Showing only the interface helps to focus on smaller technical details [20]. Practitioners must be aware of the goal of the journey map and decide accordingly. In comic visualization, facial expressions are very important to show character emotions. In storyboarding, the priority of facial expression is not high as emotions are expressed in textual expression. In our case, both facial expressions and text are suggested as in journey maps end user feelings, and pain points are elements of the vertical axis.

Level of detail. Practitioners need to decide the number of objects and actors for a particular frame. Our recommendation is to provide only the essential level of detail necessary to demonstrate the system features (see Fig. 1). Including excessive details can lead to a loss of focus and result in wasted time [20].

Inclusion of text. Text can be included in panels either tagline narration or as speech and thought bubbles or levels as signs within individual panels (see Fig. 1). Practitioners need to be conscious about using text as it often influences the audience [20].

Touchpoints & User action. A journey map lists all possible touchpoints a persona may encounter while performing a specific task. Touchpoints are typically outlined in the horizontal axis, following the order of events over time. In our linear comic visualization, each panel represents a touchpoint and the necessary steps taken by the end user (user action). The vertical axis of the journey map can be more complex. There are many arguments on what the vertical axis should uphold. Some experts believe it should focus on empathy while others believe it should represent design opportunities, branding opportunities, and tasks of employees to meet customer expectations or customer objectives [22]. Our focus is on usability and user experience, so our vertical axis should illustrate end user action, pain points, opportunities, and end user feelings. As we will be utilizing the concept of storyboarding and comic visualization, our journey map may not include an explicit vertical axis. Instead, it will incorporate the elements typically represented along the axis (see Fig. 1).

B. How to Use the Lean UCD Canvas

The canvas is very simple and straightforward. It can be used either as an online tool in Miro or as a printed version. There are two steps in this canvas:

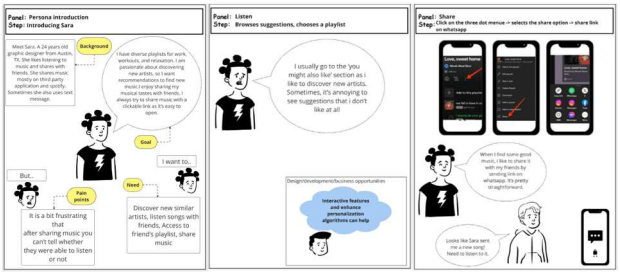


Fig. 1. Inclusion of actor(left), inclusion of text(middle), level of details(right)

Step 1 (Creating a persona). Practitioners begin by getting familiar with the persona concept and its key attributes. After the introduction, the canvas provides an example of persona based on Spotify. Using the example as a reference, users can create their persona using a predefined template. They can enter or write text directly into the template boxes or use sticky notes (in the case of the printed version). Additionally, users have the option to modify their persona image using resources from the material section. In the printed version, this can be done by cutting out images and icons from the material section and attaching them to their persona template. In the online version, users can simply copy and paste elements to their persona template. There is a hint section that provides practitioners with this information (see Fig. 2).

Step 2 (Creating a journey map). In this step, practitioners create a journey map for the persona they created in step 1. Like step 1, the process starts with an introduction to the journey map concept. Following the introduction, practitioners are provided with a detailed example of a journey map, which focuses on sharing music on Spotify. The example was created based on Spotify's actual journey map. There are many yellow-highlighted tips and instructions to guide practitioners through the process of creating their journey map. Using the provided example as a reference, practitioners are tasked with creating their journey map using a predefined template. Like step 1, they can enter or write text directly into the template boxes, use sticky notes (for printed version), and add icons or alter the persona character from the material section based on their needs (see Fig. 3).

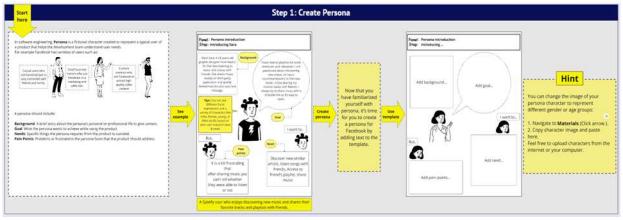


Fig. 2. Step1(Create persona)

C. Canvas Evaluation

We conducted two evaluation sessions involving a total of 16 Scrum practitioners. In the first session, we conducted a survey to gather feedback, while in the second session, we used think-aloud protocol and semi-structured interviews to gain deeper insights. The first session was a pilot test with four practitioners. In this session, practitioners were provided with a link to the Miro board containing the canvas and given detailed instructions. Practitioners were tasked with creating a persona and journey map using canvas. The pilot test was conducted entirely unsupervised, with feedback collected via a survey. Practitioners were asked about the design, clarity of instructions, and overall concept of the canvas. The pilot test

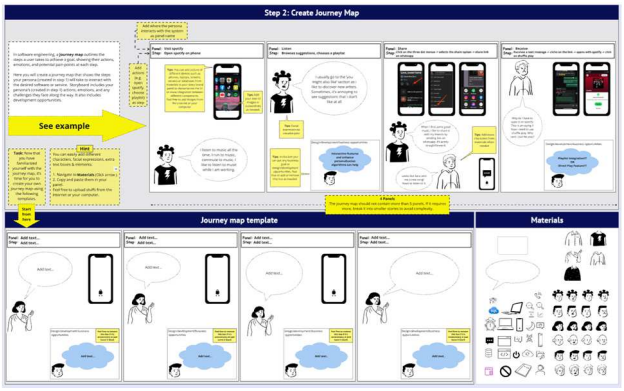


Fig. 3. Step2(Create journey map)

was mainly conducted to gather initial insights and refine our design for further evaluation. Next, we conducted the second evaluation session with twelve practitioners. This session was supervised by one researcher and followed by semi-structured interviews. Practitioners did the same task as before (creating a persona & journey map) and the questions covered similar

areas as those in the previous session. The researcher began the session with a brief background overview and guided practitioners through the task but didn't interfere if there was no question or confusion. We performed think-aloud protocols in the second session. Think-aloud protocols are a popular method in software engineering to evaluate the usability of software, interfaces, websites, and instructions. The main idea of this method is that a user needs to perform a set of tasks with the artifact tested and consistently articulate their thoughts aloud while performing the task. There are two types: 1) concurrent and 2) retrospective think-aloud protocol [32]. In our case, we used concurrent think-aloud protocols which are performing tasks and verbalizing at the same time. The session and interviews were conducted online using the Miro board and MS Teams. Think aloud observation and interviews were recorded by one researcher. The second session lasted from 30-45 minutes.

*a) Participants:*The 16 Scrum practitioners were chosen through personal networks, recommendations, and direct outreach via LinkedIn. All selected practitioners had practical experience from 1 to 15 years working in a Scrum environment as developers, senior developers, IT professional, Scrum master, project coordinator, and test automation engineer. Some also mentioned having experience working with the waterfall model. More information can be found in the table (see Table I).

TABLE I. DESCRIPTION OF PRACTITIONERS

No. of Professionals	Role	Experience (years)	Education
12	Software developer	1-10	Bachelor's and master's degree
1	QA-engineer	2	Master's degree
1	IT professional	15	Master's degree
1	Scrum master & developer	5	Master's degree
1	Project coordinator	3	Master's degree

*b) Data Analysis:*In the first session (pilot test), Webropol survey was used to gather and analyze feedback from practitioners. In the second session, recordings were collected, transcribed, and analyzed with inductive thematic analysis [30]. The collected data was reviewed and coded to identify themes in a software program for thematic analysis called NVivo. NVivo was used to organize and sort the dataset effectively. First, the recorded transcripts were reviewed and generated initial codes. Overall, 70 codes were generated. The next step was to sort and collate potentially related codes into themes. In NVivo, a parent node was created that represents a theme and grouped all relevant codes under the parent node as child nodes. Example: we created a parent node (or theme) "Instruction and example clarity" which consists of codes related to positive feedback on the clarity of the canvas's instructions and examples. Finally, we reviewed all generated themes and merged several themes into a broader theme aligning with our research question. Example: we combined "Instruction and example clarity" with another theme "Ease of use" to form a comprehensive theme titled "Usability" (see Fig. 4).

IV. RESULTS

Here we present the results we obtained from our two evaluation sessions. The pilot test was conducted mainly to refine our canvas design decisions. In the second evaluation session, we conducted inductive thematic analysis and through this process we derived four themes: Usability, design, relevance, effectiveness.

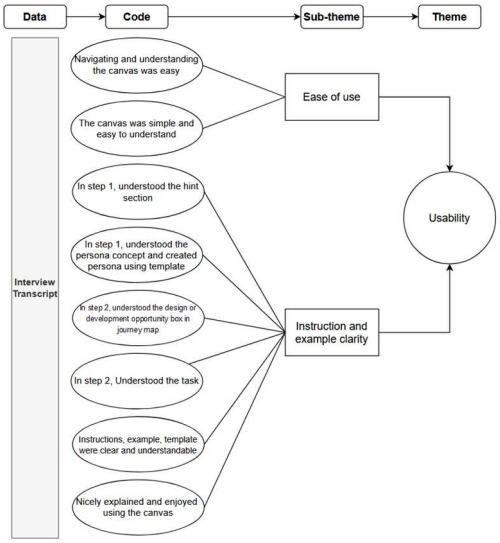


Fig. 4. Generating theme from codes

A. Canvas Design (Initial version)

From our pilot test, we tried to explore practitioners' perceptions of the canvas design, and the clarity of the instructions, examples, and templates provided within it. The first version of the canvas had a separate background and instruction section to explain the purpose and effectiveness through a story line. We also included extensive guidelines for creating a persona and journey map (see Fig. 5). However, feedback revealed that excessive text overwhelmed practitioners, leading to confusion, loss of interest, and distraction from the primary goal. Suggestions for improvement included minimizing text, simplifying instructions, and ensuring consistency between tasks. Based on this feedback, we made significant changes to the canvas design. Some of the major changes included removing the background and instructions section and removing extensive storyboard guidelines to reduce the amount of text, introducing numbered steps to ensure a clear and logical sequence. A major addition was examples of persona and journey maps (see Fig. 2 & Fig. 3). The new version of the canvas has two steps. Step 1: create a persona, step 2: create a journey map. The refined version was tested in the second session.

B. Usability and Design (Refined version)

We assessed the usability and design of the refined version by observing how practitioners navigated it, how often they encountered confusion or asked for assistance, and how they interacted with it. We also conducted think-aloud protocols and asked open-ended questions. We noticed that most of the

practitioners were able to navigate around the canvas without confusion. Our analysis showed that 10 out of 12 practitioners understood where to start the tasks and 9 out of 12 practitioners understood both tasks without any guidance. We also asked open-ended questions from practitioners. Almost all the practitioners mentioned that the instructions, examples, and templates were clear and nicely explained, and half of the practitioners replied that the canvas was very simple and easy to understand. However, a few practitioners were still having confusions. After analyzing, four common issues were found. While two of these issues were detected by half of the practitioners (6 out of 12), the other two issues were faced by two or three practitioners. The first issue is that half of the practitioners needed guidance on how to use the design/development/business opportunities box in step 2. The tips inside the box were unnoticed because they were at the bottom. We updated the design by putting the tips at the top. Another issue experienced by half of the practitioners was ignoring the hint section in step 1 due to lack of visibility. Some assumed it was meant to be used only after creating the persona; others found it unnecessary. We updated the design and instructions to make it more visible and understandable. Lastly, three practitioners needed small guidance to understand the journey map example and only two practitioners needed extensive guidance to understand step 1.

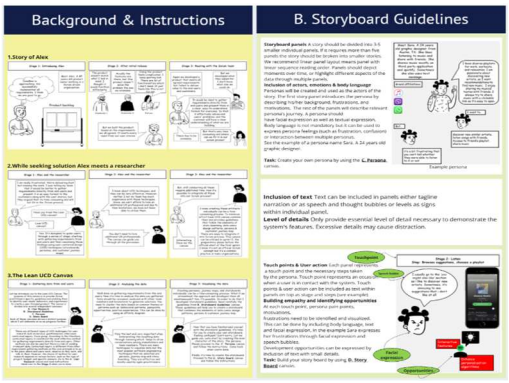


Fig. 5. Extensive guidelines & text from the initial version of the canvas

C. Relevance

We assessed relevance mainly in the second session. We asked practitioners how they think the canvas supports someone without a design background in understanding and applying UCD techniques. Most of the practitioners (8 out of 12) agreed that the canvas can be used by anyone. One practitioner was uncertain, stating: *"Difficult to say. But I think this is somewhat useful."* He further explained that he had never used a persona and journey map before and was unsure about the canvas application. We also asked about the usefulness of the canvas output. There were many positive opinions, and we categorized the answers into two themes, 1) practical application of the canvas output; 2) practical application of the canvas (see Table II).

TABLE II. USEFULLNESS OF THE CANVAS OUTPUT

Practical application of the canvas output	Practical application of the canvas
"I can easily create UML diagram from the journey map and that will help me go into the development process faster."	*"Before design, before implementation the canvas is a very good way for developer."*
"Would help to figure out user experience much better."	*"With my work, or if I want to open a business, this can be a good start"*
"It really helps to understand developer as developer thinks or at least I think there is only one type of user"	*"I think this kind of canvas will reduce gap between designers and developers."*
"This kind of thing gives you like different perspective because the user story is kind of dry."	*"I might start using and implementing it in my personal life."*

D. Effectiveness

Most of the practitioners from both sessions stated that the canvas provided them with all the necessary guidelines to create a persona and journey map. One practitioner shared his experience of working with these artifacts before where he struggled to grasp the concept. However, examples provided within the canvas significantly clarified ideas and helped in better understanding. Another practitioner expressed appreciation for the canvas approach, first creating a persona and then mapping out the journey of that persona. We also received positive feedback on comic visualization, which we have summarized in the table below (see Table III).

E. Time Efficiency

We tracked how long practitioners took to complete tasks in both sessions. In the first session with the initial version, practitioners reported on the survey that on average it took 1 hour and 30 minutes. In the second session with the refined canvas, the average time dropped to 26 minutes. The maximum time recorded was 38 minutes, while the minimum time was 11 minutes. The time varied among practitioners, as some engaged in discussions or asked questions, while others completed the tasks quietly.

TABLE III. PRACTITIONER FEEDBACK ON COMIC VISUALIZATION

Advantages of comic visualization	Practitioner quotes
Software development	*"It (The canvas) can play a vital role in software, like improvements"* *"this kind of output is very effective for development because developer can get clear message what actually needs to build."* *"(The canvas) can be effective in guiding designers and developers to create their product"*
Gathering feedback and engaging with non-technical stakeholders	*"(canvas can be used) when you want to actually get the feedback or want to get the understanding from the real end users"* *"Sometimes we have to represent our work to non-technical persons, in that case this can be an easier way"*

V. DISCUSSION

Our primary research question was: *"How can a lean UCD canvas assist non-designers in effectively adopting and*

applying UCD techniques within time-constrained scenarios?" The pilot session played a crucial role in refining our design decisions, helping us simplify instructions, reduce textual overload, and enhance the clarity of visual elements. Analysis from the second evaluation session showed that the refined canvas successfully enabled non-designers to create persona and journey maps by providing step-by-step guidance with visual aids. The usability and design of the canvas were effective with most practitioners navigating and completing the task without difficulty.

One of the primary motivations behind developing the canvas was to ensure it could be easily used by anyone regardless of their background or design expertise. We evaluated whether the canvas effectively addresses the problem it was designed to solve, while also assessing its practicality and overall usefulness. We asked practitioners whether the canvas helped them to understand what a persona and journey map are, and how to create them. Our analysis shows that practitioners from diverse backgrounds were able to use the canvas, got necessary guidelines to create UCD artifacts, and acknowledged its practical usefulness.

One of our hypotheses was that creating UCD artifacts by using the canvas would be time efficient. The average time of 26 minutes (including discussion between facilitator and practitioner) appears reasonable, considering that the concept of persona and journey map was new for many practitioners.

Comic visualization was a key element of canvas design. Practitioners found comic visualization not only effective but also highly satisfying, as reflected in their feedback. Although we didn't have any specific questions about satisfaction, the level of satisfaction was reflected in the practitioner feedback: *"it feels more interactive. So, people I'm sure, will love it."*; *"I personally prefer it this way than like some table with some text because this is more fun and easier to read"*; *"it (the canvas) felt interesting. If you would only give me writings without photos, I would feel boring."*; *"I felt like I got into the mood really."* It aligns with prior research that comic visualization is effective in visualizing data and interacting with non-domain experts [16].

Data collection and analysis were done by one researcher and reviewed by two other researchers. Some findings might have been shaped by the researcher's assumptions, biases, and perspectives. Although our study suggests that the canvas is useful, enjoyable, and time-efficient, its applicability in real-life projects was not assessed. Additionally, our results show that the canvas offers an easy and efficient way to create persona and journey maps. However, we did not directly compare the canvas's performance against traditional methods for creating these artifacts.

VI. Conclusion

Our goal is to provide Scrum practitioners with a tool (the canvas) that is capable of producing UCD artifacts in a lightweight manner without any prior knowledge in time-constraint situations. Through two evaluation sessions with 16 Scrum practitioners, we conclude that the research has effectively demonstrated the canvas' usability, design, relevance, effectiveness, and satisfaction in assisting practitioners in producing UCD artifacts. We identified several challenges after the first evaluation session, which were addressed through design improvements. Additionally, our analysis revealed that the average time required to use the canvas and create UCD artifacts is reasonable, especially for first-time users.

Although our study primarily focused on non-designers, it does not suggest that the role of UX specialists should be overlooked in the development process. Our findings suggest that basic UX practices such as producing UCD artifacts do not necessarily require dedicated UX specialists or designers. The canvas provides a structured yet lightweight approach that allows developers and other non-designers to consider usability/UX aspects in their development tasks. Meaning that user-centered decision-making in software development is possible without disrupting Scrum workflows.

Our research extends existing studies on integrating UCD/UX activities in Agile environment and combining persona, journey map, storyboard, comic visualization by introducing a lightweight canvas solution for non-designers. Unlike previous studies that focus on UI design [15] or evaluating a product after development [29], [27], [7], our canvas is designed to support the requirement gathering and analysis phase by helping teams incorporate end user needs early in the process and gather feedback from stakeholders. Persona, journey map, and storyboards have been combined in other studies [11], [12], [13], [14], but not in the way we have integrated them. Our approach uniquely represents these artifacts using comic visualization.

However, it is too early to say if the canvas fits into the Scrum sprint or helps improve the usability/UX of a product in a Scrum environment. While the canvas seems effective in a controlled setting, its efficiency in real-life projects remains unexplored. Therefore, future work will investigate its effectiveness in organizational settings by deploying it in a Scrum project. This investigation will analyze practitioners' adoption of the canvas, identify in which stage/Scrum event the canvas is useful, assess whether the generated artifacts help developers better consider end user needs and evaluate if the overall process leads to software with improve usability/UX.

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